EEGNet: A Compact Convolutional Neural Network for EEG-based Brain-Computer Interfaces

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May 17, 2018

Abstract

Objective: Brain computer interfaces (BCI) enable direct communication with a computer, using neural activity as the control signal. This neural signal is generally chosen from a variety of well-studied electroencephalogram (EEG) signals. For a given BCI paradigm, feature extractors and classifiers are tailored to the distinct characteristics of its expected EEG control signal, limiting its application to that specific signal. Convolutional Neural Networks (CNNs), which have been used in computer vision and speech recognition to perform automatic feature extraction and classification, have successfully been applied to EEG-based BCIs; however, they have mainly been applied to single BCI paradigms and thus it remains unclear how these architectures generalize to other paradigms. Here, we ask if we can design a single CNN architecture to accurately classify EEG signals from different BCI paradigms, while simultaneously being as compact as possible (defined as the number of parameters in the model). *Approach*: In this work we introduce EEGNet, a compact convolutional neural network for EEG-based BCIs. We introduce the use of depthwise and separable convolutions to construct an EEG-specific model which encapsulates well-known EEG feature extraction concepts for BCI. We compare EEGNet, both for within-subject and cross-subject classification, to current state-of-the-art approaches across four BCI paradigms: P300 visual-evoked potentials, error-related negativity responses (ERN), movement-related cortical potentials (MRCP), and sensory motor rhythms (SMR). *Results*: We show that EEGNet generalizes across paradigms better than, and achieves comparably high performance to, the reference algorithms when only limited training data is available. We also show that EEGNet effectively generalizes to both ERP and oscillatory-based BCIs. In addition, we demonstrate three different approaches to visualize the contents of a trained EEGNet model to enable interpretation of the learned features. *Significance*: Our results suggest that EEGNet is robust enough to learn a wide variety of interpretable features over a range of BCI tasks, suggesting that the observed performances were not due to artifact or noise sources in the data. Our models can be found at: https://github.com/vlawhern/arl-eegmodels.

EEGNet：一种用于脑机接口的紧凑型卷积神经网络

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2018 年 5 月 17 日

摘要

目标：脑机接口（BCI）能够通过神经活动作为控制信号直接与计算机通信。这种神经信号通常从多种经过充分研究的脑电图（EEG）信号中选择。对于给定的 BCI 范式，特征提取器和分类器都是根据其预期的 EEG 控制信号的不同特征进行定制的，这限制了其仅适用于该特定信号的应用。卷积神经网络（CNN），已在计算机视觉和语音识别中用于执行自动特征提取和分类，已成功应用于基于 EEG 的 BCI；然而，它们主要应用于单个 BCI 范式，因此这些架构如何泛化到其他范式仍不清楚。在这里，我们提出是否可以设计一个单一的 CNN 架构，以准确分类来自不同 BCI 范式的 EEG 信号，同时尽可能紧凑（定义为模型中的参数数量）。方法：在这项工作中，我们介绍了 EEGNet，一个用于基于 EEG 的 BCI 的紧凑卷积神经网络。 我们介绍了深度卷积和可分离卷积的使用，以构建一个针对脑电图（EEG）的特定模型，该模型封装了用于脑机接口（BCI）的已知 EEG 特征提取概念。我们比较了 EEGNet 在受试者内和受试者间分类方面的表现，与当前四个 BCI 范式（P300 视觉诱发电位、错误相关负反应（ERN）、运动相关皮质电位（MRCP）和感觉运动节律（SMR））中的最先进方法进行了对比。结果：我们表明，当只有有限训练数据可用时，EEGNet 在范式间的泛化能力优于参考算法，并实现了与参考算法相当的高性能。我们还表明，EEGNet 能有效泛化到基于脑电事件相关电位（ERP）和振荡的 BCI。此外，我们展示了三种不同的方法来可视化训练好的 EEGNet 模型的内容，以实现对学习到的特征的解释。意义：我们的结果表明，EEGNet 足够稳健，能够在各种 BCI 任务中学习多种可解释特征，这表明观察到的性能并非由于数据中的伪影或噪声源所致。

我们的模型可以在以下网址找到：https://github.com/vlawhern/arl-eegmodels。

Keywords: Brain-Computer Interface, EEG, Deep Learning, Convolutional Neural Network, P300, Error-Related Negativity, Sensory Motor Rhythm

1 Introduction

A Brain-Computer Interface (BCI) enables direct communication with a machine via brain signals [1]. Traditionally, BCIs have been used for medical applications such as neural control of prosthetic artificial limbs [2]. However, recent research has opened up the possibility for novel BCIs focused on enhancing performance of healthy users, often with noninvasive approaches based on electroencephalography (EEG) [3–5]. Generally speaking, a BCI consists of five main processing stages [6]: a data collection stage, where neural data is recorded; a signal processing stage, where the recorded data is preprocessed and cleaned; a feature extraction stage, where meaningful information is extracted from the neural data; a classification stage, where a decision is interpreted from the data; and a feedback stage, where the result of that decision is provided to the user. While these stages are largely the same across BCI paradigms, each paradigm relies on manual specification of signal processing [7], feature extraction [8] and classification methods [9], a process which often requires significant subject-matter expertise and/or *a priori* knowledge about the expected EEG signal. It is also possible that, because the EEG signal preprocessing steps are often very specific to the EEG feature of interest (for example, band-pass filtering to a specific frequency range), that other potentially relevant EEG features could be excluded from analysis (for example, features outside of the band-pass frequency range). The need for robust feature extraction techniques will only continue to increase as BCI technologies evolve into new application domains [3–5, 10–12].

Deep Learning has largely alleviated the need for manual feature extraction, achieving state-of-the-art performance in fields such as computer vision and speech recognition [13, 14]. Specifically, the use of deep convolutional neural networks (CNNs) has grown due in part to their success in many challenging image classification problems [15–19], surpassing methods relying on hand-crafted features (see [14] and [20] for recent reviews). Although the majority of BCI systems still rely on the use of handcrafted features, many recent works have explored the application of Deep Learning to EEG signals. For example, CNNs have been used for epilepsy prediction and monitoring [21–25], for auditory music retrieval [26, 27], for detection of visual-evoked responses [28–31] and for motor imagery classification [32], while Deep Belief Networks (DBNs) have been used for sleep stage detection [33], anomaly detection [34] and in motion-onset visual-evoked potential classification [35]. CNNs using time-frequency transforms of EEG data were used for mental workload classification [36] and for motor imagery classification [37–39]). Restricted Boltzman Machines (RBMs) have been used for motor imagery [40]. An adaptive method based on stacked denoising autoencoders has been proposed for mental workload classification [41]). These studies focused primarily on classification in a single BCI task, often times using task-specific knowledge in designing the network architecture. In addition, the amount of data used to train these networks varied significantly across studies, in part due to the difficulty in collecting data under different experimental designs. Thus, it remains unclear how these previous deep learning approaches would generalize both to other BCI tasks as well as to variable training data sizes.

In this work we introduce *EEGNet*, a compact CNN for classification and interpretation of

关键词：脑机接口，脑电图，深度学习，卷积神经网络，P300，错误相关负波，感觉运动节律

1 引言

脑机接口（BCI）可以通过脑电信号直接与机器进行通信[1]。传统上，BCI 已被用于医疗应用，如假肢的神经控制[2]。然而，最近的研究为新型 BCI 开辟了可能性，这些 BCI 专注于提升健康用户的表现，通常采用基于脑电图（EEG）的非侵入性方法[3–5]。一般来说，BCI 由五个主要处理阶段组成[6]：数据采集阶段，用于记录神经数据；信号处理阶段，对记录的数据进行预处理和清理；特征提取阶段，从神经数据中提取有意义的信息；分类阶段，从数据中解释决策；以及反馈阶段，将决策结果提供给用户。虽然这些阶段在不同 BCI 范式下基本相同，但每种范式都依赖于对信号处理[7]、特征提取[8]和分类方法[9]的手动指定，这一过程通常需要大量的专业知识或对预期 EEG 信号的先验知识。 也有可能，由于 EEG 信号预处理步骤通常非常针对感兴趣的 EEG 特征（例如，进行特定频率范围的带通滤波），因此其他潜在相关的 EEG 特征可能会被排除在分析之外（例如，带通频率范围之外的特征）。随着 BCI 技术向新的应用领域发展，对鲁棒特征提取技术的需求将只会持续增加[3–5, 10–12]。

深度学习在很大程度上减轻了手动特征提取的需求，在计算机视觉和语音识别等领域取得了最先进的性能[13, 14]。具体来说，深度卷积神经网络（CNN）的使用增长部分归因于它们在许多具有挑战性的图像分类问题中的成功[15–19]，超越了依赖手工特征的那些方法（有关最新综述，请参见[14]和[20]）。尽管大多数 BCI 系统仍然依赖手工特征，但许多近期研究探索了深度学习在脑电图（EEG）信号中的应用。例如，CNN 已被用于癫痫预测和监测[21–25]、听觉音乐检索[26, 27]、视觉诱发电位检测[28–31]和运动想象分类[32]，而深度信念网络（DBNs）则被用于睡眠阶段检测[33]、异常检测[34]以及运动起始视觉诱发电位分类[35]。使用 EEG 数据时频变换的 CNN 被用于心理负荷分类[36]和运动想象分类[37–39]。 受限玻尔兹曼机（RBMs）已被用于运动想象[40]。提出了一种基于堆叠降噪自编码器的自适应方法用于脑力负荷分类[41]。这些研究主要集中于单一 BCI 任务的分类，通常在设计中使用任务特定的知识构建网络架构。此外，用于训练这些网络的数据量在不同研究中差异显著，部分原因是收集不同实验设计下的数据存在困难。因此，这些先前的深度学习方法如何泛化到其他 BCI 任务以及可变训练数据大小仍不明确。

在这项工作中，我们引入了 EEGNet，一种用于分类和解释的紧凑型卷积神经网络。

EEG-based BCIs. We introduce the use of *Depthwise* and *Separable* convolutions, previously used in computer vision [42], to construct an EEG-specific network that encapsulates several well-known EEG feature extraction concepts, such as optimal spatial filtering and filter-bank construction, while simultaneously reducing the number of trainable parameters to fit when compared to existing approaches. We evaluate the generalizability of EEGNet on EEG datasets collected from four different BCI paradigms: P300 visual-evoked potential (P300), error-related negativity (ERN), movement-related cortical potential (MRCP) and the sensory motor rhythm (SMR), representing a spectrum of paradigms based on classification of Event-Related Potentials (P300, ERN, MRCP) as well as classification of oscillatory components (SMR). In addition, each of these data collections contained varying amounts of data, allowing us to explore the efficacy of EEGNet on various training data sizes. Our results are as follows: We show that EEGNet achieves improved classification performance over an existing paradigm-agnostic EEG CNN model across nearly all tested paradigms when limited training data is available. In addition, we show that EEGNet effectively generalizes across all tested paradigms. We also show that EEGNet performs just as well as a more paradigm-specific EEG CNN model, but with two orders of magnitude fewer parameters to fit, representing a more efficient use of model parameters (an aspect that has been explored in previous deep learning literature, see [42, 43]). Finally, through the use of feature visualization and model ablation analysis, we show that neurophysiologically interpretable features can be extracted from the EEGNet model. This is important as CNNs, despite their ability for robust and automatic feature extraction, often produce hard to interpret features. For neuroscience practitioners, the ability to derive insights into CNN-derived neurophysiological phenomena may be just as important as achieving good classification performance, depending on the intended application. We validate our architecture's ability to extract neurophysiologically interpretable signals on several well-studied BCI paradigms to show that the network performance is not being driven by noise or artifact signals in the data.

The remainder of this manuscript is structured as follows. Section 2.1 gives a brief description of the four datasets used to validate our CNN model. Section 2.2 describes our EEGNet model as well as other BCI models (both CNN and non-CNN based models) used in our model comparison. Section 3 presents the results of both within-subject and cross-subject classification performance, as well as results of our feature explainability analysis. We discuss our findings in more detail in the Discussion.

2 Materials and Methods

2.1 Data Description

BCIs are generally categorized into two types, depending on the EEG feature of interest [44]: event-related and oscillatory. *Event-Related Potential* (ERP) BCIs are designed to detect a high amplitude and low frequency EEG response to a known, time-locked external stimulus. They are generally robust across subjects and contain well-stereotyped waveforms, enabling the time course of the ERP to be modeled through machine learning efficiently [45]. In contrast to ERP-based BCIs, which rely mainly on the detection of the ERP waveform from some external event or stimulus,

基于脑电图（EEG）的脑机接口（BCI）。我们引入了在计算机视觉中先前使用过的深度卷积和可分离卷积[42]，以构建一个针对 EEG 的网络，该网络封装了几个众所周知的 EEG 特征提取概念，例如最优空间滤波和滤波器组构建，同时与现有方法相比减少了可训练参数的数量。我们在从四种不同的 BCI 范式收集的 EEG 数据集上评估了 EEGNet 的泛化能力：P300 视觉诱发电位（P300）、错误相关负波（ERN）、运动相关皮质电位（MRCP）和感觉运动节律（SMR），这些范式代表了基于事件相关电位（P300、ERN、MRCP）分类以及振荡成分（SMR）分类的多种范式。此外，这些数据集包含不同数量的数据，使我们能够探索 EEGNet 在各种训练数据大小上的有效性。我们的结果如下：我们表明，当训练数据有限时，EEGNet 在几乎所有测试的范式上都实现了比现有的范式无关 EEG CNN 模型更好的分类性能。 此外，我们展示了 EEGNet 在所有测试范式中都表现出良好的泛化能力。我们还表明，EEGNet 的性能与一个更针对特定范式的 EEG CNN 模型相当，但参数数量减少了两个数量级，这代表了对模型参数更高效的使用（这一方面在之前的深度学习文献中已有探讨，参见[42, 43]）。最后，通过特征可视化和模型消融分析，我们展示了可以从 EEGNet 模型中提取神经生理学上可解释的特征。这一点很重要，因为尽管 CNN 具有强大的自动特征提取能力，但它们往往产生难以解释的特征。对于神经科学从业者而言，从 CNN 衍生的神经生理学现象中得出见解的能力，可能与应用目标相关的良好分类性能同等重要。我们通过在几个经过充分研究的 BCI 范式上验证我们架构提取神经生理学可解释信号的能力，表明网络性能并非由数据中的噪声或伪信号驱动。

本文档的其余部分结构如下。2.1 节简要介绍了用于验证我们 CNN 模型的四个数据集。2.2 节描述了我们的 EEGNet 模型以及其他 BCI 模型（包括基于 CNN 和非 CNN 的模型）在模型比较中使用的模型。第 3 节展示了受试内和受试间分类性能的结果，以及我们的特征可解释性分析结果。我们在讨论部分将更详细地讨论我们的发现。

2 材料与方法

2.1 数据描述

BCI 通常根据感兴趣的脑电图特征分为两类[44]：事件相关和振荡。事件相关电位（ERP）BCI 旨在检测对已知、时间锁定的外部刺激的高幅低频脑电图反应。它们通常具有跨受试的鲁棒性，并包含典型的波形，能够通过机器学习高效地建模 ERP 的时间进程[45]。与主要依赖检测某些外部事件或刺激的 ERP 波形的事件相关电位 BCI 不同，

Paradigm	Feature Type	Bandpass Filter	# of Subjects	Trials per Subject	# of Classes	Class Imbalance?
P300	ERP	1-40Hz	15	~ 2000	2	Yes, $\sim 5.6{:}1$
ERN	ERP	1-40Hz	26	340	2	Yes, $\sim 3.4{:}1$
MRCP	ERP/Oscillatory	0.1-40Hz	13	~ 1100	2	No
SMR	Oscillatory	4-40Hz	9	288	4	No

Table 1: Summary of the data collections used in this study. Class imbalance, if present, is given as odds; i.e.: an odds of 2:1 means the class imbalance is 2/3 of the data for class 1 to 1/3 of the data for class 2. For the P300 and ERN datasets, the class imbalance is subject-dependent; therefore, the odds is given as the average class imbalance over all subjects.

Oscillatory BCIs use the signal power of specific EEG frequency bands for external control and are generally asynchronous [46]. When oscillatory signals are time-locked to an external stimulus, they can be represented through event-related spectral perturbation (ERSP) analyses [47]. Oscillatory BCIs are more difficult to train, generally due to the lower signal-to-noise ratio (SNR) as well as greater variation across subjects [46]. A summary of the data used in this manuscript can be found in Table 1

2.1.1 Dataset 1: P300 Event-Related Potential (P300)

The P300 event-related potential is a stereotyped neural response to novel visual stimuli [48]. It is commonly elicited with the visual oddball paradigm, where participants are shown repetitive "non-target" visual stimuli that are interspersed with infrequent "target" stimuli at a fixed presentation rate (for example, 1 Hz). Observed over the parietal cortex, the P300 waveform is a large positive deflection of electrical activity observed approximately 300 ms post stimulus onset, the strength of the observed deflection being inversely proportional to the frequency of the target stimuli. The P300 ERP is one of the strongest neural signatures observable by EEG, especially when targets are presented infrequently [48]. When the image presentation rate increases to 2 Hz or more, it is commonly referred to as rapid serial visual presentation (RSVP), which has been used to develop BCIs for large image database triage [49–51].

The EEG data used here have been previously described in [50]; a brief description is given below. 18 participants volunteered for an RSVP BCI study. Participants were shown images of natural scenery at 2 Hz rate, with images either containing a vehicle or person (target), or with no vehicle or person present (non-target). Participants were instructed to press a button with their dominant hand when a target image was shown. The target/non-target ratio was 20%/80%. Data from 3 participants were excluded from the analysis due to excessive artifacts and/or noise within the EEG data. Data from the remaining 15 participants (9 male and 14 right-handed) who ranged in age from 18 to 57 years (mean age 39.5 years) were further analyzed. EEG recordings were digitally sampled at 512 Hz from 64 scalp electrodes arranged in a 10-10 montage using a BioSemi Active Two system (Amsterdam, The Netherlands). Continuous EEG data were referenced offline to the average of the left and right earlobes, digitally bandpass filtered, using an FIR filter implemented in EEGLAB [52], to 1-40 Hz and downsampled to 128 Hz. EEG trials of target and non-target conditions were extracted at $[0, 1]s$ post stimulus onset, and used for a two-class classification.

范式	特征类型	带通滤波器	受试者数量	每个受试者的试验次数	类别数量	类别不平衡？
P300	ERP	1-40Hz	15	～2000	2	是，～5.6:1
ERN	ERP	1-40Hz	26	340	2	是，～3.4:1
MRCP	ERP/振荡	0.1-40Hz	13	～1100	2	否
SMR	振荡	4-40Hz	9	288	4	否

表 1：本研究中使用的数据集摘要。如果存在类别不平衡，则给出优势比；即：优势比为 2:1 表示类别 1 的数据占 2/3，类别 2 的数据占 1/3。对于 P300 和 ERN 数据集，类别不平衡受受试者影响；因此，优势比给出的是所有受试者的平均类别不平衡。

振荡脑机接口使用特定脑电波频段的信号功率进行外部控制，通常是非同步的[46]。当振荡信号与外部刺激时间锁定时，可以通过事件相关频谱扰动（ERSP）分析来表示[47]。振荡脑机接口更难训练，通常是由于信噪比（SNR）较低以及受试者间差异较大[46]。本文使用的数据摘要见表 1。

2.1.1 数据集 1：P300 事件相关电位（P300）

P300 事件相关电位是对新颖视觉刺激的一种标准化神经反应[48]。它通常通过视觉奇偶范式诱发出，在该范式中，参与者会看到重复的“非目标”视觉刺激，这些刺激以固定的呈现速率（例如，1 Hz）穿插着不频繁的“目标”刺激。在顶叶皮层观察到，P300 波形是刺激开始后约 300 毫秒观察到的一个大的正向偏转，观察到的偏转强度与目标刺激的频率成反比。P300 ERP 是 EEG 可观察到的最强的神经特征之一，尤其是在目标呈现不频繁时[48]。当图像呈现速率增加到 2 Hz 或更高时，通常称为快速序列视觉呈现（RSVP），它已被用于开发用于大型图像数据库筛选的脑机接口[49-51]。

这里使用的脑电图（EEG）数据已在参考文献[50]中有所描述；以下给出简要说明。18 名参与者自愿参与了一项快速视觉搜索（RSVP）脑机接口（BCI）研究。参与者以 2 Hz 的频率观看自然风景图像，图像中要么包含车辆或人物（目标），要么不包含车辆或人物（非目标）。参与者被指示在显示目标图像时用优势手按键。目标/非目标的比例为 20%/80%。由于脑电图数据中存在过多的伪影和/或噪声，3 名参与者的数据被排除在分析之外。剩余的 15 名参与者（9 名男性，14 名右撇子）年龄在 18 至 57 岁之间（平均年龄 39.5 岁）的数据进行了进一步分析。脑电图记录使用 BioSemi Active Two 系统（阿姆斯特丹，荷兰）从 64 个头皮电极以 10-10 系统排列进行数字化采样，采样频率为 512 Hz。连续脑电图数据在离线时参考了左右耳垂的平均值，并使用 EEGLAB[52]中实现的 FIR 滤波器进行数字带通滤波，滤波范围为 1-40 Hz，并下采样至 128 Hz。 在刺激开始后的[0, 1]秒内提取了目标和非目标条件下的脑电图试验数据，用于进行两类分类。

2.1.2 Dataset 2: Feedback Error-Related Negativity (ERN)

Error-Related Negativity potentials are perturbations of the EEG following an erroneous or unusual event in the subject's environment or task. They can be observed in a variety of tasks, including time interval production paradigms [53] and in forced-choice paradigms [54, 55]. Here we focus on the feedback error-related negativity (ERN), which is an amplitude perturbation of the EEG following the perception of an erroneous feedback produced by a BCI. The feedback ERN is characterized as a negative error component approximately 350ms, followed by a positive component approximately 500ms, after visual feedback (see Figure 7 of [56] for an illustration). The detection of the feedback ERN provides a mechanism to infer, and to possibly correct in real-time, the incorrect output of a BCI. This two-stage system has been proposed as a hybrid BCI in [57, 58] and has been shown to improve the performance of a P300 speller in online applications [59].

The EEG data used here comes from [56] and was used in the "BCI Challenge" hosted by Kaggle (https://www.kaggle.com/c/inria-bci-challenge); a brief description is given below. 26 healthy participants (16 for training, 10 for testing) participated in a P300 speller task, a system which uses a random sequence of flashing letters, arranged in a 6×6 grid, to elicit the P300 response [60]. The goal of the challenge was to determine whether the feedback of the P300 speller was correct or incorrect. The EEG data were originally recorded at 600Hz using 56 passive Ag/AgCl EEG sensors (VSM-CTF compatible system) following the extended 10-20 system for electrode placement. Prior to our analysis, the EEG data were band-pass filtered, using an FIR filter implemented in EEGLAB [52], to 1-40 Hz and down-sampled to 128Hz. EEG trials of correct and incorrect feedback were extracted at $[0, 1.25]s$ post feedback presentation and used as features for a two-class classification.

2.1.3 Dataset 3: Movement-Related Cortical Potential (MRCP)

Some neural activities contain both ERP as well as an oscillatory components. One particular example of this is the movement-related cortical potential (MRCP), which can be elicited by voluntary movements of the hands and feet and is observable through EEG along the central and midline electrodes, contralateral to the hand or foot movement [61–64]. The MRCP components can be seen before movement onset (a slow 0-5Hz readiness potential [65, 66] and an early desynchronization in the 10-12Hz frequency band), at movement onset (a slow motor potential [66, 67]), and after movement onset (a late synchronization of 20-30Hz activity approximately 1 second after movement execution). The MRCP has been used previously to develop motor control BCIs for both healthy and physically disabled patients [68–70]

The EEG data used here have been previously described in [71]; a brief description is given below. In this study, 13 subjects performed self-paced finger movements using the left index, left middle, right index, or right middle fingers. The data was recorded using a 256 channel BioSemi Active II system at 1024 Hz. Due to extensive signal noise present in the data, the EEG data were first processed with the PREP pipeline [72]. The data were referenced to linked mastoids, bandpass filtered, using an FIR filter implemented in EEGLAB [52], between 0.1 Hz and 40 Hz, and then downsampled to 128 Hz. We further downsampled the channel space to the standard 64 channel BioSemi montage. The index and middle finger blocks for each hand were combined for

2.1.2 数据集 2：反馈错误相关负性（ERN）

错误相关负性电位是指在受试者环境或任务中发生错误或不寻常事件后，脑电图出现的扰动。它们可以在多种任务中观察到，包括时间间隔生产范式[53]以及在强制选择范式[54, 55]中。在这里我们关注反馈错误相关负性（ERN），它是在感知到由脑机接口（BCI）产生的错误反馈后，脑电图振幅的扰动。反馈 ERN 的特征是视觉反馈后约 350 毫秒出现一个负性错误成分，随后约 500 毫秒出现一个正性成分（见图[56]的第 7 页的示意图）。检测反馈 ERN 提供了一种机制，可以推断并可能实时纠正 BCI 的错误输出。这种两阶段系统已被提议作为混合 BCI 在[57, 58]中，并已被证明可以提高 P300 拼写器在线应用中的性能[59]。

这里使用的脑电图（EEG）数据来自[56]，曾用于 Kaggle（https://www.kaggle.com/c/inria-bci-challenge）举办的“BCI 挑战赛”；以下将进行简要描述。26 名健康参与者（16 名用于训练，10 名用于测试）参与了一项 P300 拼写任务，该任务使用随机闪烁字母序列，排列成 6×6 网格，以诱发 P300 反应[60]。该挑战赛的目标是判断 P300 拼写器的反馈是否正确。脑电图（EEG）数据最初使用 56 个被动 Ag/AgCl 脑电图（EEG）传感器（兼容 VSMCTF 系统）按照扩展的 10-20 系统电极放置方案，以 600Hz 的频率进行记录。在我们进行分析之前，脑电图（EEG）数据使用 EEGLAB[52]中实现的 FIR 滤波器进行带通滤波，滤波范围为 1-40 Hz，并降采样至 128Hz。在反馈呈现后的[0, 1.25]秒内提取了正确和错误反馈的脑电图（EEG）试验，并将其用作两类分类的特征。

2.1.3 数据集 3：运动相关皮质电位（MRCP）

一些神经活动中既包含事件相关电位（ERP），也包含振荡成分。其中一个特定例子是运动相关皮质电位（MRCP），它可以通过手或脚的自主运动诱发，并通过中央和中央线电极沿对侧的脑电图观察到[61–64]。MRCP 成分可以在运动开始前（一个缓慢的 0-5Hz 准备电位[65, 66]和 10-12Hz 频段的早期去同步化）、运动开始时（一个缓慢的运动电位[66, 67]）以及运动开始后（运动执行后约 1 秒的 20-30Hz 活动的晚期同步化）观察到。MRCP 先前已被用于为健康人和身体残疾患者开发运动控制脑机接口[68–70]

这里使用的脑电图（EEG）数据已在参考文献[71]中详细描述；以下给出简要说明。在本研究中，13 名受试者使用左手食指、左手中指、右手食指或右手中指进行自我调节的手指运动。数据通过 256 通道的 BioSemi Active II 系统以 1024 Hz 的采样率进行记录。由于数据中存在大量信号噪声，首先使用 PREP 流程[72]对 EEG 数据进行处理。数据参考点连接到乳突，使用 EEGLAB[52]中实现的 FIR 滤波器进行带通滤波，频率范围为 0.1 Hz 至 40 Hz，然后下采样至 128 Hz。我们进一步将通道空间下采样至标准的 64 通道 BioSemi 布局。每只手的食指和中指区块被合并用于

binary classification of movements originating from the left or right hand. EEG trials of left and right hand finger movements were extracted at $[-.5, 1]s$ around finger movement onset and used for a two-class classification.

2.1.4 Dataset 4: Sensory Motor Rhythm (SMR)

A common control signal for oscillatory-based BCI is the sensorimotor rhythm (SMR), wherein mu (8-12Hz) and beta (18-26Hz) bands desynchronize over the sensorimotor cortex contralateral to an actual or imagined movement. The SMR is very similar to the oscillatory component of the MRCP. Although SMR-based BCIs can facilitate nuanced, endogenous BCI control, they tend to be weak and highly variable across and within subjects, conventionally demanding user-training (neurofeedback) and long calibration times (20 minutes) in order to achieve reasonable performance [44].

The EEG data used here comes from BCI Competition IV Dataset 2A [73] (called the SMR dataset for the remainder of the manuscript). The data consists of four classes of imagined movements of left and right hands, feet and tongue recorded from 9 subjects. The EEG data were originally recorded using 22 Ag/AgCl electrodes, sampled at 250 Hz and bandpass filtered between 0.5 and 100Hz. We resampled the timeseries to 128 Hz, and follow the same EEG pre-processing procedure as described in [32], using software that was provided by the authors. For both the training and test sets we epoched the data at [0.5, 2.5] seconds post cue onset (the same window which was used in [39, 44]). Note that we make predictions for only this time range on the test set. We perform a four-class classification using accuracy as the summary measure.

2.2 Classification Methods

2.2.1 EEGNet: Compact CNN Architecture

Here we introduce EEGNet, a compact CNN architecture for EEG-based BCIs that (1) can be applied across several different BCI paradigms, (2) can be trained with very limited data and (3) can produce neurophysiologically interpretable features. A visualization and full description of the EEGNet model can be found in Figure 1 and Table 2, respectively, for EEG trials, collected at 128Hz sampling rate, having C channels and T time samples. We fit the model using the Adam optimizer, using default parameters as described in [74], minimizing the categorical cross-entropy loss function. We run 500 training iterations (epochs) and perform validation stopping, saving the model weights which produced the lowest validation set loss. All models were trained on an NVIDIA Quadro M6000 GPU, with CUDA 9 and cuDNN v7, in Tensorflow [75], using the Keras API [76]. We omit the use of bias units in all convolutional layers. Note that, while all convolutions are one-dimensional, we use two-dimensional convolution functions for ease of software implementation. Our software implementation can be found at https://github.com/vlawhern/arl-eegmodels.

- In Block 1, we perform two convolutional steps in sequence. First, we fit F_1 2D convolutional filters of size $(1, 64)$, with the filter length chosen to be half the sampling rate of the data

进行左右手运动来源的二分类。在手指运动开始时的[−.5, 1]秒内提取左右手手指运动的 EEG 试验，用于二分类。

2.1.4 数据集 4：感觉运动节律（SMR）

基于振荡的脑机接口（BCI）的常用控制信号是感觉运动节律（SMR），其中 mu（8-12Hz）和 beta（18-26Hz）频段在感觉运动皮层对侧（实际或想象的运动）发生去同步。SMR 与 MRCP 的振荡成分非常相似。虽然基于 SMR 的 BCI 可以促进精细、内生的 BCI 控制，但它们往往较弱且在个体间和个体内高度可变，传统上需要用户训练（神经反馈）和较长的校准时间（20 分钟）才能达到合理性能[44]。

这里使用的脑电图（EEG）数据来自 BCI 竞赛 IV 数据集 2A [73]（在本文剩余部分称为 SMR 数据集）。这些数据包含来自 9 名受试者记录的左手、右手、双脚和舌头四种想象的运动类别。脑电图数据最初使用 22 个 Ag/AgCl 电极以 250 Hz 的采样率记录，并进行了 0.5 到 100 Hz 的带通滤波。我们将时间序列重采样为 128 Hz，并遵循[32]中描述的相同脑电图预处理步骤，使用作者提供的软件。对于训练集和测试集，我们在刺激开始后的[0.5, 2.5]秒内对数据进行分时段处理（与[39, 44]中使用的相同窗口）。请注意，我们在测试集上仅对这一时间范围进行预测。

我们使用准确率作为汇总指标，进行四分类。

2.2 分类方法

2.2.1 EEGNet：紧凑型卷积神经网络架构

这里我们介绍了 EEGNet，一种适用于基于脑电的脑机接口的紧凑型 CNN 架构，它（1）可以应用于多种不同的脑机接口范式，（2）可以用非常有限的数据进行训练，（3）可以产生神经生理学上可解释的特征。EEGNet 模型的可视化和完整描述分别可以在图 1 和表 2 中找到，这些脑电试验是在 128Hz 采样率下收集的，具有 C 个通道和 T 个时间样本。我们使用 Adam 优化器拟合模型，采用[74]中描述的默认参数，最小化分类交叉熵损失函数。我们运行了 500 次训练迭代（轮次），并执行验证停止，保存产生了最低验证集损失的模型权重。所有模型都在 NVIDIA Quadro M6000 GPU 上训练，使用 CUDA 9 和 cuDNN v7，在 Tensorflow [75]中使用 Keras API [76]。我们在所有卷积层中都省略了偏置单元。请注意，虽然所有卷积都是一维的，但我们使用二维卷积函数以便于软件实现。

我们的软件实现可以在 https://github.com/vlawhern/arl-eegmodels 找到。

- 在块 1 中，我们按顺序执行两个卷积步骤。首先，我们使用大小为（1，64）的 F2D 卷积滤波器，滤波器长度选择为数据采样率的一半

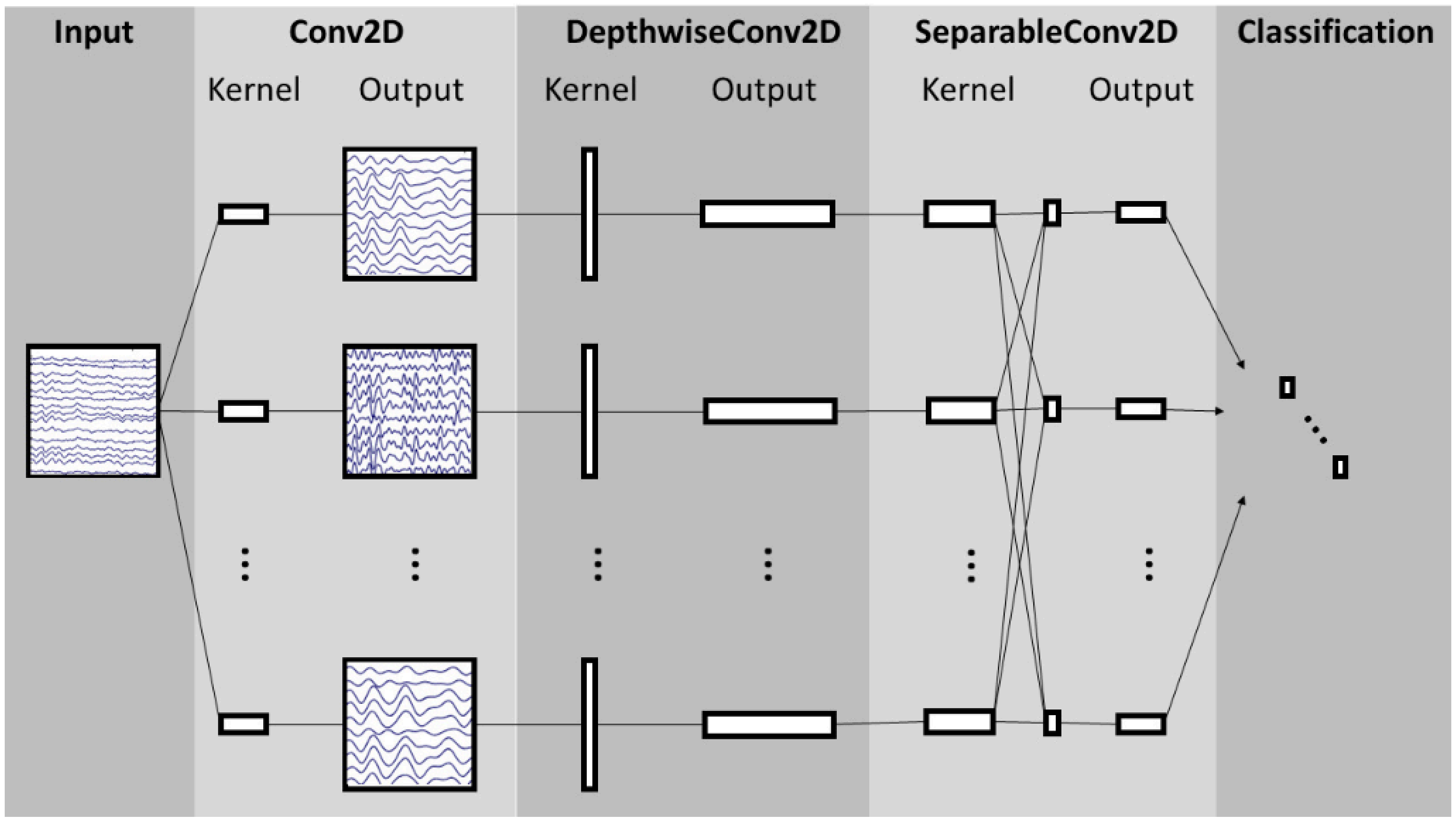
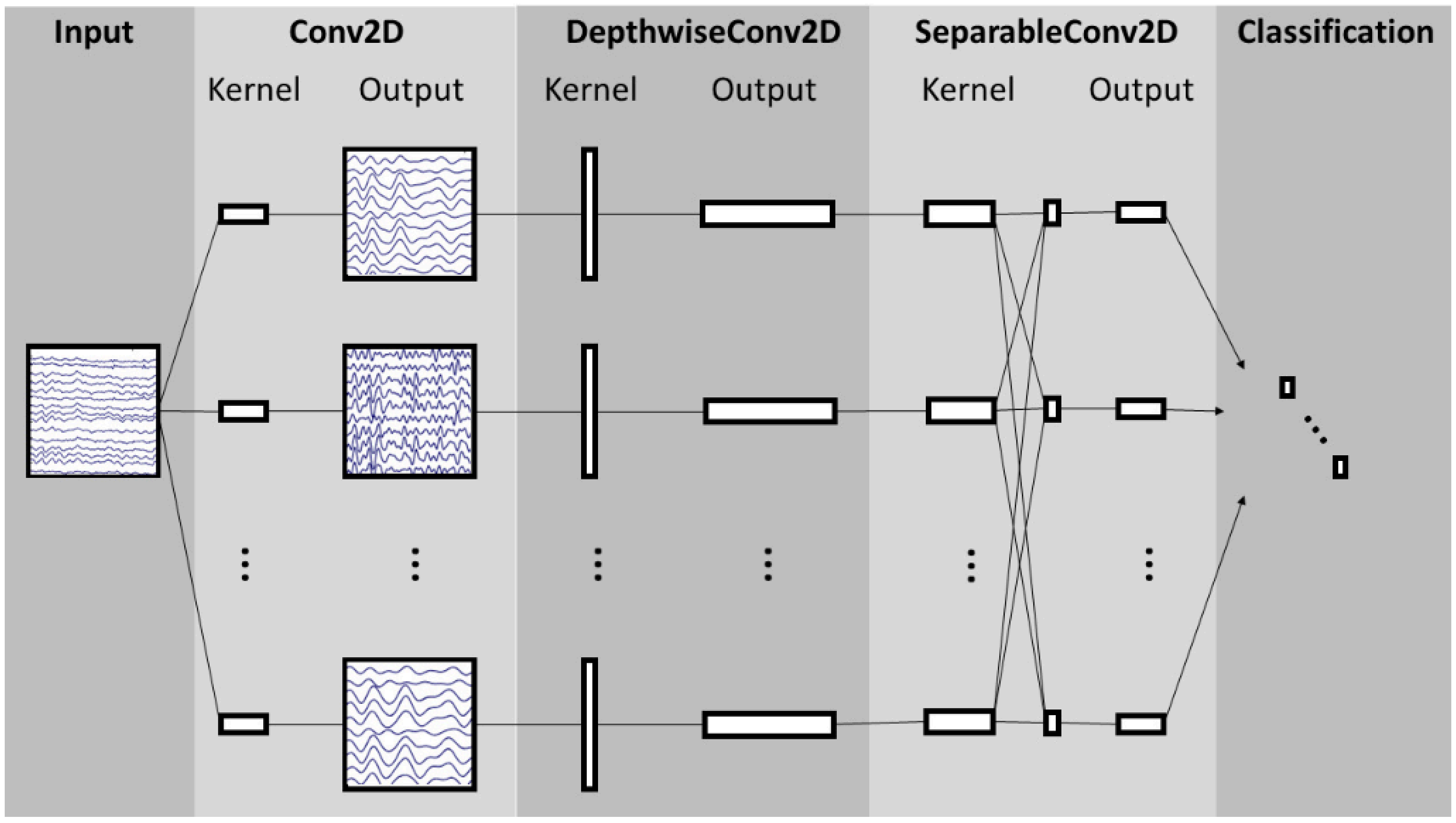


Figure 1: Overall visualization of the EEGNet architecture. Lines denote the convolutional kernel connectivity between inputs and outputs (called *feature maps*) . The network starts with a temporal convolution (second column) to learn frequency filters, then uses a depthwise convolution (middle column), connected to each feature map individually, to learn frequency-specific spatial filters. The separable convolution (fourth column) is a combination of a depthwise convolution, which learns a temporal summary for each feature map individually, followed by a pointwise convolution, which learns how to optimally mix the feature maps together. Full details about the network architecture can be found in Table 2.

(here, 128Hz), outputting F_1 feature maps containing the EEG signal at different band-pass frequencies. Setting the length of the temporal kernel at half the sampling rate allows for capturing frequency information at 2Hz and above. We then use a *Depthwise Convolution* [42] of size $(C, 1)$ to learn a spatial filter. In CNN applications for computer vision the main benefit of a depthwise convolution is reducing the number of trainable parameters to fit, as these convolutions are not fully-connected to all previous feature maps (see Figure 1 for an illustration). Importantly, when used in EEG-specific applications, this operation provides a direct way to learn spatial filters for each temporal filter, thus enabling the efficient extraction of frequency-specific spatial filters (see the middle column of Figure 1). A depth parameter D controls the number of spatial filters to learn for each feature map ($D = 1$ is shown in Figure 1 for illustration purposes). This two-step convolutional sequence is inspired in part by the Filter-Bank Common Spatial Pattern (FBCSP) algorithm [77] and is similar in nature to another decomposition technique, Bilinear Discriminant Component Analysis [78]. We keep both convolutions linear as we found no significant gains in performance when using nonlinear activations. We apply Batch Normalization [79] along the feature map dimension before applying the exponential linear unit (ELU) nonlinearity [80]. To help regularize or model, we use the Dropout technique [81]. We set the dropout probability to 0.5 for within-subject classification to help prevent over-fitting when training on small sample sizes, whereas we set the dropout probability to 0.25 in cross-subject classification, as the training set sizes



我们依次执行两个卷积步骤。图 1：EEGNet 架构的整体可视化。线条表示输入和输出（称为特征图）之间的卷积核连接。网络首先使用时间卷积（第二列）来学习频率滤波器，然后使用深度卷积（中间列），单独连接到每个特征图，来学习特定频率的空间滤波器。可分离卷积（第四列）是深度卷积和逐点卷积的组合，深度卷积单独为每个特征图学习时间摘要，逐点卷积学习如何最优地混合特征图。有关网络架构的详细信息，请参见表 2。

(此处，128Hz)，输出包含不同带通频率的 EEG 信号的 F 特征图。将时间卷积核的长度设置为采样率的一半，可以捕捉 2Hz 及以上的频率信息。然后我们使用一个大小为(C, 1)的深度卷积[42]来学习空间滤波器。在计算机视觉的 CNN 应用中，深度卷积的主要优势是减少可训练参数的数量，因为这些卷积并非完全连接到所有先前特征图（见图 1 说明）。重要的是，在 EEG 特定应用中，此操作提供了一种直接学习每个时间滤波器的空间滤波器的方法，从而能够高效地提取特定频率的空间滤波器（见图 1 中间列）。深度参数 D 控制为每个特征图学习空间滤波器的数量（图 1 中为说明目的显示了 D = 1）。这个两步卷积序列部分受到滤波器组公共空间模式（FBCSP）算法[77]的启发，并在本质上与双线性判别成分分析[78]这种分解技术相似。 我们保持两个卷积线性，因为我们发现使用非线性激活时性能没有显著提升。我们在应用指数线性单元（ELU）非线性[80]之前，沿着特征图维度应用批量归一化[79]。为了帮助正则化或建模，我们使用 Dropout 技术[81]。对于受试者内分类，我们将 dropout 概率设置为 0.5，以帮助在训练小样本时防止过拟合；而在受试者间分类中，我们将 dropout 概率设置为 0.25，因为训练集大小

Block	Layer	# filters	size	# params	Output	Activation	Options
1	Input				(C, T)		
	Reshape				(1, C, T)		
	Conv2D	F_1	(1, 64)	$64 * F_1$	(F_1, C, T)	Linear	mode = same
	BatchNorm			$2 * F_1$	(F_1, C, T)		
	DepthwiseConv2D	D * F_1	(C, 1)	$C * D * F_1$	(D * F_1, 1, T)	Linear	mode = valid, depth = D, max norm = 1
	BatchNorm			$2 * D * F_1$	(D * F_1, 1, T)		
	Activation				(D * F_1, 1, T)	ELU	
	AveragePool2D		(1, 4)		(D * F_1, 1, T // 4)		
	Dropout*				(D * F_1, 1, T // 4)		$p = 0.25$ or $p = 0.5$
2	SeparableConv2D	F_2	(1, 16)	$16 * D * F_1 + F_2 * (D * F_1)$	(F_2, 1, T // 4)	Linear	mode = same
	BatchNorm			$2 * F_2$	(F_2, 1, T // 4)		
	Activation				(F_2, 1, T // 4)	ELU	
	AveragePool2D		(1, 8)		(F_2, 1, T // 32)		
	Dropout*				(F_2, 1, T // 32)		$p = 0.25$ or $p = 0.5$
	Flatten				(F_2 * (T // 32))		
Classifier	Dense	N * (F_2 * T // 32)			N	Softmax	max norm = 0.25

Table 2: EEGNet architecture, where C = number of channels, T = number of time points, F_1 = number of temporal filters, D = depth multiplier (number of spatial filters), F_2 = number of pointwise filters, and N = number of classes, respectively. For the Dropout layer, we use $p = 0.5$ for within-subject classification and $p = 0.25$ for cross-subject classification (see Section 2.1.1 for more details)

are much larger (see Section 2.3 for more details on our within- and cross-subject analyses). We apply an average pooling layer of size (1, 4) to reduce the sampling rate of the signal to 32Hz. We also regularize each spatial filter by using a maximum norm constraint of 1 on its weights; $\|w\|^2 < 1$.

- In Block 2, we use a *Separable Convolution*, which is a Depthwise Convolution (here, of size $(1, 16)$, representing 500ms of EEG activity at 32Hz) followed by F_2 $(1, 1)$ Pointwise Convolutions [42]. The main benefits of separable convolutions are (1) reducing the number of parameters to fit and (2) explicitly decoupling the relationship within and across feature maps by first learning a kernel summarizing each feature map individually, then optimally merging the outputs afterwards. When used for EEG-specific applications this operation separates learning how to summarize individual feature maps in time (the depthwise convolution) with how to optimally combine the feature maps (the pointwise convolution). This operation is also particularly useful for EEG signals as different feature maps may represent data at different time-scales of information. In our case we first learn a 500 ms "summary" of each feature map, then combine the outputs afterwards. An Average Pooling layer of size $(1, 8)$ is used for dimension reduction.

- In the classification block, the features are passed directly to a softmax classification with N units, N being the number of classes in the data. We omit the use of a dense layer for feature aggregation prior to the softmax classification layer to reduce the number of free parameters in the model, inspired by the work in [82].

We investigate several different configurations of the EEGNet architecture by varying the number of filters, F_1, and the number of spatial filters per temporal filter, D to learn. We set $F_2 = D * F_1$

Block Layer # filters size # params Output Activation Options

1 Input (C, T) Reshape (1, C, T)

Conv2D F(1, 64) 64 ∗ F(F, C, T) Linear mode = same BatchNorm 2 ∗ F(F, C, T) DepthwiseConv2D D * F(C, 1) C ∗ D ∗ F(D * F, 1, T) Linear mode = valid, depth = D, max norm = 1 BatchNorm 2 ∗ D ∗ F(D * F, 1, T) Activation (D * F, 1, T) ELU AveragePool2D (1, 4) (D * F, 1, T // 4) Dropout* (D * F, 1, T // 4) p = 0.25 or p = 0.5 2 SeparableConv2D F(1, 16) 16 ∗ D ∗ F+ F∗ (D ∗ F) (F, 1, T // 4) Linear mode = same BatchNorm 2 ∗ F(F, 1, T // 4)

激活 (F, 1, T // 4) ELU 平均池化 2D (1, 8) (F, 1, T // 32) Dropout* (F, 1, T // 32) p = 0.25 或 p = 0.5 扁平化 (F* (T // 32)) 分类器 Dense N * (F* T // 32) N Softmax 最大范数 = 0.25

我们将 Dropout 概率设置为 0 表 2：EEGNet 架构，其中 C = 通道数，T = 时间点数，F= 时间滤波器数，D = 深度乘数（空间滤波器数），F= 点滤波器数，N = 类别数。对于 Dropout 层，我们在被试内分类时使用 p = 0.5，在跨被试分类时使用 p = 0.25（更多细节请参见第 2.1.1 节）

（见第 2.3 节了解更多关于被试内和跨被试分析的细节）。我们对信号应用了一个大小为 (1, 4) 的平均池化层，将采样率降低到 32Hz。我们还通过对其权重施加最大范数约束 1 来正则化每个空间滤波器；‖ w ‖ < 1。

- 在 Block 2 中，我们使用可分离卷积，它是一个深度卷积（此处大小为(1, 16)，代表 32Hz 下 500ms 的脑电活动）后接 F(1, 1)逐点卷积[42]。可分离卷积的主要优点是(1)减少需要拟合的参数数量和(2)通过首先学习总结每个特征图的卷积核，然后最优地合并输出，显式地解耦特征图内部和跨特征图之间的关系。当用于特定于脑电的应用时，此操作将学习如何在时间上总结单个特征图（深度卷积）与如何最优地组合特征图（逐点卷积）分离。此操作对于脑电信号也特别有用，因为不同的特征图可能代表不同时间尺度的信息。在我们的案例中，我们首先学习每个特征图的 500ms“总结”，然后合并输出。使用大小为(1, 8)的平均池化层进行降维。

- 在分类模块中，特征直接传递给具有 N 个单元的 softmax 分类器，其中 N 是数据中的类别数。我们省略了 softmax 分类器之前的密集层用于特征聚合，以减少模型中的自由参数数量，灵感来源于文献[82]的工作。

我们通过改变滤波器数量 F 和每个时间滤波器对应的空间滤波器数量 D 来研究 EEGNet 架构的几种不同配置。我们设置 F=D*F。

	Trial Length (sec)	DeepConvNet	ShallowConvNet	EEGNet-4,2	EEGNet-8,2
P300	1	174,127	104,002	**1,066**	2,258
ERN	1.25	169,927	91,602	**1,082**	2,290
MRCP	1.5	175,727	104,722	**1,098**	2,322
SMR*	2	152,219	40,644	**796**	1,716

Table 3: Number of trainable parameters per model and per dataset for all CNN-based models. We see that the EEGNet models are up to two orders of magnitude smaller than both DeepConvNet and ShallowConvNet across all datasets. Note that we use a temporal kernel length of 32 samples for the SMR dataset as the data were high-passed at 4Hz.

(the number of temporal filters along with their associated spatial filters from Block 1) for the duration of the manuscript, although in principle F_2 can take any value; $F_2 < D * F_1$ denotes a compressed representation, learning fewer feature maps than inputs, whereas $F_2 > D * F_1$ denotes an overcomplete representation, learning more feature maps than inputs. We use the notation EEGNet-F_1,D to denote the number of temporal and spatial filters to learn; i.e.: EEGNet-4,2 denotes learning 4 temporal filters and 2 spatial filters per temporal filter.

2.2.2 Comparison with existing CNN Approaches

We compare the performance of EEGNet against the DeepConvNet and ShallowConvNet models proposed by [32]; full table descriptions of both models can be found in the Appendix. We implemented these models in Tensorflow and Keras, following the descriptions found in the paper. As their architectures were originally designed for 250Hz EEG signals (as opposed to 128Hz signals used here) we divided the lengths of temporal kernels and pooling layers in their architectures by 2 to correspond approximately to the sampling rate used in our models. We train these models in the same way we train the EEGNet model (see Section 2.2.1).

The DeepConvNet architecture consists of five convolutional layers with a softmax layer for classification (see Figure 1 of [32]). The ShallowConvNet architecture consists of two convolutional layers (temporal, then spatial), a squaring nonlinearity ($f(x) = x^2$), an average pooling layer and a log nonlinearity ($f(x) = \log(x)$). We would like to emphasize that the ShallowConvNet architecture was designed specifically for oscillatory signal classification (by extracting features related to log band-power); thus, it may not work well on ERP-based classification tasks. However, the DeepConvNet architecture was designed to be a general-purpose architecture that is not restricted to specific feature types [32], and thus it serves as a more valid comparison to EEGNet. Table 3 shows the number of trainable parameters per model across all CNN models.

2.2.3 Comparison with Traditional Approaches

We also compare the performance of EEGNet to that of the best performing traditional approach for each individual paradigm. For all ERP-based data analyses (P300, ERN, MRCP) the traditional approach is the approach which won the Kaggle BCI Competition (code and documenta-

	试验长度（秒）	DeepConvNet	浅层卷积网络	EEGNet-4,2	EEGNet-8,2
P300	1	174,127	104,002	1,066	2,258
ERN	1.25	169,927	91,602	1,082	2,290
MRCP	1.5	175,727	104,722	1,098	2,322
SMR*	2	152,219	40,644	796	1,716

表 3：所有基于 CNN 的模型每个模型和数据集的可训练参数数量。我们看到 EEGNet 模型在所有数据集上比 DeepConvNet 和浅层卷积网络小一到两个数量级。注意，由于数据在 4Hz 处进行了高通滤波，我们使用 32 个样本的时间卷积核长度来处理 SMR 数据集。

(块 1 中的时间滤波器数量及其相关空间滤波器)在整个手稿期间，尽管原则上 F 可以取任何值；$F< D * F$ 表示压缩表示，学习比输入更少的特征图，而 $F> D * F$ 表示过完备表示，学习比输入更多的特征图。我们使用符号 EEGNet-F,D 表示学习的时间滤波器和空间滤波器数量；即：EEGNet-4,2 表示每个时间滤波器学习 4 个时间滤波器和 2 个空间滤波器。

2.2.2 与现有 CNN 方法的比较

我们将 EEGNet 的性能与[32]中提出的 DeepConvNet 和 ShallowConvNet 模型进行了比较；这两个模型的详细表格描述可以在附录中找到。我们按照论文中的描述，使用 Tensorflow 和 Keras 实现了这些模型。由于它们的架构最初是为 250Hz 的脑电图信号（而不是这里使用的 128Hz 信号）设计的，我们将它们架构中时间卷积核和池化层的长度除以 2，以大致对应我们模型中使用的采样率。我们以与训练 EEGNet 模型相同的方式训练这些模型（见第 2.2.1 节）。

DeepConvNet 架构由五个卷积层和一个用于分类的 softmax 层组成（见图 [32] 的图 1）。ShallowConvNet 架构由两个卷积层（先时间域再空间域）、一个平方非线性（$f(x) = x$）、一个平均池化层和一个对数非线性（$f(x) = \log(x)$）组成。我们想强调的是，ShallowConvNet 架构是专门为振荡信号分类设计的（通过提取与对数带功率相关的特征）；因此，它可能不适合基于 ERP 的分类任务。然而，DeepConvNet 架构被设计为一种通用架构，不受特定特征类型的限制 [32]，因此它更适用于与 EEGNet 进行比较。表 3 展示了所有 CNN 模型中每个模型的可训练参数数量。

2.2.3 与传统方法的比较

我们还比较了 EEGNet 在每个范式下表现最佳的传统的最佳方法的性能。对于所有基于 ERP 的数据分析（P300、ERN、MRCP），传统方法是指赢得 Kaggle BCI 竞赛的方法（代码和文档）。

tion at http://github.com/alexandrebarachant/bci-challenge-ner-2015), which uses a combination of xDAWN Spatial Filtering [83], Riemannian Geometry [84, 85], channel subset selection and L_1 feature regularization (referred to as xDAWN + RG for the remainder of the manuscript). Here we provide a summary of the approach, which is done in five steps:

1. Train two set of 5 xDAWN spatial filters, one set for each class of a binary classification task, using the ERP template concatenation method as described in [85, 86].
2. Perform EEG electrode selection through backward elimination [87] to keep only the most relevant 35 channels.
3. Project the covariance matrices onto the tangent space using the log-euclidean metric [84, 88].
4. Perform feature normalization using an L_1 ratio of 0.5, signifying an equal weight for L_1 and L_2 penalties. An L_1 penalty encourages the sum of the absolute values of the parameters to be small, whereas an L_2 penalty encourages the sum of the squares of the parameters to be small (a theoretical overview of these penalties can be found in [89]).
5. Perform classification using an Elastic Net regression.

We use the same xDAWN+RG model parameters across all comparisons (P300, ERN, MRCP) with the exception of the initial number of EEG channels to use, which was set to 56 for ERN and 64 for P300 and MRCP. While the original solution used an ensemble of bagged classifiers, for this analysis we only compared a single model with this approach to a single EEGNet model on identical training and test sets, as we expect any gains from ensemble learning to benefit both approaches equally. The original solution also used a set of "meta features" that were specific to that data collection. As the goal of this work is to investigate a general-purpose CNN model for EEG-based BCIs, we omitted the use of these features as they are specific to that particular data collection.

For oscillatory-based classification of SMR, the traditional approach is our own implementation of the One-Versus-Rest (OVR) filter-bank common spatial pattern (FBCSP) algorithm as described in [77]. Here we provide a brief summary of our approach:

1. Bandpass filter the EEG signal into 9 non-overlapping filter banks in 4Hz steps, starting at 4Hz: 4-8Hz, 8-12Hz, ..., 36-40Hz.
2. As the classification problem is multi-class, we use OVR classification, which requires that we train a classifier for all pairs of OVR combinations, which there are 4 here (class 1 vs all others, class 2 vs all others, etc). We train 2 CSP filter pairs (4 filters total) for each filter bank on the training data using the auto-covariance shrinkage method by [90]. This will give a total of 36 features (9 filter banks $\times$ 4 CSP filters) for each trial and each OVR combination.
3. Train an elastic-net logistic regression classifier [91] for each OVR combination. We set the elastic net penalty $\alpha = 0.95$.

在 http://github.com/alexandrebarachant/bci-challenge-ner-2015 上提供的方案，结合了 xDAWN 空间滤波 [83]、黎曼几何 [84, 85]、通道子集选择和 L 特征正则化（在本文剩余部分中称为 xDAWN + RG）。这里我们提供该方法的概述，分为五个步骤：

1. 使用 [85, 86] 中描述的 ERP 模板连接方法，为二元分类任务的每个类别训练两组 5 个 xDAWN 空间滤波器。
2. 通过反向消除 [87] 进行 EEG 电极选择，仅保留最相关的 35 个通道。
3. 使用对数欧几里得度量 [84,88] 将协方差矩阵投影到切空间上。
4. 使用 Lratio 为 0.5 进行特征归一化，表示对 LandLpenalties 赋予同等权重。Lpenalty 鼓励参数的绝对值之和尽可能小，而 Lpenalty 则鼓励参数的平方和尽可能小（这些惩罚的理论概述可参考[89]）。

5. 使用弹性网络回归进行分类。

我们在所有比较中（P300、ERN、MRCP）都使用相同的 xDAWN+RG 模型参数，唯一的例外是初始使用的 EEG 通道数，ERN 设置为 56，P300 和 MRCP 设置为 64。虽然原始解决方案使用了装袋分类器的集成，但在本次分析中，我们仅将采用这种方法的单个模型与在相同训练和测试集上的单个 EEGNet 模型进行比较，因为我们预期集成学习的任何收益将平等地惠及这两种方法。原始解决方案还使用了一组特定于该数据收集的“元特征”。由于本工作的目标是研究适用于基于 EEG 的 BCI 的通用 CNN 模型，我们省略了这些特征的使用，因为它们特定于该数据收集。
对于基于振荡的分类 SMR，传统方法是我们在[77]中描述的自实现的 One-Versus-Rest（OVR）滤波器组常见空间模式（FBCSP）算法。在此，我们简要概述我们的方法：

1. 将 EEG 信号以 4Hz 的步长进行带通滤波，分为 9 个非重叠的滤波器组，起始频率为 4Hz：4-8Hz、8-12Hz、...、36-40Hz。
2. 由于分类问题是多类的，我们使用 OVR 分类，这要求我们为所有 OVR 组合的对进行分类器训练，这里有 4 种组合（类别 1 与其他所有类别、类别 2 与其他所有类别等）。我们使用[90]中的自协方差收缩方法在训练数据上为每个滤波器组训练 2 对 CSP 滤波器（总共 4 个滤波器）。这将为每个试验和每个 OVR 组合提供 36 个特征（9 个滤波器组×4 个 CSP 滤波器）。
3. 为每个 OVR 组合训练弹性网络逻辑回归分类器[91]。我们设置弹性网络惩罚$\alpha = 0.95$。

4. Find the optimal λ value for the elastic-net logistic regression that maximizes the validation set accuracy by evaluating the trained classifiers on a held-out validation set. The multi-class label for each trial is the classifier that produces the highest probability among the 4 OVR classifiers.

5. Apply the trained classifiers to the test set, using the λ values obtained in Step 4.

Note that this approach differs slightly from the original technique as proposed in [77], where they use a Naive Bayes Parzen Window classifier. We opted to use an elastic net logistic regression for ease of implementation, and the fact that it has been used in existing software implementations of FBCSP (for example, in BCILAB [92]).

2.3 Data Analysis

Classification results are reported for two sets of analyses: within-subject and cross-subject. Within-subject classification uses a portion of the subjects data to train a model specifically for that subject, although cross-subject classification uses the data from other subjects to train a subject-agnostic model. While within-subject models tend to perform better than cross-subject models on a variety of tasks, there is ongoing research investigating techniques to minimize (or possibly eliminate) the need for subject-specific information to train robust systems [44, 51].

For within-subject, we use four-fold blockwise cross-validation, where two of the four blocks are chosen to be the training set, one block as the validation set, and the final block as testing. We perform statistical testing using a repeated-measures Analysis of Variance (ANOVA), modeling classification results (AUC for P300/MRCP/ERN and Classification Accuracy for SMR) as the response variable with subject number and classifier type as factors. For cross-subject analysis in P300 and MRCP we choose, at random, 4 subjects for the validation set, one subject for the test set, and all remaining subjects for the training set (see Table 1 for number of subjects per dataset). This process was repeated 30 times, producing 30 different folds. We follow the same procedure for the ERN dataset, except we use the 10 test subjects from the original Kaggle Competition as the test set for each fold. We perform statistical testing using a one-way Analysis of Variance, using classifier type as the factor. For the SMR dataset, we partitioned the data as follows: For each subject, select the training data from 5 other subjects at random to be the training set and the training data from the remaining 3 subjects to be the validation set. The test set remains the same as the original test set for the competition. Note that this enforces a fully cross-subject classification analysis as we never use the test subjects' training data. This process is repeated 10 times for each subject, creating 90 different folds. The mean and standard error of classification performance were calculated over the 90 folds. We perform statistical testing for this analysis using the same testing procedure as the within-subject analysis.

When training both the within-subject and cross-subject models, we apply a class-weight to the loss function whenever the data is imbalanced (unequal number of trials for each class). The class-weight we apply is the inverse of the proportion in the training data, with the majority class set to 1. For example, in the P300 dataset, there is a 5.6:1 odds between non-targets and targets

4. 通过在保留的验证集上评估训练好的分类器，找到弹性网络逻辑回归的最优λ值，以最大化验证集的准确率。每次试验的多类标签是四个 OVR 分类器中产生最高概率的分类器。

5. 将训练好的分类器应用于测试集，使用步骤 4 中获得的λ值。

请注意，这种方法与[77]中提出的原始技术略有不同，他们使用的是朴素贝叶斯 Parzen 窗分类器。我们选择使用弹性网络逻辑回归，因为实现起来更简单，而且它已经在现有的 FBCSP 软件实现中使用过（例如，在 BCILAB [92]中）。

2.3 数据分析

报告了两种分析集的分类结果：组内和组间。组内分类使用部分受试者数据来训练一个专门针对该受试者的模型，而组间分类则使用其他受试者的数据来训练一个不受试者影响的模型。虽然组内模型在各种任务上往往比组间模型表现更好，但仍有持续的研究在探索减少（或可能消除）训练鲁棒系统所需受试者特定信息的技术[44, 51]。

在组内分析中，我们采用四折交叉验证，其中选择其中的两折作为训练集，一折作为验证集，最后一折作为测试集。我们使用重复测量方差分析（ANOVA）进行统计检验，将分类结果（P300/MRCP/ERN 的 AUC 和 SMR 的分类准确率）建模为响应变量，以被试编号和分类器类型为因素。在 P300 和 MRCP 的跨被试分析中，我们随机选择 4 名被试作为验证集，1 名被试作为测试集，其余所有被试作为训练集（具体每个数据集的被试数量见表 1）。该过程重复进行 30 次，生成 30 个不同的折。对于 ERN 数据集，我们遵循相同的流程，但使用原始 Kaggle 竞赛中的 10 名测试被试作为每个折的测试集。我们使用单因素方差分析进行统计检验，以分类器类型为因素。 对于 SMR 数据集，我们按以下方式划分数据：对于每个受试者，从其他 5 个受试者中随机选择训练数据作为训练集，从剩余的 3 个受试者中选择训练数据作为验证集。竞赛的测试集保持与原始测试集相同。请注意，这确保了完全的跨受试者分类分析，因为我们从未使用测试受试者的训练数据。该过程对每个受试者重复 10 次，创建 90 个不同的折。在 90 个折上计算分类性能的平均值和标准误差。我们使用与受试者内分析相同的测试程序对该分析进行统计检验。

在训练受试者内和跨受试者模型时，每当数据不平衡（每个类别的试验次数不等）时，我们会对损失函数应用类别权重。我们应用的类别权重是训练数据中比例的倒数，多数类别设置为 1。例如，在 P300 数据集中，非目标和目标的概率比为 5.6:1

(Table 1) . In this case the class-weight for non-targets was set to 1, while the class-weight for targets was set to 6 (when the odds are a fraction, we take the next highest integer). This procedure was applied to the P300 and ERN datasets only, as these were the only datasets where significant class imbalance was present.

Note that for the SMR analysis, we set the temporal kernel length to be 32 samples long (as opposed to 64 samples long as given in Table 2) since the data were high-passed at 4Hz.

2.4 EEGNet Feature Explainability

The development of methods for enabling feature explainability from deep neural networks has become an active research area over the past few years, and has been proposed as an essential component of a robust model validation procedure, to ensure that the classification performance is being driven by relevant features as opposed to noise or artifacts in the data [16, 93–99]. We present three different approaches for understanding the features derived by EEGNet:

1. **Summarizing averaged outputs of hidden unit activations:** This approach focuses on summarizing the activations of hidden units at layers specified by the user. In this work we choose to summarize the hidden unit activations representing the data after the depthwise convolution (the spatial filter operation in EEGNet). Because the spatial filters are tied directly to a particular temporal filter, they provide additional insights into the spatial localization of narrow-band frequency activity. Here we summarize the spatially-filtered data by calculating the difference in averaged time-frequency representations between classes, using Morlet wavelets [100].

2. **Visualizing the convolutional kernel weights:** This approach focuses on directly visualizing and interpreting the convolutional kernel weights from the model. Generally speaking, interpreting the convolutional kernel weights is very difficult due to the cross-filter-map connectivity between any two layers. However, because EEGNet limits the connectivity of the convolutional layers (using depthwise and separable convolutions), it is possible to interpret the temporal convolution as narrow-band frequency filters and the depthwise convolution as frequency-specific spatial filters.

3. **Calculating single-trial feature *relevance* on the classification decision:** This approach focuses on calculating, on a single-trial basis, the *relevance* of individual features on the resulting classification decision. Positive values of relevance denote evidence supporting the outcome, while negative values of relevance denote evidence against the outcome. In our analysis we used DeepLIFT with the Rescale rule [97], as implemented in [98], to calculate single-trial EEG feature relevance. DeepLIFT is a gradient-based relevance attribution method that calculates relevance values per feature relative to a "reference" input (here, an input of zeros, as is suggested in [97]), and is a technique similar to Layerwise Relevance Propagation (LRP) which has been used previously for EEG analysis [101] (a summary of gradient-based relevance attribution methods can be found in [98]). This analysis can be used to elucidate feature relevance from high-confidence versus low-confidence predictions, and can

将多数类设置为 1（表 1）。在这种情况下，非目标类的类权重设置为 1，而目标类的类权重设置为 6（当概率为分数时，我们取下一个最高的整数）。此方法仅应用于 P300 和 ERN 数据集，因为这些是唯一存在显著类别不平衡的数据集。
注意，对于 SMR 分析，我们将时间核长度设置为 32 个样本长（与表 2 中给出的 64 个样本长相反），因为数据在 4Hz 处进行了高通滤波。

2.4 EEGNet 特征可解释性

近年来，开发能够从深度神经网络中实现特征可解释性的方法已成为一个活跃的研究领域，并已被提议作为稳健模型验证过程的一个基本组成部分，以确保分类性能是由相关特征驱动的，而不是数据中的噪声或伪影[16, 93–99]。我们提出了三种不同的方法来理解 EEGNet 导出的特征：

1. 概括隐藏单元激活的平均输出：这种方法专注于总结用户指定层级的隐藏单元激活。在本工作中，我们选择总结深度卷积（EEGNet 中的空间滤波操作）后表示数据的隐藏单元激活。由于空间滤波器直接与特定的时间滤波器相关联，它们为窄带频率活动的空间定位提供了额外的见解。在这里，我们通过使用 Morlet 小波[100]计算类间平均时频表示的差异来总结空间滤波数据。

2. 可视化卷积核权重：这种方法着重于直接可视化和解释模型中的卷积核权重。一般来说，由于任意两层之间的跨滤波器-特征图连接性，解释卷积核权重非常困难。然而，由于 EEGNet 限制了卷积层的连接性（使用深度卷积和分离卷积），因此可以将时间卷积解释为窄带频率滤波器，将深度卷积解释为特定频率的空间滤波器。

3. 在分类决策上计算单次试验特征相关性：这种方法专注于计算单个试验中各个特征对最终分类决策的相关性。相关性的正值表示支持结果，而负值表示反对结果。在我们的分析中，我们使用了 DeepLIFT 与 Rescale 规则[97]，如[98]中实现的方法，来计算单次试验的 EEG 特征相关性。DeepLIFT 是一种基于梯度的相关性归因方法，它计算每个特征相对于“参考”输入（此处为建议在[97]中使用的零输入）的相关性值，并且是一种类似于先前用于 EEG 分析[101]的层状相关性传播（LRP）的技术（基于梯度的相关性归因方法的总结可以在[98]中找到）。这种分析可用于阐明高置信度与低置信度预测中的特征相关性，并可以

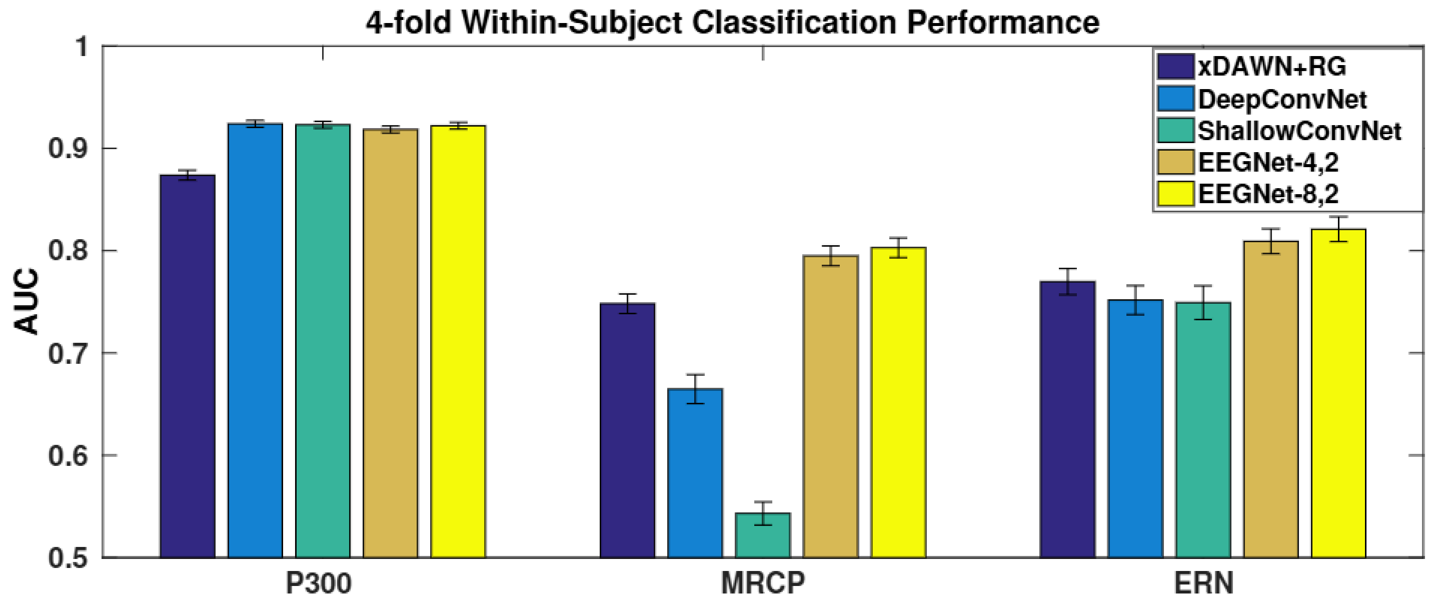


Figure 2: 4-fold within-subject classification performance for the P300, ERN and MRCP datasets for each model, averaged over all folds and all subjects. Error bars denote 2 standard errors of the mean. We see that, while there is minimal difference between all the CNN models for the P300 dataset, there are significant differences in the MRCP dataset, with both EEGNet models outperforming all other models. For the ERN dataset we also see both EEGNet models performing better than all others ($p < 0.05$).

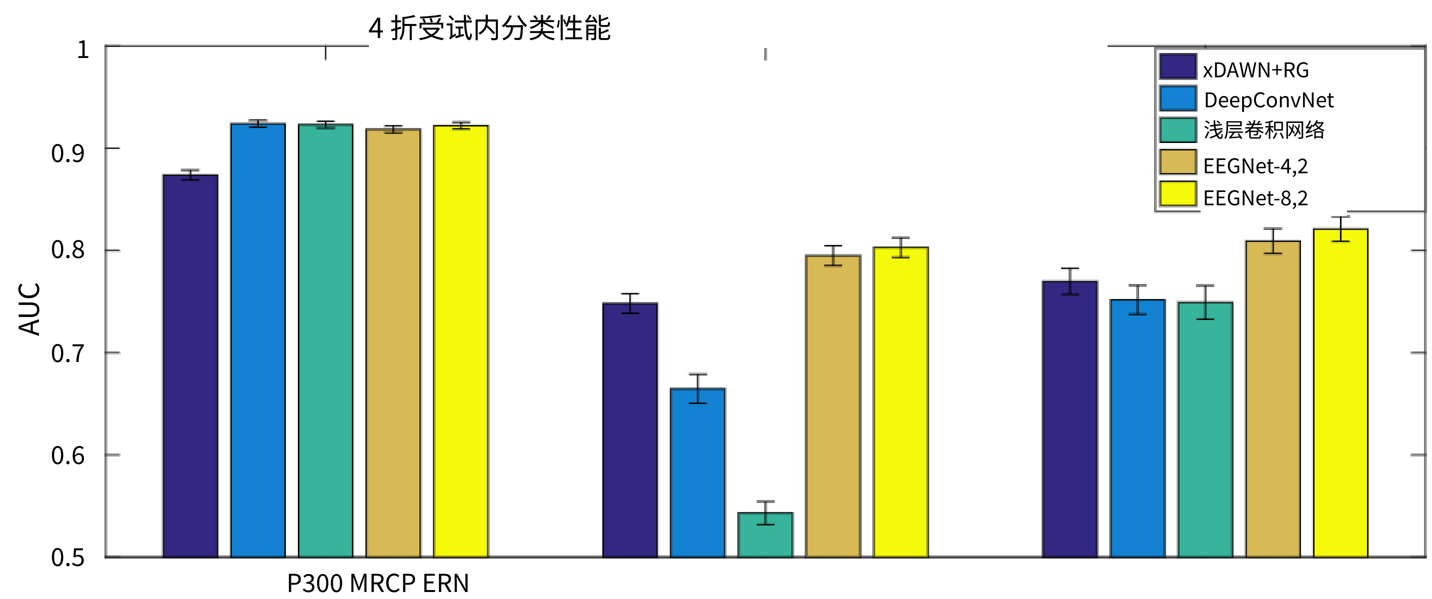
be used to confirm that the relevant features learned are interpretable, as opposed to noise or artifact features.

3 Results

3.1 Within-Subject Classification

We compare the performance of both the CNN-based reference algorithms (DeepConvNet and ShallowConvNet) and the traditional approach (xDAWN+RG for P300/MRCP/ERN and FBCSP for SMR) with EEGNet-4,2 and EEGNet-8,2. Within-subject four-fold cross-validation results across all algorithms for P300, MRCP and ERN datasets are shown in Figure 2. We observed, across all paradigms, that there was no statistically significant difference between EEGNet-4,2 and EEGNet-8,2 ($p > 0.05$), indicating that the increase in model complexity did not statistically improve classification performance. For the P300 dataset, all CNN-based models significantly outperform xDAWN+RG ($p < 0.05$) while not performing significantly different amongst themselves. For the ERN dataset, EEGNet-8,2 outperforms DeepConvNet, ShallowConvNet and xDAWN+RG ($p < 0.05$), while EEGNet-4,2 outperforms DeepConvNet and ShallowConvNet ($p < 0.05$). The biggest difference observed among all the approaches is in the MRCP dataset, where both EEGNet models statistically outperform all others by a significant margin (DeepConvNet, ShallowConvNet and xDAWN+RG, $p < 0.05$ for each comparison).

Four-fold cross-validation results for the SMR dataset are shown in Figure 3. Here we see the performances of ShallowConvNet and FBCSP are very similar, replicating previous results as reported in [32], while DeepConvNet performance is significantly lower. We also see that EEGNet-



输入全零图 2：对于每个模型，P300、ERN 和 MRCP 数据集的 4 折被试内分类性能，所有折和所有被试的平均值。误差线表示平均值的 2 个标准误差。我们看到，对于 P300 数据集，所有 CNN 模型之间差异很小，但在 MRCP 数据集中存在显著差异，EEGNet 模型均优于其他所有模型。对于 ERN 数据集，我们也看到 EEGNet 模型的表现优于所有其他模型（$p < 0.05$）。

用于确认所学习的相关特征是可解释的，而不是噪声或伪影特征。

3 结果

3.1 个体内分类

我们将基于 CNN 的参考算法（DeepConvNet 和 ShallowConvNet）以及传统方法（xdaawn+rg 用于 P300/MRCP/ERN 和 FBCSP 用于 SMR）的性能与 EEGNet-4,2 和 EEGNet-8,2 进行比较。图 2 展示了所有算法在 P300、MRCP 和 ERN 数据集上的组内四折交叉验证结果。我们观察到，在所有范式下，EEGNet-4,2 和 EEGNet-8,2 之间没有统计学上的显著差异（$p > 0.05$），表明模型复杂性的增加并没有统计学上改善分类性能。对于 P300 数据集，所有基于 CNN 的模型都显著优于 xdaawn+rg（$p < 0.05$），而它们彼此之间没有显著差异。对于 ERN 数据集，EEGNet-8,2 优于 DeepConvNet、ShallowConvNet 和 xdaawn+rg（$p < 0.05$），而 EEGNet-4,2 优于 DeepConvNet 和 ShallowConvNet（$p < 0.05$）。所有方法中观察到的最大差异是在 MRCP 数据集上，其中两个 EEGNet 模型都显著优于其他所有方法（DeepConvNet、ShallowConvNet 和 xdaawn+rg，每个比较的 $p < 0.05$）。

SMR 数据集的四折交叉验证结果如图 3 所示。在这里我们看到 ShallowConvNet 和 FBCSP 的性能非常相似，复制了之前在[32]中报告的结果，而 DeepConvNet 的性能明显较低。我们还看到 EEGNet-

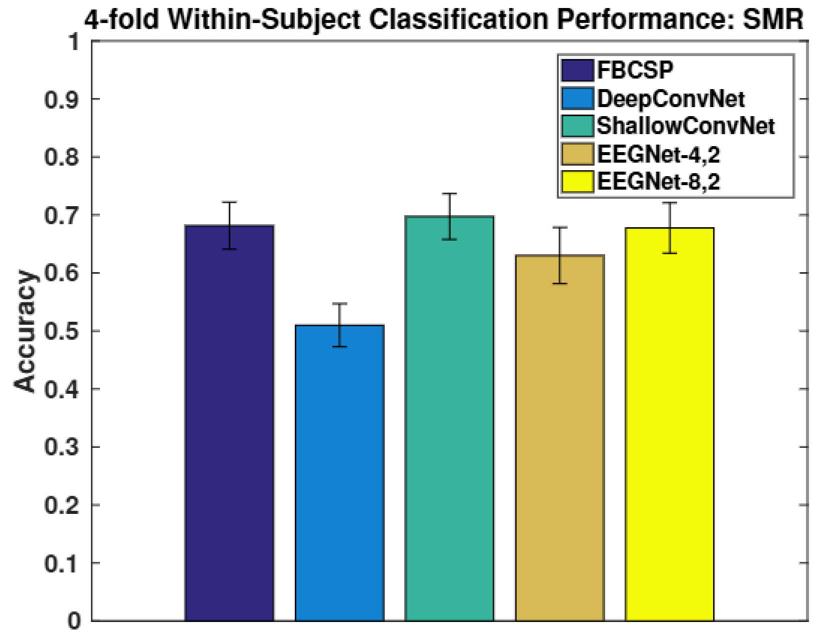


Figure 3: 4-fold within-subject classification performance for the SMR dataset for each model, averaged over all folds and all subjects. Error bars denote 2 standard errors of the mean. Here we see DeepConvNet statistically performed worse than all other models ($p < 0.05$). ShallowConvNet and EEGNet-8,2 performed similarly to that of FBCSP.

8,2 performance is similar to FBCSP as well.

3.2 Cross-Subject Classification

Cross-subject classification results across all algorithms for P300, MRCP and ERN datasets are shown in Figure 4. Similar to the within-subject analysis, we observed no statistical difference between EEGNet-4,2 and EEGNet-8,2 across all datasets ($p > 0.05$). For the P300 dataset, all CNN-based models significantly outperform xDAWN+RG ($p < 0.05$) while not performing sig-

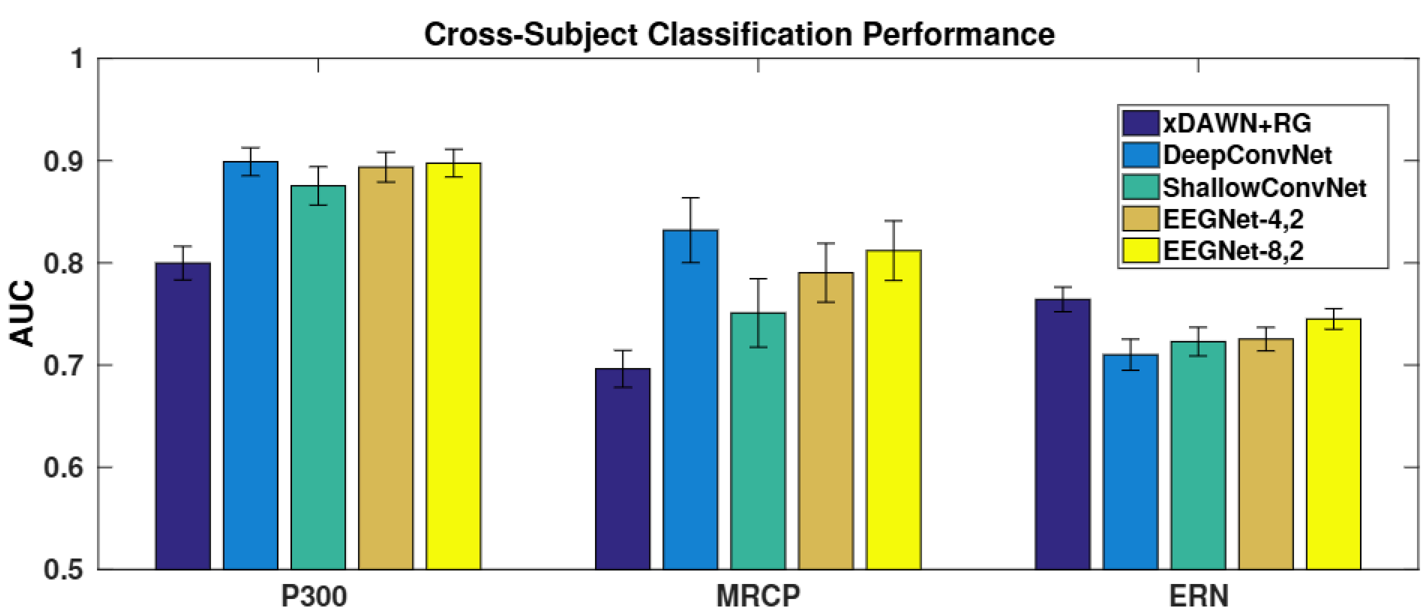


Figure 4: Cross-Subject classification performance for the P300, ERN and MRCP datasets for each model, averaged for 30 folds. Error bars denote 2 standard errors of the mean. For the P300 and MRCP datasets there is minimal difference between the DeepConvNet and the EEGNet models, with both models outperforming ShallowConvNet. For the ERN dataset the reference algorithm (xDAWN + RG) significantly outperforms all other models.

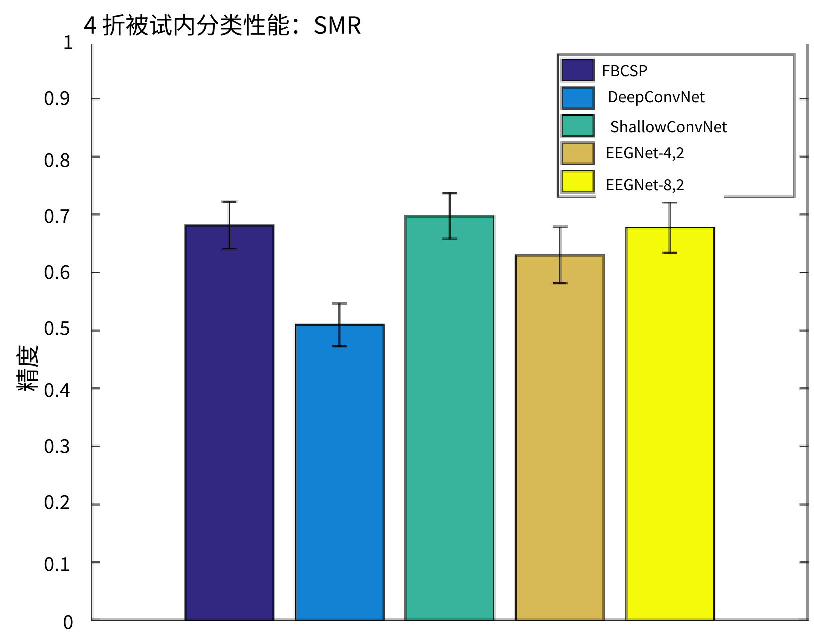


图 3：SMR 数据集上每个模型的 4 折被试内分类性能，所有折和所有被试的平均值。误差线表示平均值的 2 个标准误。我们看到 DeepConvNet 在统计上表现比所有其他模型差（$p < 0.05$）。ShallowConvNet 和 EEGNet-8,2 的表现与 FBCSP 相似。

8,2 的表现也与 FBCSP 相似。

3.2 跨被试分类

所有算法在 P300、MRCP 和 ERN 数据集上的跨主题分类结果如图 4 所示。类似于被试内分析，我们在所有数据集上观察到 EEGNet-4,2 和 EEGNet-8,2 之间没有统计学差异（$p > 0.05$）。对于 P300 数据集，所有基于 CNN 的模型显著优于 xDAWN+RG（$p < 0.05$），但表现并不显著...

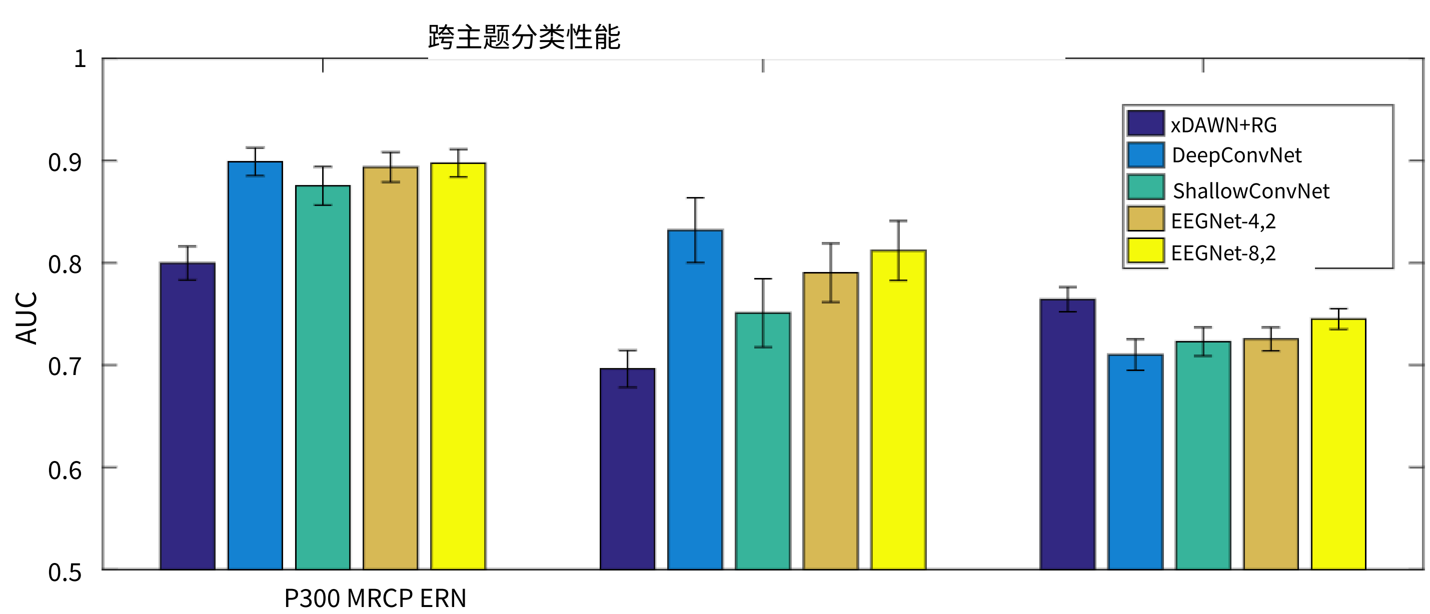


图 4：每个模型在 P300、ERN 和 MRCP 数据集上的跨被试分类性能，平均计算 30 折的结果。误差线表示平均值的 2 个标准误差。对于 P300 和 MRCP 数据集，DeepConvNet 和 EEGNet 模型之间差异很小，这两个模型都优于 ShallowConvNet。对于 ERN 数据集，参考算法（xdaawn + rg）显著优于所有其他模型。

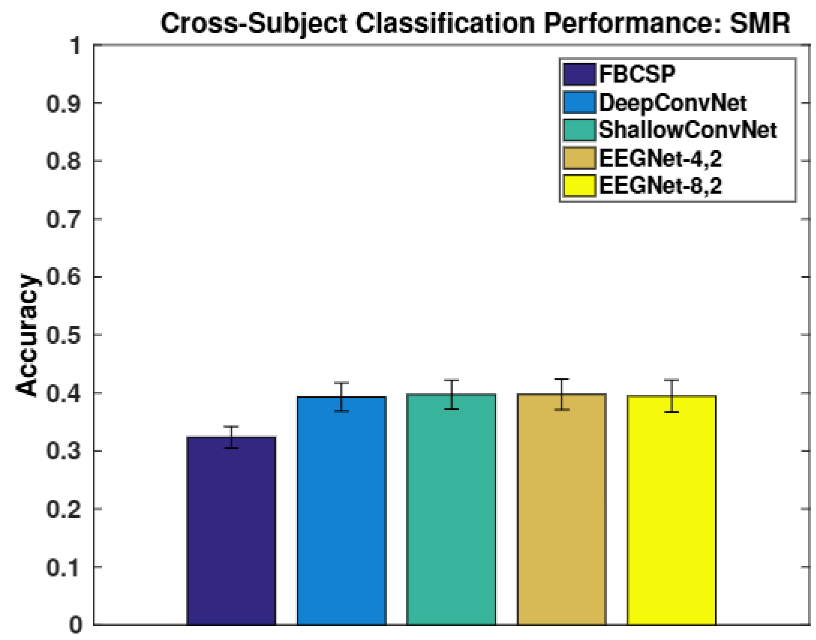


Figure 5: Cross-Subject classification performance for the SMR for each model, averaged over all folds and all subjects. Error bars denote 2 standard errors of the mean. We see that all CNN-based models perform similarly, while slightly outperforming FBCSP.

nificantly different amongst themselves. For the MRCP dataset EEGNet-8,2 and DeepConvNet significantly outperform ShallowConvNet ($p < 0.05$). We also see that both DeepConvNet and ShallowConvNet performance is better when compared to its within-subject performance for the MRCP dataset. For the ERN dataset, xDAWN + RG outperforms all CNN models ($p < 0.05$). Cross-subject classification results for the SMR dataset are shown in Figure 5, where we found no significant difference in performance across all CNN-based models ($p > 0.05$).

3.3 EEGNet Feature Characterization

We illustrate three different approaches to characterize the features learned by EEGNet: (1) Summarizing averaged outputs of hidden unit activations, (2) visualizing convolutional kernel weights, and (3) calculating single-trial feature relevances on classification decision. We illustrate Approach 1 on the P300 dataset for a cross-subject trained EEGNet-4,1 model. We chose to analyze the filters from the P300 dataset due to the fact that multiple neurophysiological events occur simultaneously: participants were told to press a button with their dominant hand whenever a target image appeared on the screen. Because of this, target trials contain both the P300 event-related potential as well as the alpha/beta desynchronizations in contralateral motor cortex due to button presses. Here we were interested in whether or not the EEGNet architecture was capable of separating out these confounding events. We were also interested in quantifying the classification performance of the architecture whenever specific filters were removed from the model.

Figure 6 shows the spatial topographies of the four filters along with an average wavelet time-frequency difference, calculated using Morlet wavelets [100], between all target trials and all non-target trials. Here we see four distinct filters appear. The time-frequency analysis of Filter 1 shows an increase in low-frequency power approximately 500ms after image presentation, followed by desynchronizations in alpha frequency. As nearly all subjects in the P300 dataset are right-handed, we also see significant activity along the left motor cortex. Time-frequency analysis of Filter 2

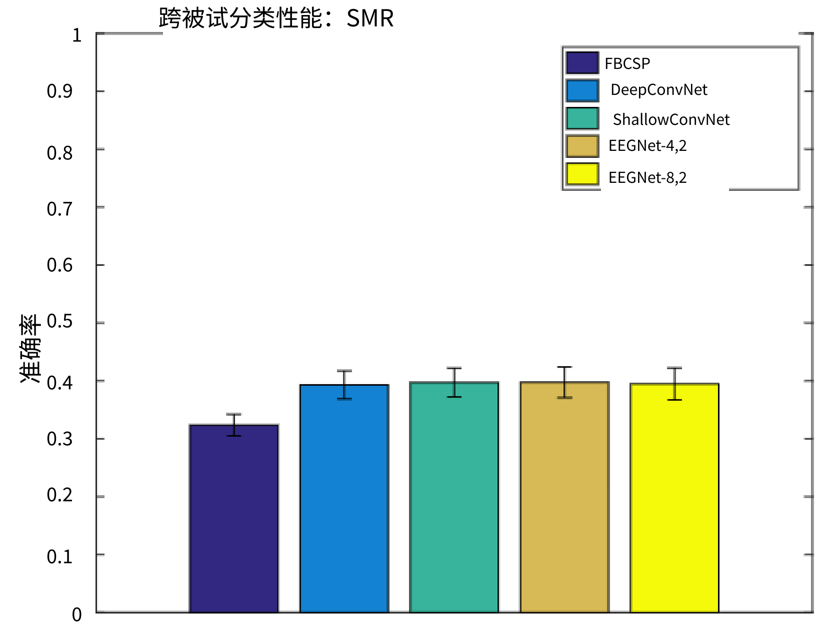


图 5：每个模型对 SMR 的跨主题分类性能，所有折叠和所有主题的平均值。误差线表示平均值的 2 个标准误差。我们看到所有基于 CNN 的模型表现相似，略微优于 FBCSP。

它们彼此之间没有显著差异。对于 MRCP 数据集，EEGNet-8.2 和 DeepConvNet 显著优于 ShallowConvNet（$p < 0.05$）。我们还看到，与 MRCP 数据集的组内性能相比，DeepConvNet 和 ShallowConvNet 的性能更好。对于 ERN 数据集，xDAWN + RG 优于所有 CNN 模型（$p < 0.05$）。SMR 数据集的跨主题分类结果如图 5 所示，我们发现所有基于 CNN 的模型之间性能没有显著差异（$p > 0.05$）。

3.3 EEGNet 特征表征

我们展示了三种不同的方法来表征 EEGNet 学习到的特征：(1) 汇总隐藏单元激活的平均输出，(2) 可视化卷积核权重，以及(3) 计算单次试验特征在分类决策中的相关性。我们在 P300 数据集上展示了方法 1，使用的是跨被试训练的 EEGNet-4.1 模型。我们选择分析 P300 数据集的过滤器，因为同时发生了多种神经生理事件：参与者被告知在屏幕上出现目标图像时用优势手按键。因此，目标试验既包含 P300 事件相关电位，也包含由于按键导致对侧运动皮层的 alpha/beta 去同步。在这里，我们感兴趣的是 EEGNet 架构是否能够分离出这些混杂事件。我们还感兴趣于在从模型中移除特定过滤器时，量化该架构的分类性能。

图 6 展示了四个滤波器的空间拓扑结构，以及使用 Morlet 小波[100]计算的所有目标试验与所有非目标试验之间的平均小波时频差异。这里我们可以看到四个不同的滤波器。滤波器 1 的时频分析显示，在图像呈现后约 500ms 时低频功率增加，随后出现 alpha 频率的去同步现象。由于 P300 数据集中的几乎所有受试者都是右撇子，我们也观察到左侧运动皮层存在显著活动。滤波器 2 的时频分析

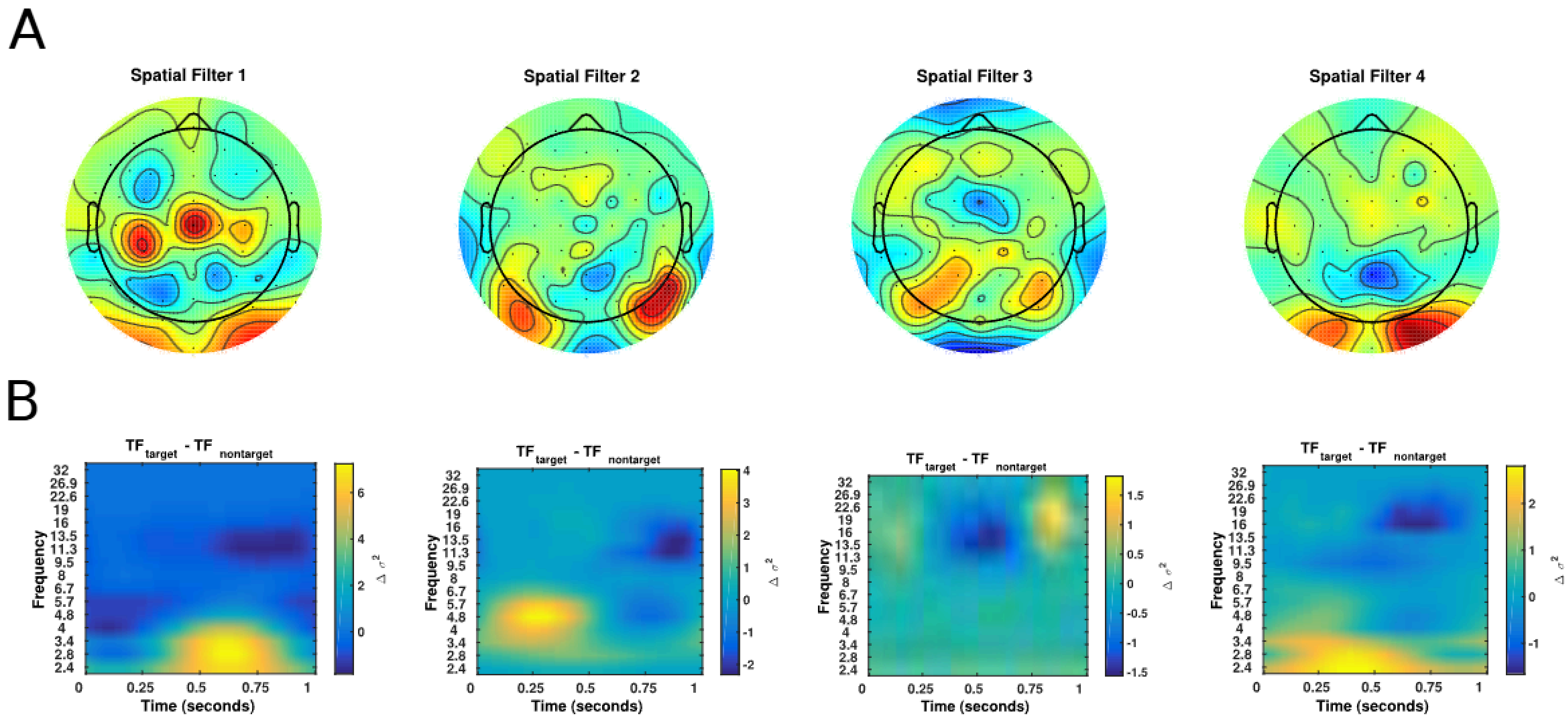
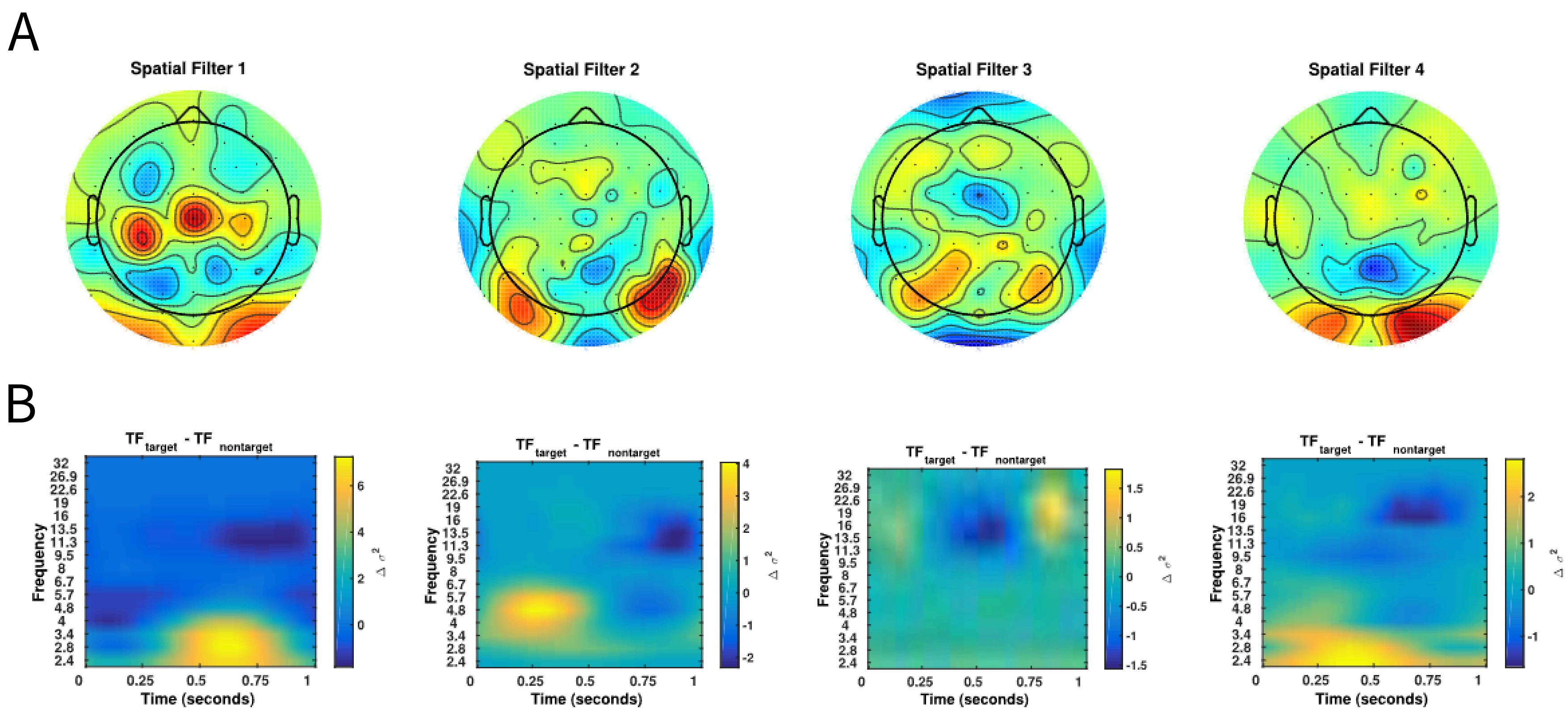


Figure 6: Visualization of the features derived from an EEGNet-4,1 model configuration for one particular cross-subject fold in the P300 dataset. (A) Spatial topoplots for each spatial filter. (B) The mean wavelet time-frequency difference between target and non-target trials for each individual filter.

appears to show a significant theta-beta relationship; while increases in theta activity have been previously noted in the P300 literature in response to targets [102], a relationship between theta and beta has not previously been noted. The time-frequency difference for Filter 4 appears to correspond with the P300, with an increase low-frequency power approximately 350ms after image presentation.

Filters Removed	Test Set AUC
(1)	0.8866
(2)	**0.9076**
(3)	0.8910
(4)	0.8747
(1, 2)	0.8875
(1, 3)	0.8593
(1, 4)	0.8325
(2, 3)	**0.8923**
(2, 4)	0.8721
(3, 4)	0.8206
(1, 2, 3)	**0.8637**
(1, 2, 4)	0.8202
(1, 3, 4)	0.7108
(2, 3, 4)	0.7970
None	**0.9054**

Table 4: Performance of a cross-subject trained EEGNet-4,1 model when removing certain filters from the model, then using the model to predict the test set for one randomly chosen fold of the P300 dataset. AUC values in bold denote the best performing model when removing 1, 2 or 3 filters at a time. As the number of filters removed increases, we see decreases in classification performance, although the magnitude of the decrease depends on which filters are removed.



滤波器 1 的时间-频率分析显示，在图像呈现后约 500ms 时，低频功率有所增加（图 6：P300 数据集中特定跨受试者折选的 EEGNet-4,1 模型配置导出的特征可视化。（A）每个空间滤波器的空间拓扑图。（B）每个独立滤波器在目标与非目标试验之间的平均小波时间-频率差异）。

似乎显示出显著的 theta-beta 关系；虽然 theta 活动的增加在 P300 文献中已被注意到是对目标的反应[102]，但此前并未注意到 theta 与 beta 之间的关系。滤波器 4 的时间-频率差异似乎与 P300 相对应，在图像呈现后约 350ms 时，低频功率有所增加。

移除滤波器测试集 AUC
(1) 0.8866
(2) 0.9076
(3) 0.8910
(4) 0.8747
(1, 2) 0.8875
(1, 3) 0.8593
(1, 4) 0.8325
(2, 3) 0.8923
(2, 4) 0.8721
(3, 4) 0.8206
(1, 2, 3) 0.8637
(1, 2, 4) 0.8202
(1, 3, 4) 0.7108
(2, 3, 4) 0.7970
无 0.9054

表 4：在移除模型中某些滤波器后，使用交叉学科训练的 EEGNet-4,1 模型对 P300 数据集随机选择的一个折进行测试集预测的性能表现。加粗的 AUC 值表示在每次移除 1、2 或 3 个滤波器时表现最佳的模型。随着移除滤波器数量的增加，分类性能下降，但下降的幅度取决于移除的是哪些滤波器。

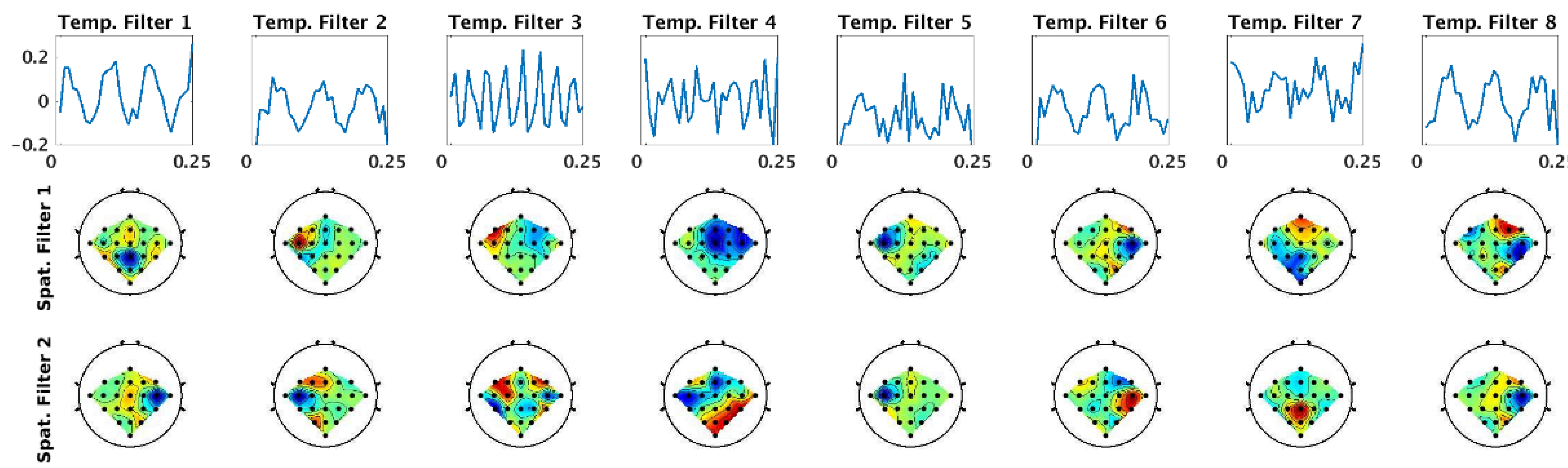


Figure 7: Visualization of the features derived from a within-subject trained EEGNet-8,2 model for Subject 3 of the SMR dataset. Each of the 8 columns shows the learned temporal kernel for a 0.25 second window (top) with its two associated spatial filters (bottom two). We see that, while many of the temporal filters are isolating slower-wave activity, the network identifies a higher-frequency filter at approximately 32Hz (Temp. Filter 3, which shows 8 cycles in a 0.25 s window).

We also conducted a feature ablation study, where we iteratively removed a set of filters (by replacing the filters with zeros) and re-applied the model to predict trials in the test set. We do this for all combinations of the four filters. Classification results for this ablation study are shown in Table 4. We see that test set performance is minimally impacted by the removal of any single filter, with the largest decrease occurring when removing Filter 4. As expected, when removing pairs of filters the decrease in performance is more pronounced, with the largest decrease observed when removing Filters 3 and 4. Removing Filters 2 and 3 results in practically no change in classification performance when compared to the full model, suggesting that the most important features in this task are being captured by Filters 1 and 4. This finding is further reinforced when looking at classification performance when three filters are removed; a model that contains only Filter 4 (0.8637 AUC) performs fairly well when compared to models that contain only Filter 2 (0.7108 AUC) or Filter 1 (0.7970 AUC).

Figure 7 shows the filters learned for the EEGNet-8,2 model for a within-subject classification of Subject 3 for the SMR dataset. Each column of this figure denotes the learned temporal kernel (top row) with its two associated spatial filters (bottom two rows). Note that we are learning temporal filters of length 32 samples, which correspond to 0.25 seconds in time; hence, we estimate the frequency for each temporal filter as four times the number of observed cycles. Here we see that EEGNet-8,2 learns both slow-frequency activity at approximately 12Hz (Filters 1, 2, 6 and 8, which show three cycles in a 0.25 s window) and high-frequency activity at approximately 32Hz (Filter 3, which show 8 cycles). Figure 8 compares the spatial filters associated with 8-12Hz frequency band learned by EEGNet-8,2 with the spatial filters learned by FBCSP in the 8-12Hz filter-bank for each of the four OVR combinations. For ease of description we will use the notation X-Y to denote the row-column filter. Here we see many of the filters are strongly positively correlated across models (i.e.: the 1-1 filter of EEGNet-8,2 with the 3-1 filter for FBCSP ($\rho = 0.93$) and the 2-1 filter of EEGNet-8,2 with the 3-4 filter of FBCSP ($\rho = 0.83$)), while some are strongly negatively correlated (the 3-1 filter of EEGNet-8,2 with the 1-1 filter of FBCSP ($\rho = -0.93$)), indicating a similar filter up to a sign ambiguity. This suggests that EEGNet, through the use of depthwise convolutions, is

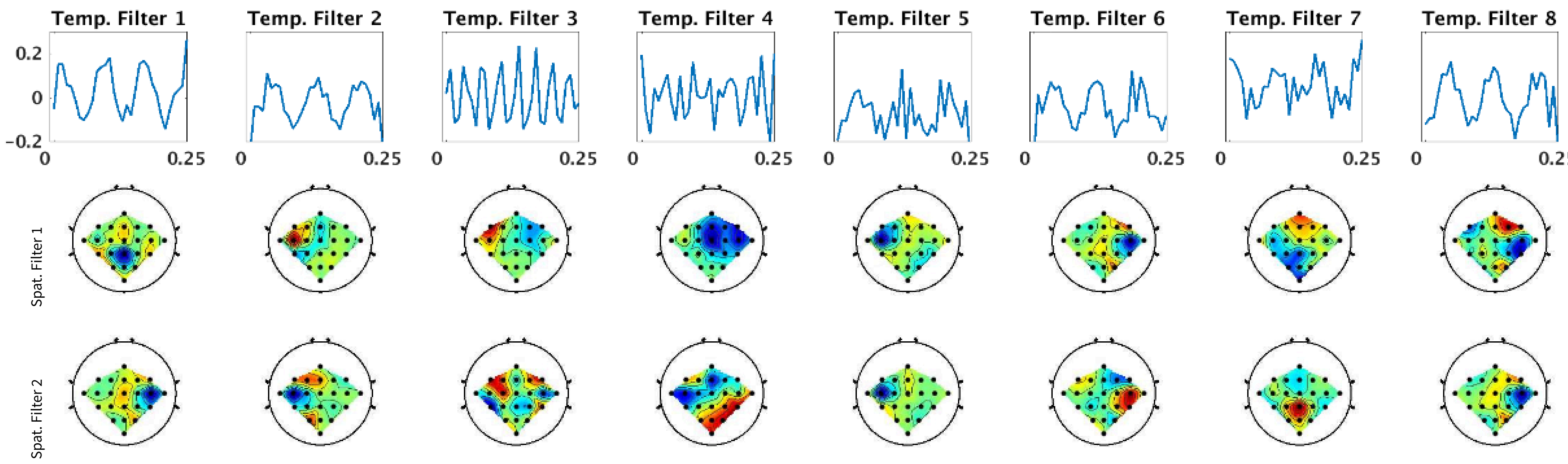


图 7：SMR 数据集中受试者 3 的 EEGNet-8,2 模型内受试者训练所导出的特征可视化。每列展示了一个 0.25 秒窗口的学习时序核（顶部）及其相关的两个空间滤波器（底部两个）。我们看到，虽然许多时序滤波器在隔离慢波活动，但网络在约 32Hz 处识别了一个更高频的滤波器（时序滤波器 3，在 0.25 秒窗口中显示 8 个周期）。

我们还进行了一项特征消融研究，其中我们迭代地移除一组滤波器（通过将滤波器替换为零），并将模型重新应用于预测测试集中的试验。我们对所有四个滤波器的组合都进行了这样的操作。这项消融研究的分类结果如表 4 所示。我们发现，移除任何一个滤波器对测试集性能的影响最小，当移除滤波器 4 时，性能下降最大。正如预期的那样，当移除滤波器对时，性能下降更加明显，当移除滤波器 3 和 4 时，观察到性能下降最大。与完整模型相比，移除滤波器 2 和 3 时，分类性能几乎没有变化，这表明这项任务中最重要的特征是由滤波器 1 和 4 捕获的。当移除三个滤波器时，这一发现得到了进一步的证实；仅包含滤波器 4 的模型（0.8637 AUC）与仅包含滤波器 2（0.7108 AUC）或滤波器 1（0.7970 AUC）的模型相比，性能表现相当好。

图 7 展示了用于 SMR 数据集中受试者 3 的受试者内分类的 EEGNet-8,2 模型所学习的滤波器。该图的每一列表示一个学习到的时域核（顶行），以及与之相关的两个空间滤波器（底两行）。请注意，我们学习的时域滤波器长度为 32 个样本，对应于 0.25 秒的时间；因此，我们将每个时域滤波器的频率估计为观测到的周期数的四倍。在这里我们看到，EEGNet-8,2 学习了大约 12Hz 的慢频率活动（滤波器 1、2、6 和 8，它们在 0.25 秒的窗口中显示三个周期）和大约 32Hz 的高频活动（滤波器 3，显示 8 个周期）。图 8 比较了 EEGNet-8,2 学习的与 8-12Hz 频率带相关的空间滤波器与 FBCSP 在 8-12Hz 滤波器组中为每个 OVR 组合学习的空间滤波器。为了便于描述，我们将使用 X-Y 表示法来表示行-列滤波器。 在这里我们看到许多滤波器在各个模型中呈强正相关（即：EEGNet-8,2 的 1-1 滤波器与 FBCSP 的 3-1 滤波器（$\rho = 0.93$）以及 EEGNet-8,2 的 2-1 滤波器与 FBCSP 的 3-4 滤波器（$\rho = 0.83$）），而有些则呈强负相关（EEGNet-8,2 的 3-1 滤波器与 FBCSP 的 1-1 滤波器（$\rho = -0.93$）），这表明滤波器相似，只是存在符号模糊。这表明 EEGNet 通过使用深度卷积，是

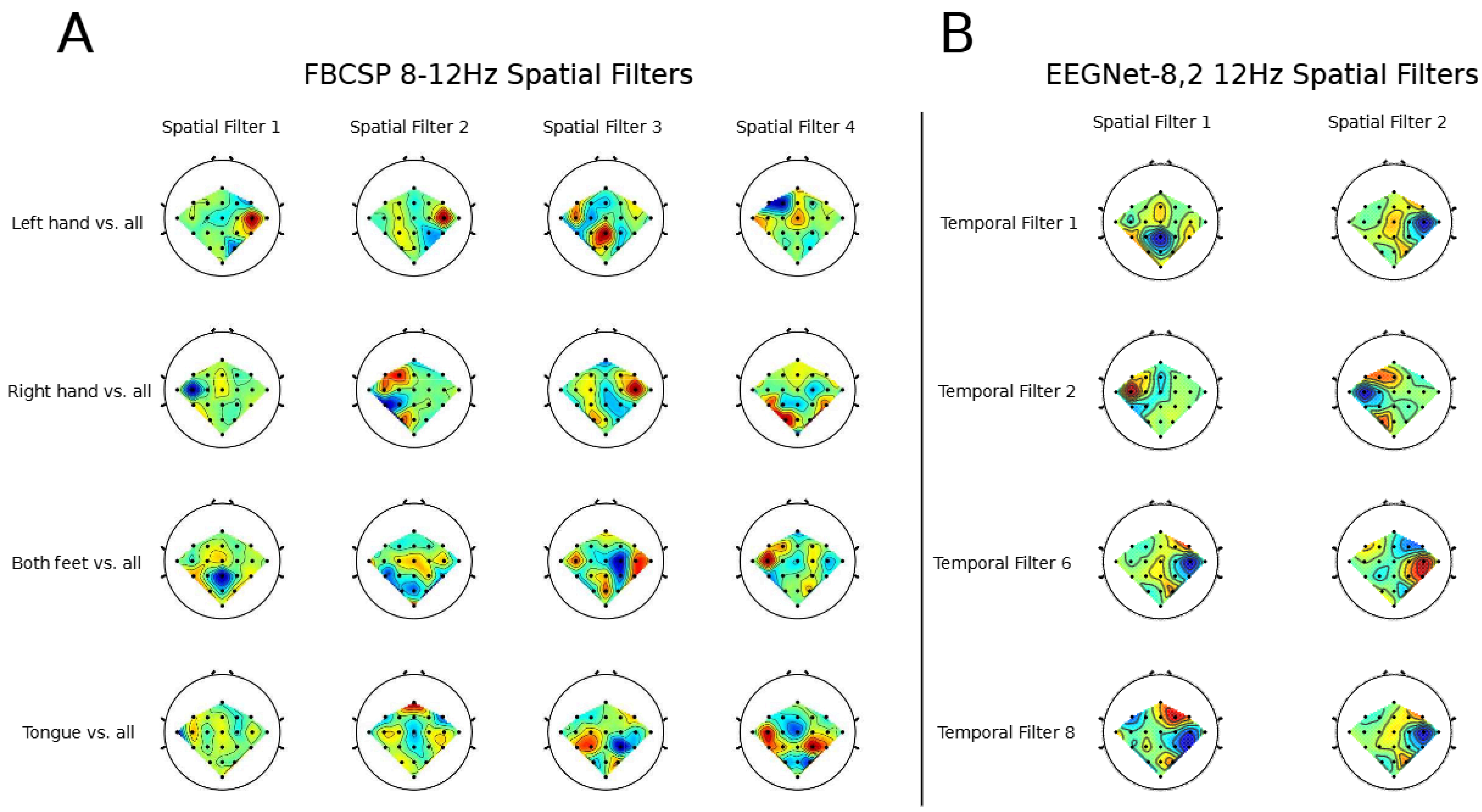
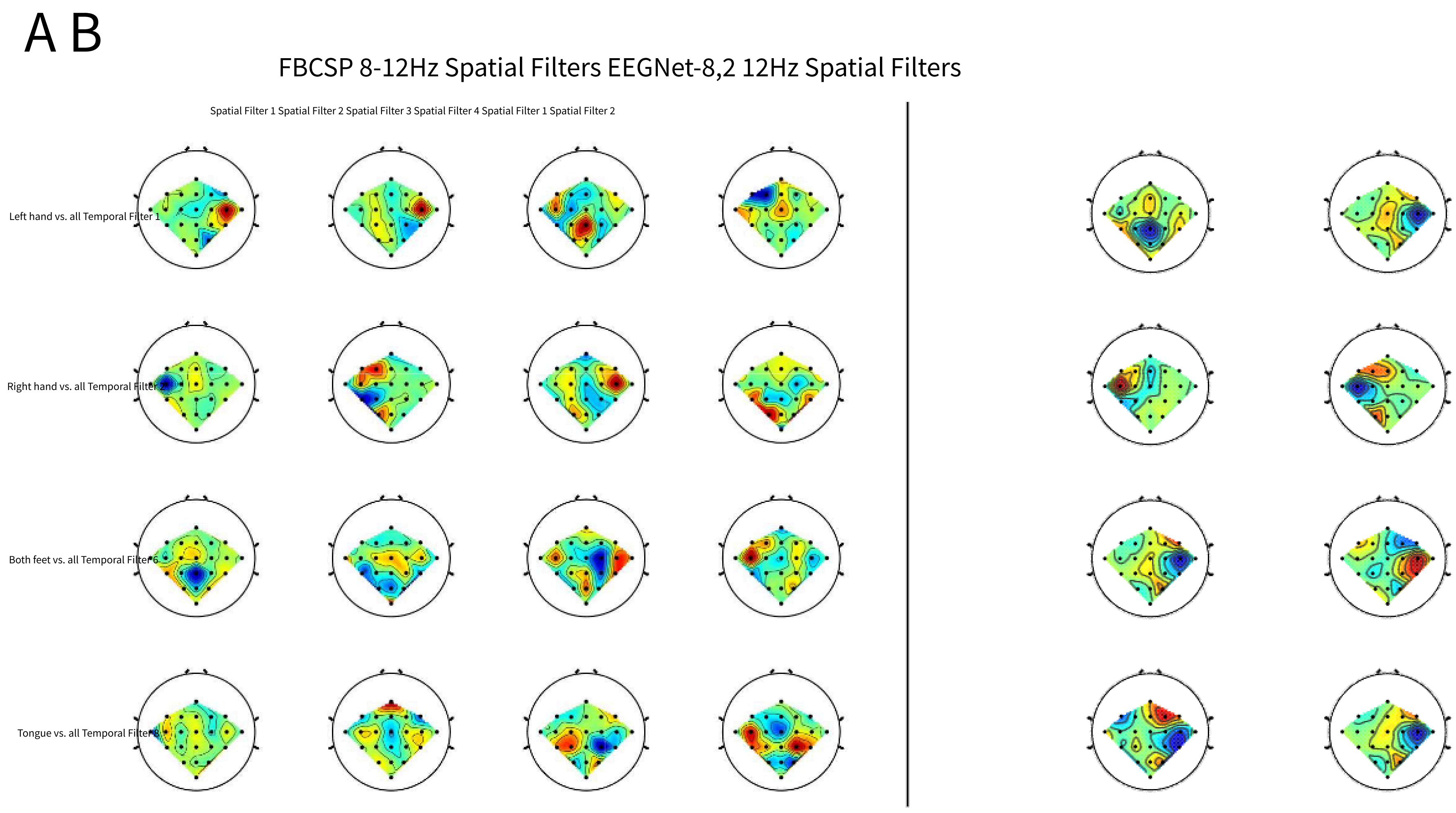


Figure 8: Comparison of the 4 spatial filters learned by FBCSP in the 8-12Hz filter bank for each OVR class combination (A) with the spatial filters learned by EEGNet-8,2 (B) for 4 temporal filters that capture 12Hz frequency activity for Subject 3 of the SMR dataset (Temporal Filters 1, 2, 6 and 8, see Figure 7). We see that similar filters appear across both FBCSP and EEGNet-8,2.

capable of learning band-specific spatial filters in a similar manner as FBCSP.

Figure 9 shows the single-trial feature relevances for EEGNet-8,2, calculated using DeepLIFT, for three three different test trials for one cross-subject fold of the MRCP dataset. Here we see that the high-confidence predictions (Figure 9A and Figure 9B, for left and right finger movement, respectively) both correctly show the contralateral motor cortex relevance as expected, whereas for a low-confidence prediction (Figure 9C), the feature relevance is more broadly distributed, both in time and in space on the scalp.

Figure 10 shows an additional example of using DeepLIFT to analyze feature relevance for a cross-subject trained EEGNet-4,2 model for one test subject of the ERN dataset. Margaux et. al. [56] previously noted that the average ERP for correct feedback trials has an earlier peak positive potential, corresponding to approximately 325 ms, whereas the positive peak potential for incorrect trials occurs slightly later, approximately 475 ms. Here we see the same temporal difference in the timing of the peak positive potential for incorrect feedback trials (vertical line in top row of Figure 10) and correct feedback trials (vertical line in bottom row of Figure 10). We also see the DeepLIFT feature relevances align very closely to that of the peak positive potential for both classes, suggesting that the network has focused on the peak positive potential as the relevant feature for ERN classification. This finding supports results previously reported in [56], where they showed a strong positive correlation between the amplitude of the peak positive potential and the accuracy of error detection.



93)) 图 8：FBCSP 在 8-12Hz 滤波器组中为每个 OVR 类别组合学习的 4 个空间滤波器（A）与 EEGNet-8,2 学习的空间滤波器（B）的比较，后者用于捕获 SMR 数据集第 3 位受试者的 12Hz 频率活动的 4 个时间滤波器（时间滤波器 1、2、6 和 8，见图 7）。我们看到 FBCSP 和 EEGNet-8,2 中出现了相似的滤波器。

capable of learning band-specific spatial filters in a similar manner as FBCSP.

图 9 展示了使用 DeepLIFT 计算的 EEGNet-8,2 的单次试验特征相关性，针对 MRCP 数据集的一个跨受试者折中的一三个不同测试试验。这里我们可以看到，高置信度预测（图 9A 和图 9B，分别对应左手和右手运动）都正确地显示了预期的对侧运动皮层相关性，而对于一个低置信度预测（图 9C），特征相关性在时间和头皮空间上都更广泛地分布。

图 10 展示了一个使用 DeepLIFT 分析特征相关性的额外示例，该示例针对 ERN 数据集的一个测试受试者，使用了一个跨主题训练的 EEGNet-4.2 模型。Margaux 等人[56]先前指出，正确反馈试验的平均 ERP 具有更早的峰值正电位，对应大约 325 毫秒，而错误试验的正峰值电位则稍晚，大约为 475 毫秒。在这里，我们看到错误反馈试验（图 10 顶行中的垂直线）和正确反馈试验（图 10 底行中的垂直线）的峰值正电位时间差异相同。我们还看到 DeepLIFT 特征相关性与两类中的峰值正电位非常吻合，这表明网络将峰值正电位作为 ERN 分类的相关特征。这一发现支持了先前在[56]中报告的结果，他们展示了峰值正电位的振幅与错误检测准确率之间存在强烈的正相关。

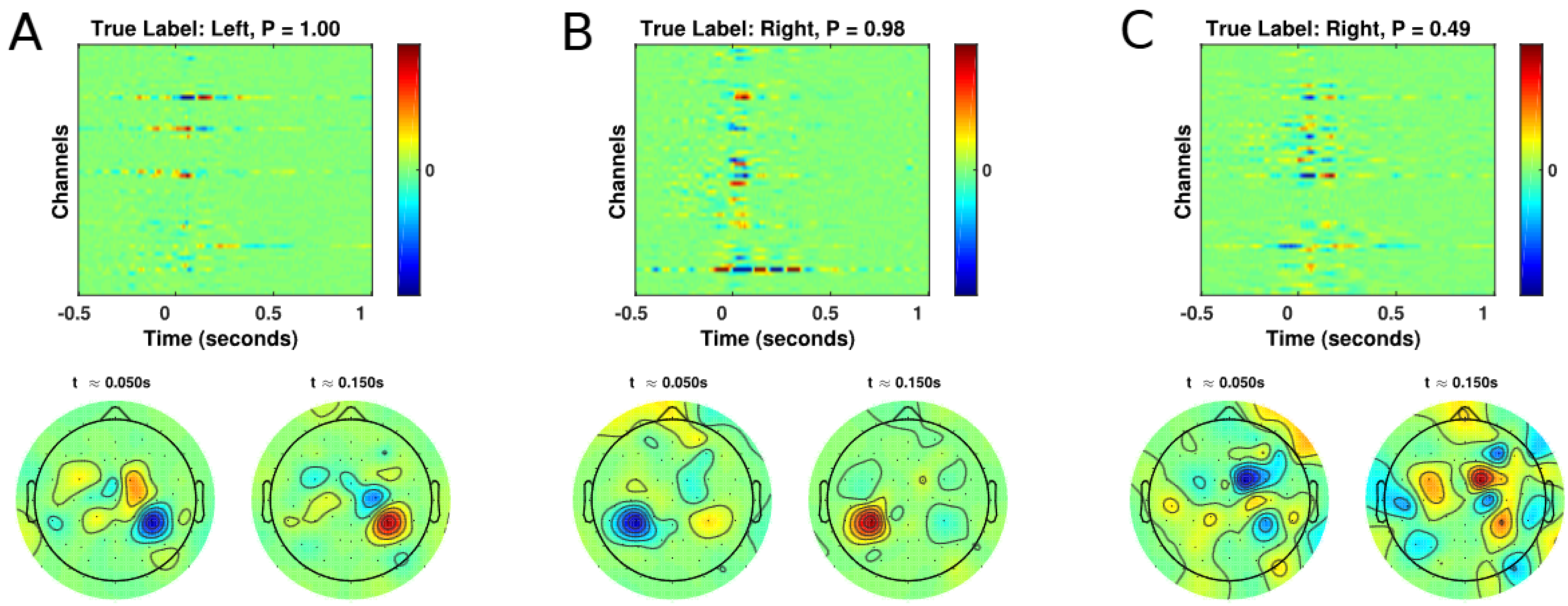


Figure 9: (Top row) Single-trial EEG feature relevance for a cross-subject trained EEGNet-8,2 model, using DeepLIFT, for three different test trials of the MRCP dataset: (A) a high-confidence, correct prediction of left finger movement, (B) a high-confidence, correct prediction of right finger movement and (C) a low-confidence, incorrect prediction of left finger movement. Titles include the true class label and the predicted probability of that label. (Bottom row) Spatial topoplots of the relevances at two time points: approximately 50 ms and 150 ms after button press. As expected, the high-confidence trials show the correct relevances corresponding to contralateral motor cortex for left (A) and right (B) button presses, respectively. For the low-confidence trial we see the relevances are more mixed and broadly distributed, without a clear spatial localization to motor cortices.

4 Discussion

In this work we proposed *EEGNet*, a compact convolutional neural network for EEG-based BCIs that can generalize across different BCI paradigms in the presence of limited data and can produce interpretable features. We evaluated EEGNet against the state-of-the-art approach for both ERP and Oscillatory-based BCIs across four EEG datasets: P300 visual-evoked potentials, Error-Related Negativity (ERN), Movement-Related Cortical Potentials (MRCP) and Sensory Motor Rhythms (SMR). To the best of our knowledge, this represents the first work that has validated the use of a single network architecture across multiple BCI datasets, each with their own feature characteristics and data set sizes. Our work introduced the use of Depthwise and Separable Convolutions [42] for EEG signal classification, and showed that they can be used to construct an EEG-specific model which encapsulates well-known EEG feature extraction concepts. Finally, through the use of feature visualization and ablation analysis, we show that neurophysiologically interpretable features can be extracted from the EEGNet model, providing further validation and evidence that the network performance is not being driven by noise or artifact signals in the data. This last finding is particularly important, as it is a critical component to understanding the validity and robustness of CNN model architectures not just for EEG [32,101], but for CNN architectures in general [16,94,99].

The learning capacity of CNNs comes in part from their ability to automatically extract intricate feature representations from raw data. However, since the features are not hand-designed by

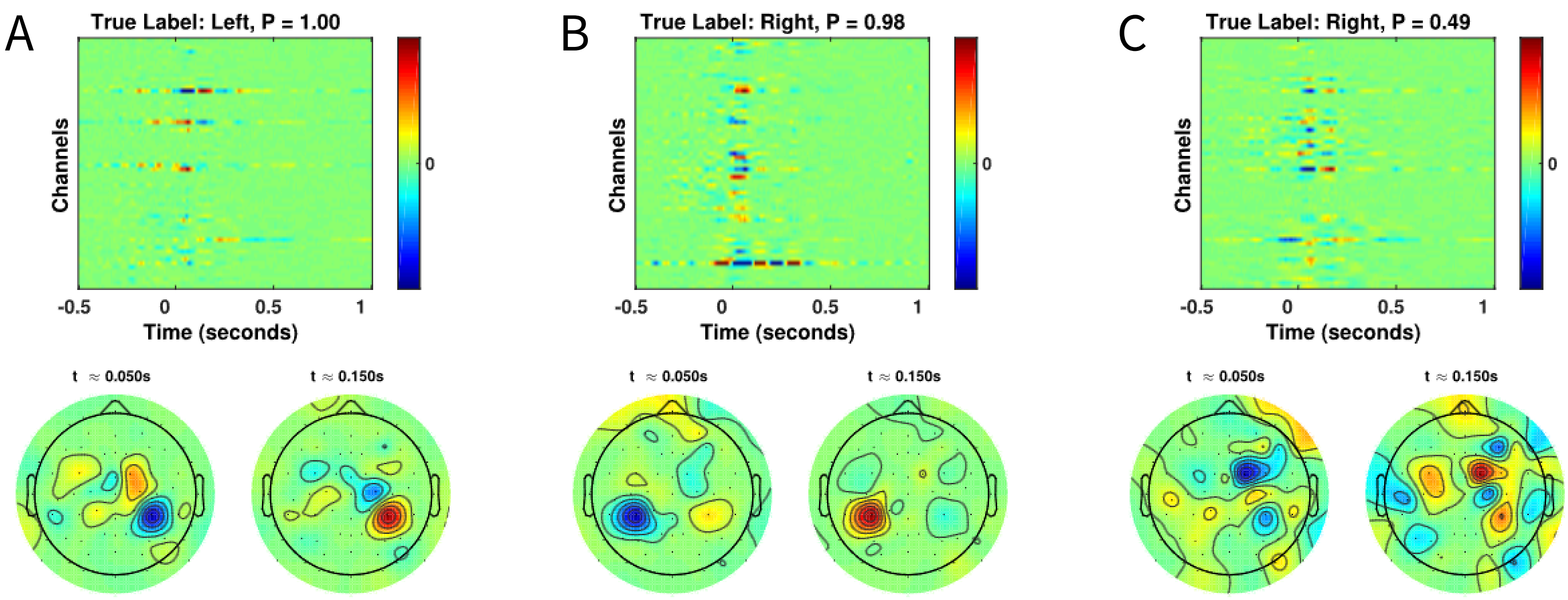


图 9：（顶行）跨被试训练的 EEGNet-8,2 模型使用 DeepLIFT 的单次试验 EEG 特征相关性，针对 MRCP 数据集的三种不同测试试验：（A）高置信度正确预测左手运动，（B）高置信度正确预测右手运动和（C）低置信度错误预测左手运动。标题包括真实类别标签和该标签的预测概率。（底行）两个时间点的相关性空间拓扑图：按钮按下后约 50 ms 和 150 ms。正如预期，高置信度试验显示了与左侧（A）和右侧（B）按钮按下相对应的正确相关性，分别对应对侧运动皮层。对于低置信度试验，我们看到相关性更加混杂且广泛分布，没有明确的运动皮层空间定位。

4 Discussion

在这项工作中，我们提出了 EEGNet，一种用于基于脑电图（EEG）的脑机接口（BCI）的紧凑型卷积神经网络，该网络能够在有限数据的情况下泛化到不同的 BCI 范式，并能产生可解释的特征。我们使用四个脑电图（EEG）数据集——P300 视觉诱发电位、错误相关负性（ERN）、运动相关皮质电位（MRCP）和感觉运动节律（SMR）——评估了 EEGNet 在脑电图诱发电位（ERP）和基于振荡的 BCI 方面的性能，并与当前最先进的方法进行了比较。据我们所知，这是首次验证单一网络架构在多个 BCI 数据集上的应用，每个数据集都具有其独特的特征特性和数据集大小。我们的工作引入了深度卷积和可分离卷积[42]在脑电图信号分类中的应用，并表明它们可以用于构建一个专门针对脑电图的模型，该模型封装了著名的脑电特征提取概念。 最后，通过特征可视化和消融分析，我们展示了可以从 EEGNet 模型中提取神经生理学可解释的特征，这进一步验证并提供了证据，证明网络性能并非由数据中的噪声或伪信号驱动。这一最新发现尤为重要，因为它对于理解 CNN 模型架构的有效性和鲁棒性至关重要，不仅适用于 EEG[32,101]，也适用于 CNN 架构的普遍情况[16,94,99]。

The learning capacity of CNNs comes in part from their ability to automatically extract intricate feature representations from raw data. However, since the features are not hand-designed by

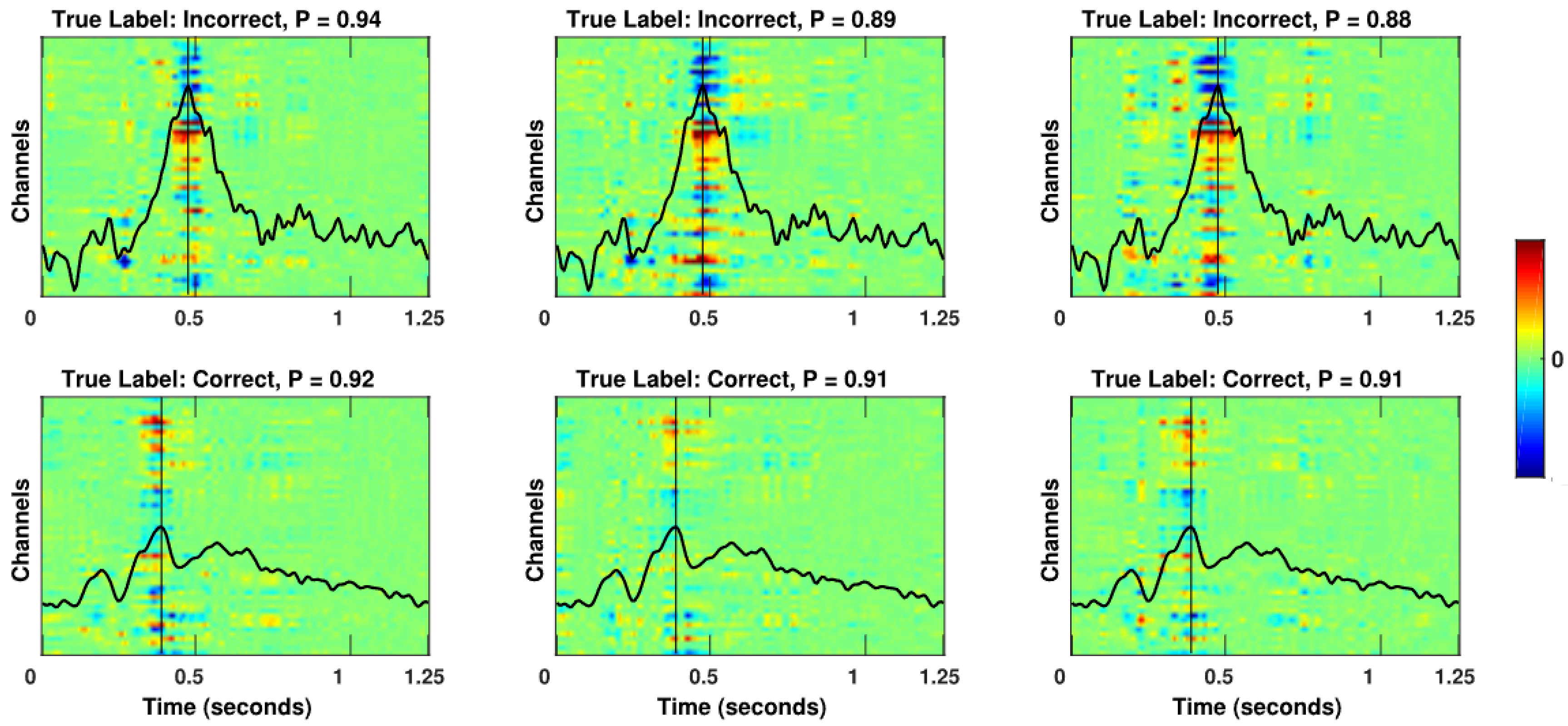


Figure 10: Single-trial EEG feature relevance for a cross-subject trained EEGNet-4,2 model, using DeepLIFT, for the one test subject of the ERN dataset. (Top Row) Feature relevances for three correctly predicted trials of incorrect feedback, along with its predicted probability P. (Bottom Row) Same as the top row but for three correctly predicted trials of correct feedback. The black line denotes the average ERP, calculated at channel Cz, for incorrect feedback trials (top row) and for correct feedback trials (bottom row). The thin vertical line denotes the positive peak of the average ERP waveform. Here we see feature relevances coincide strongly with the positive peak of the average ERP waveform for each trial. We also see the positive peak occurring slightly earlier for correct feedback trials versus incorrect feedback trials, consistent with the results in [56].

human engineers, understanding the meaning of those features poses a significant challenge in producing interpretable models [95]. This is especially true when CNNs are used for the analysis of EEG data where features from neural signals are often non-stationary and corrupted by noise artifacts [103, 104]. In this study, we illustrated three different approaches for visualizing the features learned by EEGNet: (1) analyzing spatial filter outputs, averaged over trials, on the P300 dataset, (2) visualizing the convolutional kernel weights on the SMR dataset and comparing them to the weights learned by FBCSP, and (3) performing single-trial relevance analysis on the MRCP and SMR datasets. For the ERN dataset we compared single-trial feature relevances to averaged ERPs and showed that relevant features coincided with the peak of the positive potential for correct and incorrect feedback trials, which has been shown in previous literature to be positively correlated to classifier performance [56]. In addition, we conducted a feature ablation study to understand the impact of a classification decision on the presence or absence of a particular feature on the P300 dataset. In each of these analyses, we showed that EEGNet was capable of extracting interpretable features that generally corresponded to known neurophysiological phenomena. These results suggest that the classification performances we observed were not due to artifact or noise sources in the data.

Our results showed that the spatial filters learned by EEGNet for temporal kernels around 12Hz were significantly correlated to the spatial filters learned by FBCSP in the 8-12Hz filter bank for the SMR dataset. This is interesting to note, as the optimization criterion for CSP

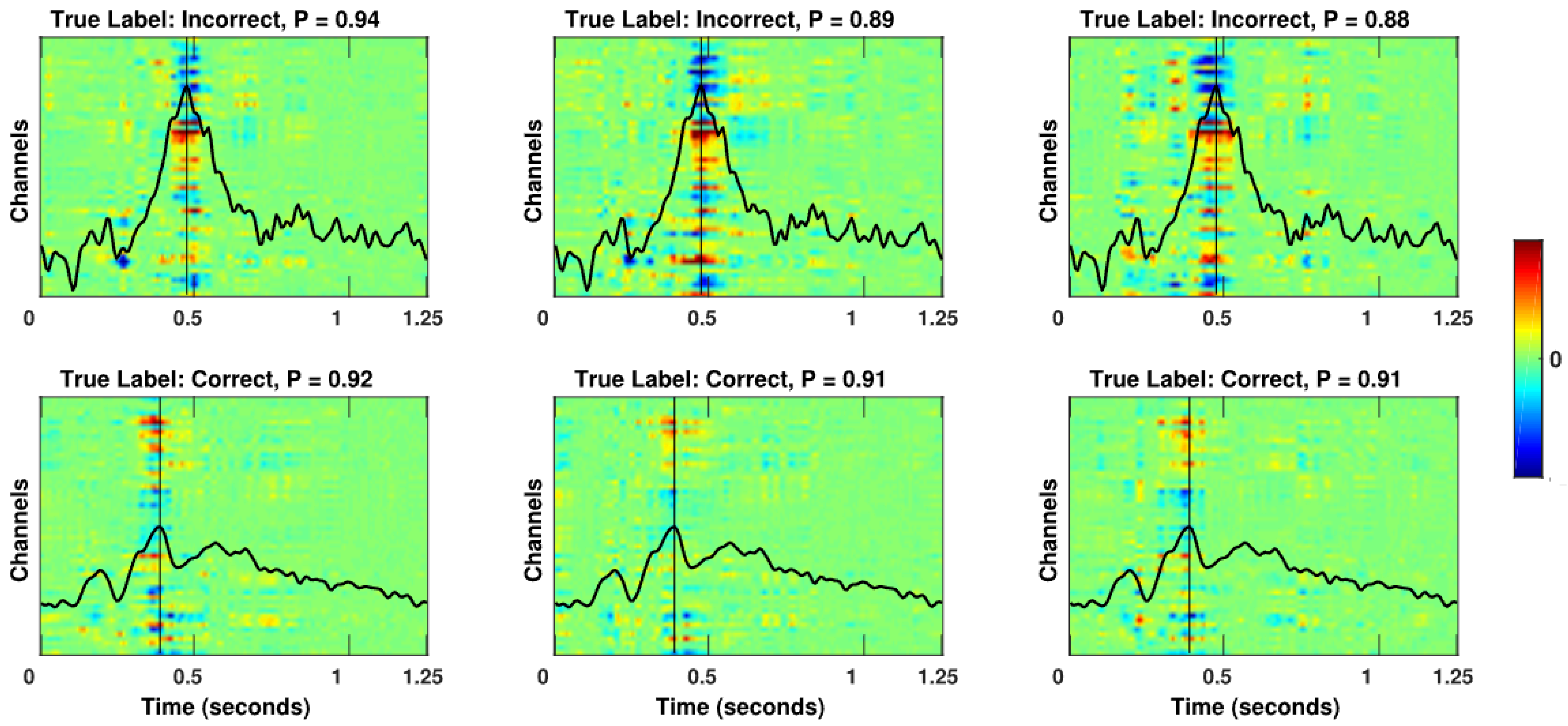


图 10：使用 DeepLIFT 对跨被试训练的 EEGNet-4,2 模型进行单次试验 EEG 特征相关性的分析，针对 ERN 数据集的一个测试被试。（顶行）正确预测但反馈错误的三个试验的特征相关性及其预测概率 P。（底行）与顶行相同，但针对正确预测且反馈正确的三个试验。黑线表示错误反馈试验（顶行）和正确反馈试验（底行）在 Cz 通道计算出的平均 ERP。细垂直线表示平均 ERP 波形的正峰值。我们看到特征相关性与每个试验的平均 ERP 波形的正峰值高度吻合。我们还观察到正确反馈试验的正峰值出现时间比错误反馈试验稍早，这与文献[56]中的结果一致。

人类工程师理解这些特征的意义，在构建可解释模型时面临重大挑战[95]。当使用卷积神经网络（CNN）分析脑电图（EEG）数据时，尤其如此，因为神经信号的特征通常是时变的，且受到噪声伪影的干扰[103, 104]。在本研究中，我们展示了三种可视化 EEGNet 学习到的特征的方法：（1）在 P300 数据集上分析试验平均后的空间滤波器输出；（2）在 SMR 数据集上可视化卷积核权重，并与 FBCSP 学习到的权重进行比较；（3）在 MRCP 和 SMR 数据集上进行单试验相关性分析。对于 ERN 数据集，我们比较了单试验特征的相关性与平均事件相关电位（ERP），并表明相关特征与正确和错误反馈试验的正向电位峰值相吻合，正如先前文献所示，这与分类器性能呈正相关[56]。此外，我们进行了特征消融研究，以理解分类决策对 P300 数据集上特定特征存在与否的影响。 在每项分析中，我们都表明 EEGNet 能够提取可解释的特征，这些特征通常与已知的神经生理现象相对应。这些结果表明，我们观察到的分类性能并非由于数据中的伪影或噪声源所致。

我们的结果表明，EEGNet 学习的 12Hz 附近时间核的空间滤波器与 SMR 数据集在 8-12Hz 滤波器组中学习的 FBCSP 空间滤波器显著相关。这一点值得关注，因为 CSP 的优化准则

(optimal variance separation) is different than the optimization criterion for EEGNet (minimum cross-entropy loss). Because of this, it is not guaranteed that the learned filters from these methods would be comparable. It was encouraging to see that many of the filters did in fact overlap (up to a sign ambiguity), suggesting that EEGNet is learning a similar feature representation to that of FBCSP. This analysis is directly enabled by EEGNet's use of depthwise convolutions to tie spatial filters directly to a temporal filter, an aspect that is unique to this model.

Generally speaking, the classification performance of DeepConvNet and EEGNet were similar across all cross-subject analyses, whereas DeepConvNet performance was lower across nearly all within-subject analyses (with the exception of P300). One possible explanation for this discrepancy is the amount of training data used to train the model; in cross-subject analyses the training set sizes were about 10-15 times larger than that of within-subject analyses. This suggests that DeepConvNet is more data-intensive compared to EEGNet, an unsurprising result given that the model size of DeepConvNet is two orders of magnitude larger than EEGNet (see Table 3). We believe this intuition is consistent with the findings originally reported by the developers of DeepConvNet [32], where they state that a training data augmentation strategy was needed to obtain good classification performance on the SMR dataset. In contrast to their work, we show that EEGNet performed well across all tested datasets without the need for data augmentation, making the model simpler to use in practice.

In general we found that, both in within- and cross-subject analyses, that ShallowConvNet tended to perform worse on the ERP BCI datasets than on the oscillatory BCI dataset (SMR), while the opposite behavior was observed with DeepConvNet. We believe this is due to the fact that the ShallowConvNet architecture was designed specifically to extract log bandpower features; in situations where the dominant feature is signal amplitude (as is the case in many ERP BCIs), ShallowConvNet performance tended to suffer. The opposite situation occurred with DeepConvNet; as its architecture was not designed to extract frequency features, its performance was lower in situations where frequency power is the dominant feature. In contrast, we found that EEGNet performed just as well as ShallowConvNet in SMR classification and just as well as DeepConvNet in ERP classification (and outperforming in the case of within-subject MRCP, ERN and SMR classifications), suggesting that EEGNet is robust enough to learn a wide variety of features over a range of BCI tasks.

The severe underperformance of ShallowConvNet on within-subject MRCP classification was unexpected, given the similarity in neural responses between the MRCP and SMR, and the fact that ShallowConvNet performed well on SMR. This discrepancy in performance is not due to the amount of training data used, as within-subject MRCP classification has approximately 700 training trials, evenly split among left and right finger movements, whereas the SMR dataset has only 192 training trials, evenly split among four classes. In addition, we did not observe large deviations in ShallowConvNet performance on the other datasets (P300 and ERN). In fact, ShallowConvNet performed fairly well on within-subject ERN classification, even though this dataset is the smallest among all datasets used in this study (only having 170 training trials total). Determining the underlying source of this phenomena will be explored in future research.

Deep Learning models for EEG generally employ one of three input styles, depending on their targeted application: (1) the EEG signal of all available channels, (2) a transformed EEG signal

（最优方差分离）与 EEGNet 的优化标准（最小交叉熵损失）不同。由于这一点，不能保证这些方法学习到的滤波器会具有可比性。看到许多滤波器实际上确实重叠（直到符号模糊），这令人鼓舞，表明 EEGNet 正在学习与 FBCSP 相似的特征表示。这种分析直接得益于 EEGNet 使用深度卷积将空间滤波器直接与时间滤波器关联起来，这是该模型独有的一个方面。

一般来说，在所有跨被试分析中，DeepConvNet 和 EEGNet 的分类性能相似，而在几乎所有单被试分析中（P300 除外），DeepConvNet 的性能较低。这种差异的一个可能解释是用于训练模型的数据量；在跨被试分析中，训练集的大小比单被试分析大约大 10-15 倍。这表明与 EEGNet 相比，DeepConvNet 更依赖数据量，考虑到 DeepConvNet 的模型大小比 EEGNet 大两个数量级（见表 3），这一结果并不令人意外。我们认为这一直觉与 DeepConvNet 的开发者最初报告的发现一致[32]，他们指出需要数据增强策略才能在 SMR 数据集上获得良好的分类性能。相比之下，我们表明 EEGNet 在所有测试数据集上都表现良好，无需数据增强，使得该模型在实际应用中更简单。

总的来说，我们发现无论是在被试内还是被试间分析中，ShallowConvNet 在 ERP BCI 数据集上的表现通常比在振荡 BCI 数据集（SMR）上的表现要差，而 DeepConvNet 则表现出相反的行为。我们认为这是因为 ShallowConvNet 的架构专门设计用于提取对数带宽特征；在主导特征是信号幅度（如许多 ERP BCI 的情况）的情况下，ShallowConvNet 的表现往往会下降。DeepConvNet 则出现了相反的情况；由于其架构并非设计用于提取频率特征，在频率功率为主导特征的情况下，其表现较低。相比之下，我们发现 EEGNet 在 SMR 分类上的表现与 ShallowConvNet 相当，在 ERP 分类上的表现与 DeepConvNet 相当（并且在被试内 MRCP、ERN 和 SMR 分类中表现更优），这表明 EEGNet 足够稳健，能够在各种 BCI 任务中学习广泛多样的特征。

ShallowConvNet 在受试者内 MRCP 分类上的严重表现不佳出乎意料，考虑到 MRCP 和 SMR 的神经响应相似性，以及 ShallowConvNet 在 SMR 上表现良好这一事实。这种性能差异并非由于训练数据量，因为受试者内 MRCP 分类大约有 700 个训练试验，均匀分配在左右手指运动之间，而 SMR 数据集只有 192 个训练试验，均匀分配在四个类别之间。此外，我们没有观察到 ShallowConvNet 在其他数据集（P300 和 ERN）上的性能出现较大偏差。事实上，ShallowConvNet 在受试者内 ERN 分类上的表现相当不错，尽管这个数据集是本研究中使用的数据集中规模最小的（总共只有 170 个训练试验）。未来研究将探索这种现象的根本原因。

深度学习模型在处理脑电图（EEG）时，通常根据其目标应用采用三种输入方式之一：(1) 所有可用通道的脑电图信号，(2) 转换后的脑电图信号

(generally a time-frequency decomposition) of all available channels [36] or (3) a transformed EEG signal of a subset of channels [37]. Models that fall in (2) generally see a significant increase in data dimensionality, thus requiring either more data or more model regularization (or both) to learn an effective feature representation. This introduces more hyperparameters that must be learned, increasing the potential variability in model performance due to hyperparameter misspecification. Models that fall in (3) generally require *a priori* knowledge about the channels to select. For example, the model proposed in [37] uses the time-frequency decomposition of channels Cz, C3 and C4 as the inputs for a motor imagery classification task. This channel selection is intentional, given the fact that neural responses to motor actions (the sensory motor rhythm) are observed strongest at those channels and are easily observed through a time-frequency analysis. Also, by only working with three channels, the authors reduce the significant increase in dimensionality of the data. While this approach works well if the feature of interest is known beforehand, this approach is not guaranteed to work well in other applications where the features are not observed at those channels, limiting the overall utility of this approach. We believe models that fall in (1), such as EEGNet and others [28, 30, 31], offer the best tradeoff between input dimensionality and the flexibility to discover relevant features by providing all available channels. This is especially important as BCI technologies evolve into novel application spaces, as the features needed for these future BCIs may not be known beforehand [3–5, 10–12].

Acknowledgments

This project was sponsored by the U.S. Army Research Laboratory under ARL-H70-HR52, ARL-74A-HRCYB and through Cooperative Agreement Number W911NF-10-2-0022. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Government purposes notwithstanding any copyright notation herein.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

(通常是一种时频分解)所有可用通道[36]或(3)一部分通道的转换 EEG 信号[37]。属于(2)类模型的通常会显著增加数据维度，因此需要更多数据或更多模型正则化（或两者）来学习有效的特征表示。这引入了更多需要学习的超参数，由于超参数指定不当，增加了模型性能的潜在可变性。属于(3)类模型的通常需要关于通道的先验知识来选择。例如，文献[37]中提出的模型使用通道 Cz、C3 和 C4 的时频分解作为运动想象分类任务的输入。鉴于神经对运动动作（感觉运动节律）的响应在这些通道上最强且易于通过时频分析观察到，这种通道选择是故意的。此外，仅使用三个通道，作者减少了数据维度的显著增加。 如果感兴趣的特征是事先已知的，这种方法效果很好，但在其他应用中，如果特征在这些通道上未被观察到，这种方法并不能保证效果良好，从而限制了这种方法的总体效用。我们认为属于（1）类模型，如 EEGNet 和其他模型[28, 30, 31]，在输入维度和发现相关特征的灵活性之间提供了最佳权衡，通过提供所有可用通道。随着 BCI 技术发展进入新的应用领域，这一点尤为重要，因为这些未来 BCI 所需的特征可能事先未知[3–5, 10–12]。

致谢

该项目由美国陆军研究实验室资助，项目编号为 ARL-H70-HR52、ARL74A-HRCYB，并通过合作协议编号 W911NF-10-2-0022。本文所含观点和结论均为作者个人意见，不应被视为代表美国政府官方政策，无论是否明确表述。美国政府有权出于政府目的复制和分发本文副本，尽管其中可能包含版权声明。

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作者声明，本研究在开展过程中未涉及任何可能被视为潜在利益冲突的商业或财务关系。

5 Appendix

5.1 DeepConvNet and ShallowConvNet architectures

The DeepConvNet and ShallowConvNet architectures are given in Tables 5 and 6, respectively. The DeepConvNet was designed to be a general-purpose architecture that is not restricted to specific feature types, whereas ShallowConvNet is designed specifically for oscillatory signal classification.

Layer	# filters	size	# params	Activation	Options
Input		(C, T)			
Reshape		(1, C, T)			
Conv2D	25	(1, 5)	150	Linear	mode = valid, max norm = 2
Conv2D	25	(C, 1)	25 * 25 * C + 25	Linear	mode = valid, max norm = 2
BatchNorm			2 * 25		epsilon = 1e-05, momentum = 0.1
Activation				ELU	
MaxPool2D		(1, 2)			
Dropout					p = 0.5
Conv2D	50	(1, 5)	25 * 50 * C + 50	Linear	mode = valid, max norm = 2
BatchNorm			2 * 50		epsilon = 1e-05, momentum = 0.1
Activation				ELU	
MaxPool2D		(1, 2)			
Dropout					p = 0.5
Conv2D	100	(1, 5)	50 * 100 * C + 100	Linear	mode = valid, max norm = 2
BatchNorm			2 * 100		epsilon = 1e-05, momentum = 0.1
Activation				ELU	
MaxPool2D		(1, 2)			
Dropout					p = 0.5
Conv2D	200	(1, 5)	100 * 200 * C + 200	Linear	mode = valid, max norm = 2
BatchNorm			2 * 200		epsilon = 1e-05, momentum = 0.1
Activation				ELU	
MaxPool2D		(1, 2)			
Dropout					p = 0.5
Flatten					
Dense	N			softmax	max norm = 0.5

Table 5: DeepConvNet architecture, where C = number of channels, T = number of time points and N = number of classes, respectively.

5 Appendix

5.1 DeepConvNet and ShallowConvNet architectures

DeepConvNet 和 ShallowConvNet 架构分别给出在表 5 和表 6 中。DeepConvNet 被设计为一种通用架构，不受特定特征类型的限制，而 ShallowConvNet 则专门为振荡信号分类而设计。

Layer # filters size # params Activation Options

输入（C，T）重塑（1，C，T）卷积 2D 25（1，5）150 线性模式=有效，最大范数=2 卷积 2D 25（C，1）25 * 25 * C + 25 线性模式=有效，最大范数=2 批归一化 2 * 25 epsilon = 1e-05，动量=0.1 激活 ELU 最大池化 2D（1，2）Dropout p = 0.5 卷积 2D 50（1，5）25 * 50 * C + 50 线性模式=有效，最大范数=2 批归一化 2 * 50 epsilon = 1e-05，动量=0.1 激活 ELU 最大池化 2D（1，2）Dropout p = 0.5 卷积 2D 100（1，5）50 * 100 * C + 100 线性模式=有效，最大范数=2 批归一化 2 * 100 epsilon = 1e-05，动量=0.1 激活 ELU 最大池化 2D（1，2）Dropout p = 0.5 卷积 2D 200（1，5）100 * 200 * C + 200 线性模式=有效，最大范数=2 批归一化 2 * 200 epsilon = 1e-05，动量=0.1 激活 ELU 最大池化 2D（1，2）Dropout p = 0.5 扁平化 Dense N softmax 最大范数=0.5

Table 5: DeepConvNet architecture, where C = number of channels, T = number of time points and N = number of classes, respectively.

Layer	# filters	size	# params	Activation	Options
Input		(C, T)			
Reshape		(1, C, T)			
Conv2D	40	(1, 13)	560	Linear	mode = same, max norm = 2
Conv2D	40	(C, 1)	40 * 40 * C	Linear	mode = valid, max norm = 2
BatchNorm			2 * 40		epsilon = 1e-05, momentum = 0.1
Activation				square	
AveragePool2D		(1, 35), stride (1, 7)			
Activation				log	
Flatten					
Dropout					p = 0.5
Dense	N			softmax	max norm = 0.5

Table 6: ShallowConvNet architecture, where C = number of channels, T = number of time points and N = number of classes, respectively. Here, the 'square' and 'log' activation functions are given as $f(x) = x^2$ and $f(x) = \log(x)$, respectively. Note that we clip the log function such that the minimum input value is a very small number ($\epsilon = 10e^{-7}$) for numerical stability.

层 #滤波器大小 #参数 激活选项

输入 (C, T)

Reshape (1, C, T) Conv2D 40 (1, 13) 560 Linear 模式 = 相同, 最大范数 = 2 Conv2D 40 (C, 1) 40 * 40 * C Linear 模式 = 有效, 最大范数 = 2 BatchNorm 2 * 40 epsilon = 1e-05, 动量 = 0.1 激活平方 AveragePool2D (1, 35), 步长 (1, 7) 激活对数 Flatten Dropout p = 0.5 Dense N softmax 最大范数 = 0.5

表 6：浅层卷积网络架构，其中 C 为通道数，T 为时间点数，N 为类别数。这里，“平方”和“对数”激活函数分别表示为 f(x) = x 和 f(x) = log(x)。注意，我们限制对数函数的输入值，使得最小输入值是一个非常小的数（ε = 10e）以确保数值稳定性。

References

[1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control." *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*, vol. 113, no. 6, pp. 767–91, jun 2002. [Online]. Available: http://www.ncbi.nlm.nih.gov/pubmed/12048038

[2] A. B. Schwartz, X. T. Cui, D. Weber, and D. W. Moran, "Brain-controlled interfaces: Movement restoration with neural prosthetics," *Neuron*, vol. 52, no. 1, pp. 205 – 220, 2006.

[3] J. van Erp, F. Lotte, and M. Tangermann, "Brain-Computer Interfaces: Beyond Medical Applications," *Computer*, vol. 45, no. 4, pp. 26–34, Apr. 2012.

[4] S. Saproo, J. Faller, V. Shih, P. Sajda, N. R. Waytowich, A. Bohannon, V. J. Lawhern, B. J. Lance, and D. Jangraw, "Cortically coupled computing: A new paradigm for synergistic human-machine interaction," *Computer*, vol. 49, no. 9, pp. 60–68, Sept 2016.

[5] B. J. Lance, S. E. Kerick, A. J. Ries, K. S. Oie, and K. McDowell, "Brain-computer interface technologies in the coming decades," *Proceedings of the IEEE*, vol. 100, no. Special Centennial Issue, pp. 1585–1599, May 2012.

[6] L. F. Nicolas-Alonso and J. Gomez-Gil, "Brain computer interfaces, a review," *Sensors*, vol. 12, no. 2, p. 1211, 2012.

[7] A. Bashashati, M. Fatourechi, R. K. Ward, and G. E. Birch, "A survey of signal processing algorithms in brain–computer interfaces based on electrical brain signals," *Journal of Neural Engineering*, vol. 4, no. 2, p. R32, 2007. [Online]. Available: http://stacks.iop.org/1741-2552/4/i=2/a=R03

[8] D. J. McFarland, C. W. Anderson, K. R. Muller, A. Schlogl, and D. J. Krusienski, "Bci meeting 2005-workshop on bci signal processing: feature extraction and translation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 14, no. 2, pp. 135–138, June 2006.

[9] F. Lotte, M. Congedo, A. Lécuyer, F. Lamarche, and B. Arnaldi, "A review of classification algorithms for eeg-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 4, no. 2, p. R1, 2007. [Online]. Available: http://stacks.iop.org/1741-2552/4/i=2/a=R01

[10] T. O. Zander and C. Kothe, "Towards passive brain–computer interfaces: applying brain–computer interface technology to human–machine systems in general," *Journal of Neural Engineering*, vol. 8, no. 2, p. 025005, 2011. [Online]. Available: http://stacks.iop.org/1741-2552/8/i=2/a=025005

[11] B. Blankertz, M. Tangermann, C. Vidaurre, S. Fazli, C. Sannelli, S. Haufe, C. Maeder, L. E. Ramsey, I. Sturm, G. Curio, and K. R. Müeller, "The berlin brain-computer interface: Non-medical uses of bci technology," *Frontiers in Neuroscience*, vol. 4, no. 198, 2010.

[12] S. M. Gordon, M. Jaswa, A. J. Solon, and V. J. Lawhern, "Real world bci: Cross-domain learning and practical applications," in *Proceedings of the 2017 ACM Workshop on An Application-oriented Approach to BCI out of the Laboratory*, ser. BCIforReal '17. New York, NY, USA: ACM, 2017, pp. 25–28. [Online]. Available: http://doi.acm.org/10.1145/3038439.3038444

[13] G. Hinton, L. Deng, D. Yu, G. E. Dahl, A. r. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T. N. Sainath, and B. Kingsbury, "Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups," *IEEE Signal Processing Magazine*, vol. 29, no. 6, pp. 82–97, Nov 2012.

[14] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[15] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, F. Pereira, C. J. C. Burges, L. Bottou, and K. Q. Weinberger, Eds., 2012, pp. 1097–1105. [Online]. Available: http://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf

[16] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *CoRR*, vol. abs/1409.1556, 2014. [Online]. Available: http://arxiv.org/abs/1409.1556

[17] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. E. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," *CoRR*, vol. abs/1409.4842, 2014. [Online]. Available: http://arxiv.org/abs/1409.4842

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *CoRR*, vol. abs/1512.03385, 2015. [Online]. Available: http://arxiv.org/abs/1512.03385

参考文献

[1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, “脑机接口”
通信和控制接口。”临床神经生理学：国际临床神经生理学联合会官方期刊，第 113 卷，第 6 期，第 767–91 页，2002 年 6 月。[在线]. 可获得：http://www.ncbi.nlm.nih.gov/pubmed/12048038 [2] A. B. Schwartz, X. T. Cui, D. Weber, 和 D. W. Moran, “脑控接口：神经假体恢复运动,” Neuron, 第 52 卷，第 1 期，第 205 – 220 页，2006 年。

[3] J. van Erp, F. Lotte, 和 M. Tangermann, “脑机接口：超越医疗应用,” Computer, 卷. 45, 第. 4, 页. 26–34, 2012 年 4 月.

[4] S. Saproo, J. Faller, V. Shih, P. Sajda, N. R. Waytowich, A. Bohannon, V. J. Lawhern, B. J. Lance, and D. Jangraw, “皮层耦合计算：一种协同人机交互的新范式,”

计算机，第 49 卷，第 9 期，第 60-68 页，2016 年 9 月。

[5] B. J. Lance, S. E. Kerick, A. J. Ries, K. S. Oie, 和 K. McDowell, “未来十年的脑机接口技术,” IEEE 会刊, 卷. 100, 特刊: 百年纪念, pp. 1585–1599, 2012 年 5 月. [6] L. F. Nicolas-Alonso 和 J. Gomez-Gil, “脑机接口综述,” 传感器, 卷. 12, 第 2 期, p. 1211,

2012.

[7] A. Bashashati, M. Fatourechi, R. K. Ward, 和 G. E. Birch, “基于脑电信号的脑机接口信号处理算法综述,” 神经工程杂志, 卷. 4, 第 2 期, p.
R32, 2007. [在线]. 可获得: http://stacks.iop.org/1741-2552/4/i=2/a=R03 [8] D. J. McFarland, C. W. Anderson, K. R. Muller, A. Schlogl, 和 D. J. Krusienski, “BCI 会议 2005-脑机接口信号处理研讨会: 特征提取和转换,” IEEE 神经系统和康复工程汇刊, 卷. 14, 第 2 期, pp. 135–138, 2006 年 6 月.

[9] F. Lotte, M. Congedo, A. L´ecuyer, F. Lamarche, 和 B. Arnaldi, “基于 EEG 的脑机接口分类算法综述,” 神经工程杂志, 卷. 4, 第 2 期, p. R1, 2007. [在线].

可获取：http://stacks.iop.org/1741-2552/4/i=2/a=R01 [10] T. O. Zander 和 C. Kothe，“迈向无源脑机接口：应用脑机接口技术在人机系统中的普遍应用，”《神经工程杂志》，第 8 卷，第 2 期，第 025005 页，2011 年。[在线]. 可获得：http://stacks.iop.org/1741-2552/8/i=2/a=025005 [11] B. Blankertz, M. Tangermann, C. Vidaurre, S. Fazli, C. Sannelli, S. Haufe, C. Maeder, L. E. Ramsey, I. Sturm, G. Curio, 和 K. R. M¨ueller, “柏林脑机接口：脑机接口技术的非医疗应用,”

神经科学前沿，第 4 卷，第 198 期，2010 年。

[12] S. M. Gordon, M. Jaswa, A. J. Solon, 和 V. J. Lawhern, “现实世界脑机接口：跨领域学习与
实际应用，”在 2017 年 ACM 关于 BCI 实际应用方法的研讨会论文集中，系列 BCIforReal’17。纽约，纽约，美国：ACM，2017 年，第 25-28 页。[在线]。

获取：http://doi.acm.org/10.1145/3038439.3038444 [13] G. Hinton, L. Deng, D. Yu, G. E. Dahl, A. r. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T. N.

Sainath,以及 B. Kingsbury，“用于语音识别中声学建模的深度神经网络：四个研究小组的共同观点，”IEEE 信号处理杂志，第 29 卷，第 6 期，第 82-97 页，2012 年 11 月。

[14] Y. LeCun, Y. Bengio,以及 G. Hinton，“深度学习，”自然，第 521 卷，第 436-444 页，2015 年。

[15] A. Krizhevsky, I. Sutskever, 和 G. E. Hinton, “使用深度卷积神经网络进行 ImageNet 分类,” 在《神经信息处理系统进展》中, F. Pereira, C. J. C.
Burges, L. Bottou, 和 K. Q. Weinberger, 编, 2012, 第 1097-1105 页. [在线]. 可获得: http: //papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf [16] K. Simonyan 和 A. Zisserman, “用于大规模图像识别的非常深的卷积网络,” CoRR,
vol. abs/1409.1556, 2014. [在线]. 可获得: http://arxiv.org/abs/1409.1556 [17] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. E. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and
A. 拉宾诺维奇，“通过卷积进一步深入”，arXiv，编号 abs/1409.4842，2014。[在线]. 可获得：http://arxiv.org/abs/1409.4842 [18] K. He, X. Zhang, S. Ren, 和 J. Sun，“用于图像识别的深度残差学习”，arXiv，编号

abs/1512.03385, 2015. [在线]. 可获得: http://arxiv.org/abs/1512.03385

[19] G. Huang, Z. Liu, K. Q. Weinberger, and L. van der Maaten, “Densely connected convolutional networks,” *CoRR*, vol. abs/1608.06993, 2016. [Online]. Available: http://arxiv.org/abs/1608.06993

[20] J. Schmidhuber, “Deep learning in neural networks: An overview,” *arXiv*, vol. abs/1404.7828, 2014. [Online]. Available: http://arxiv.org/abs/1404.7828

[21] A. Antoniades, L. Spyrou, C. C. Took, and S. Sanei, “Deep learning for epileptic intracranial eeg data,” in *2016 IEEE 26th International Workshop on Machine Learning for Signal Processing (MLSP)*, Sept 2016, pp. 1–6.

[22] J. Liang, R. Lu, C. Zhang, and F. Wang, “Predicting seizures from electroencephalography recordings: A knowledge transfer strategy,” in *2016 IEEE International Conference on Healthcare Informatics (ICHI)*, Oct 2016, pp. 184–191.

[23] A. Page, C. Shea, and T. Mohsenin, “Wearable seizure detection using convolutional neural networks with transfer learning,” in *2016 IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2016, pp. 1086–1089.

[24] P. Mirowski, D. Madhavan, Y. LeCun, and R. Kuzniecky, “Classification of patterns of {EEG} synchronization for seizure prediction,” *Clinical Neurophysiology*, vol. 120, no. 11, pp. 1927 – 1940, 2009.

[25] P. Thodoroff, J. Pineau, and A. Lim, “Learning robust features using deep learning for automatic seizure detection,” *CoRR*, vol. abs/1608.00220, 2016. [Online]. Available: http://arxiv.org/abs/1608.00220

[26] S. Stober, D. J. Cameron, and J. A. Grahn, “Using convolutional neural networks to recognize rhythm stimuli from electroencephalography recordings,” in *Advances in Neural Information Processing Systems 27*, Z. Ghahramani, M. Welling, C. Cortes, N. D. Lawrence, and K. Q. Weinberger, Eds. Curran Associates, Inc., 2014, pp. 1449–1457.

[27] S. Stober, A. Sternin, A. M. Owen, and J. A. Grahn, “Deep feature learning for EEG recordings,” *CoRR*, vol. abs/1511.04306, 2015. [Online]. Available: http://arxiv.org/abs/1511.04306

[28] H. Cecotti and A. Graser, “Convolutional neural networks for p300 detection with application to brain-computer interfaces,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 3, pp. 433–445, March 2011.

[29] R. Manor and A. Geva, “Convolutional neural network for multi-category rapid serial visual presentation bci,” *Frontiers in Computational Neuroscience*, vol. 9, no. 146, 2015.

[30] J. Shamwell, H. Lee, H. Kwon, A. R. Marathe, V. Lawhern, and W. Nothwang, “Single-trial eeg rsvp classification using convolutional neural networks,” pp. 983 622–983 622–10, 2016. [Online]. Available: http://dx.doi.org/10.1117/12.2224172

[31] H. Cecotti, M. P. Eckstein, and B. Giesbrecht, “Single-trial classification of event-related potentials in rapid serial visual presentation tasks using supervised spatial filtering,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 25, no. 11, pp. 2030–2042, Nov 2014.

[32] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, “Deep learning with convolutional neural networks for eeg decoding and visualization,” *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017. [Online]. Available: http://dx.doi.org/10.1002/hbm.23730

[33] M. Längkvist, L. Karlsson, and A. Loutfi, “Sleep stage classification using unsupervised feature learning,” *Adv. Artif. Neu. Sys.*, vol. 2012, pp. 5:5–5:5, Jan. 2012. [Online]. Available: http://dx.doi.org/10.1155/2012/107046

[34] D. F. Wulsin, J. R. Gupta, R. Mani, J. A. Blanco, and B. Litt, “Modeling electroencephalography waveforms with semi-supervised deep belief nets: fast classification and anomaly measurement,” *Journal of Neural Engineering*, vol. 8, no. 3, p. 036015, 2011.

[35] T. Ma, H. Li, H. Yang, X. Lv, P. Li, T. Liu, D. Yao, and P. Xu, “The extraction of motion-onset vep bci features based on deep learning and compressed sensing,” *Journal of Neuroscience Methods*, vol. 275, pp. 80 – 92, 2017. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0165027016302680

[36] P. Bashivan, I. Rish, M. Yeasin, and N. Codella, “Learning representations from EEG with deep recurrent-convolutional neural networks,” *CoRR*, vol. abs/1511.06448, 2015. [Online]. Available: http://arxiv.org/abs/1511.06448

[37] Y. R. Tabar and U. Halici, “A novel deep learning approach for classification of eeg motor imagery signals,” *Journal of Neural Engineering*, vol. 14, no. 1, p. 016003, 2017. [Online]. Available: http://stacks.iop.org/1741-2552/14/i=1/a=016003

2015 [19] 黄葛，刘铮，魏克强，和范德马滕，“密集连接卷积网络”，CoRR，vol. abs/1608.06993，2016。[在线]。可访问：http://arxiv.org/abs/1608.06993 [20] 贾斯明・施密德胡伯，“神经网络的深度学习：概述”，arXiv，vol. abs/1404.7828，2014。[在线]。

可用：http://arxiv.org/abs/1404.7828 [21] A. Antoniades, L. Spyrou, C. C. Took, 和 S. Sanei, “深度学习用于癫痫性颅内脑电图数据,” 在 2016 IEEE 第 26 届机器学习信号处理研讨会 (MLSP), 2016 年 9 月, 第 1–6 页。

[22] 李建，陆荣，张超，王峰，“从脑电图记录中预测癫痫发作：”

知识转移策略，”在 2016 年 IEEE 国际医疗信息学会议（ICHI）中，2016 年 10 月，第 184-191 页。

[23] A. Page, C. Shea, 和 T. Mohsenin, “使用迁移学习的卷积神经网络进行可穿戴癫痫检测,” 在 2016 年 IEEE 国际电路与系统会议 (ISCAS), 2016 年 5 月, pp.
1086–1089.

[24] P. Mirowski, D. Madhavan, Y. LeCun, 和 R. Kuzniecky, “用于癫痫预测的 {EEG} 同步模式分类,” 临床神经生理学, 第 120 卷, 第 11 期, pp. 1927 – 1940, 2009.

[25] P. Thodoroff, J. Pineau, 和 A. Lim, “使用深度学习学习鲁棒特征以实现自动癫痫

“检测”，CoRR，卷 abs/1608.00220，2016。[在线]. 可获得：http://arxiv.org/abs/1608.00220 [26] S. Stober, D. J. Cameron 和 J. A. Grahn, “使用卷积神经网络识别节奏刺激

来自脑电图记录，”载于《神经信息处理系统进展》第 27 卷，Z. Ghahramani、M. Welling、C. Cortes、N. D. Lawrence 和 K. Q. Weinberger 编，Curran Associates, Inc.，2014 年，第 pp.
1449–1457.

[27] S. Stober, A. Sternin, A. M. Owen, 和 J. A. Grahn, “用于脑电图记录的深度特征学习,” CoRR, vol.

abs/1511.04306, 2015. [在线]. 可获得: http://arxiv.org/abs/1511.04306 [28] H. Cecotti 和 A. Graser, “卷积神经网络用于 p300 检测及其在脑机接口中的应用”

接口，” IEEE 模式分析与机器智能汇刊，第 33 卷，第 3 期，第 433–445 页，2011 年 3 月。

[29] R. Manor 和 A. Geva，“用于多类别快速序列视觉呈现脑机接口的卷积神经网络，”
计算神经科学前沿，第 9 卷，第 146 期，2015 年。

[30] J. Shamwell, H. Lee, H. Kwon, A. R. Marathe, V. Lawhern, 和 W. Nothwang, “单次试验脑电图

使用卷积神经网络的 rsvp 分类，”第 983 622–983 622–10 页，2016 年。[在线]。可访问：http://dx.doi.org/10.1117/12.2224172 [31] H. Cecotti, M. P. Eckstein 和 B. Giesbrecht，“单次试验的事件相关电位分类在快速

使用监督空间滤波的序列视觉呈现任务，”IEEE 神经网络与学习系统汇刊，第 25 卷，第 11 期，第 2030-2042 页，2014 年 11 月。

[32] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann,

F. Hutter, W. Burgard 和 T. Ball, “使用卷积神经网络进行脑电图解码和可视化深度学习,” Human Brain Mapping, 卷. 38, 第 11 期, pp. 5391–5420, 2017. [在线]. 可获得: http://dx.doi.org/10.1002/hbm.23730 [33] M. L¨angkvist, L. Karlsson 和 A. Loutfi, “使用无监督特征学习进行睡眠阶段分类,” Adv.

人工神经网络系统，第 2012 卷，第 5:5–5:5 页，2012 年 1 月。[在线]. 可获得：http://dx.doi.org/10.1155/2012/107046 [34] D. F. Wulsin, J. R. Gupta, R. Mani, J. A. Blanco, 和 B. Litt, “脑电图波形建模”

使用半监督深度信念网络的表单：快速分类和异常测量，《神经工程杂志》，第 8 卷，第 3 期，第 036015 页，2011 年。

[35] T. Ma, H. Li, H. Yang, X. Lv, P. Li, T. Liu, D. Yao, 和 P. Xu, “运动起始 VEP 脑机接口的提取”

基于深度学习和压缩感知的特征，”《神经科学方法杂志》，第 275 卷，第 80-92 页，2017 年。[在线]。可获取：http://www.sciencedirect.com/science/article/pii/S0165027016302680 [36] P. Bashivan, I. Rish, M. Yeasin, 和 N. Codella, “从脑电图学习表示，”

深度递归卷积神经网络，” CoRR, 卷. abs/1511.06448, 2015. [在线]. 可获得: http://arxiv.org/abs/1511.06448

[37] Y. R. Tabar 和 U. Halici，“一种用于脑电图运动分类的新型深度学习方法”
"图像信号"，《神经工程杂志》，第 14 卷，第 1 期，第 016003 页，2017 年。[在线]。可获取：http://stacks.iop.org/1741-2552/14/i=1/a=016003

[38] X. An, D. Kuang, X. Guo, Y. Zhao, and L. He, *A Deep Learning Method for Classification of EEG Data Based on Motor Imagery.* Cham: Springer International Publishing, 2014, pp. 203–210. [Online]. Available: http://dx.doi.org/10.1007/978-3-319-09330-7_25

[39] S. Sakhavi, C. Guan, and S. Yan, "Parallel convolutional-linear neural network for motor imagery classification," in *2015 23rd European Signal Processing Conference (EUSIPCO)*, Aug 2015, pp. 2736–2740.

[40] N. Lu, T. Li, X. Ren, and H. Miao, "A deep learning scheme for motor imagery classification based on restricted boltzmann machines," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 6, pp. 566–576, June 2017.

[41] Z. Yin and J. Zhang, "Cross-session classification of mental workload levels using eeg and an adaptive deep learning model," *Biomedical Signal Processing and Control*, vol. 33, pp. 30 – 47, 2017. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S1746809416302014

[42] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," *CoRR*, vol. abs/1610.02357, 2016. [Online]. Available: http://arxiv.org/abs/1610.02357

[43] Z. Yang, M. Moczulski, M. Denil, N. d. Freitas, A. Smola, L. Song, and Z. Wang, "Deep fried convnets," in *2015 IEEE International Conference on Computer Vision (ICCV)*, Dec 2015, pp. 1476–1483.

[44] F. Lotte, "Signal Processing Approaches to Minimize or Suppress Calibration Time in Oscillatory Activity-Based Brain-Computer Interfaces," *Proceedings of the IEEE*, vol. 103, no. 6, pp. 871–890, Jun. 2015.

[45] R. Fazel-Rezai, B. Z. Allison, C. Guger, E. W. Sellers, S. C. Kleih, and A. Kübler, "P300 brain computer interface: current challenges and emerging trends," *Frontiers in Neuroengineering*, vol. 5, no. 14, 2012.

[46] G. Pfurtscheller and C. Neuper, "Motor imagery and direct brain-computer communication," *Proceedings of the IEEE*, vol. 89, no. 7, pp. 1123–1134, Jul 2001.

[47] S. Makeig, "Auditory event-related dynamics of the {EEG} spectrum and effects of exposure to tones," *Electroencephalography and Clinical Neurophysiology*, vol. 86, no. 4, pp. 283 – 293, 1993.

[48] J. Polich, "Updating p300: An integrative theory of {P3a} and {P3b}," *Clinical Neurophysiology*, vol. 118, no. 10, pp. 2128 – 2148, 2007.

[49] P. Sajda, E. Pohlmeyer, J. Wang, L. C. Parra, C. Christoforou, J. Dmochowski, B. Hanna, C. Bahlmann, M. K. Singh, and S. F. Chang, "In a blink of an eye and a switch of a transistor: Cortically coupled computer vision," *Proceedings of the IEEE*, vol. 98, no. 3, pp. 462–478, March 2010.

[50] A. R. Marathe, V. J. Lawhern, D. Wu, D. Slayback, and B. J. Lance, "Improved neural signal classification in a rapid serial visual presentation task using active learning," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 24, no. 3, pp. 333–343, March 2016.

[51] N. Waytowich, V. Lawhern, A. Bohannon, K. Ball, and B. Lance, "Spectral transfer learning using information geometry for a user-independent brain-computer interface," *Frontiers in Neuroscience*, vol. 10, p. 430, 2016. [Online]. Available: http://journal.frontiersin.org/article/10.3389/fnins.2016.00430

[52] A. Delorme and S. Makeig, "Eeglab: an open source toolbox for analysis of single-trial {EEG} dynamics including independent component analysis," *Journal of Neuroscience Methods*, vol. 134, no. 1, pp. 9 – 21, 2004.

[53] W. H. R. Miltner, C. H. Braun, and M. G. H. Coles, "Event-related brain potentials following incorrect feedback in a time-estimation task: Evidence for a "generic" neural system for error detection," *Journal of Cognitive Neuroscience*, vol. 9, no. 6, pp. 788–798, 1997.

[54] W. J. Gehring, B. Goss, M. G. H. Coles, D. E. Meyer, and E. Donchin, "A neural system for error detection and compensation," *Psychological Science*, vol. 4, no. 6, pp. 385–390, 1993. [Online]. Available: http://pss.sagepub.com/content/4/6/385.abstract

[55] M. Falkenstein, J. Hohnsbein, J. Hoormann, and L. Blanke, "Effects of crossmodal divided attention on late {ERP} components. ii. error processing in choice reaction tasks," *Electroencephalography and Clinical Neurophysiology*, vol. 78, no. 6, pp. 447 – 455, 1991.

[56] P. Margaux, M. Emmanuel, D. Sébastien, B. Olivier, and M. Jérémie, "Objective and subjective evaluation of online error correction during p300-based spelling," *Advances in Human-Computer Interaction*, vol. 2012, p. 4, 2012.

[57] T. O. Zander, C. Kothe, S. Welke, and M. Roetting, *Utilizing Secondary Input from Passive Brain-Computer Interfaces for Enhancing Human-Machine Interaction.* Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 759–771.

[38] X. An, D. Kuang, X. Guo, Y. Zhao, and L. He, 一种用于脑电图数据分类的深度学习方法
基于运动意象。Cham: Springer International Publishing, 2014, 第 203-210 页。[在线]. 可获得:
http://dx.doi.org/10.1007/978-3-319-09330-7 25 [39] S. Sakhavi, C. Guan, 和 S. Yan, "用于运动意象分类的并行卷积-线性神经网络,"

2015 年 23 届欧洲信号处理会议（EUSIPCO），2015 年 8 月，第 2736–2740 页。

[40] N. Lu, T. Li, X. Ren, and H. Miao, "基于受限的深度学习方案用于运动想象分类"
玻尔兹曼机，"IEEE 神经系统和康复工程汇刊，第 25 卷，第 6 期，第 566-576 页，2017 年 6 月。

[41] Z. Yin 和 J. Zhang，"使用 EEG 和自适应深度
学习模型，"生物医学信号处理与控制，第 33 卷，第 30-47 页，2017 年。[在线]. 可获得：
http://www.sciencedirect.com/science/article/pii/S1746809416302014 [42] F. Chollet, "Xception: 深度可分离卷积的深度学习，" CoRR, 第 abs/1610.02357 卷。

2016. [在线]. 获取方式: http://arxiv.org/abs/1610.02357 [43] Z. Yang, M. Moczulski, M. Denil, N. d. Freitas, A. Smola, L. Song, 和 Z. Wang, "Deep fried convnets," 在 2015 IEEE 国际计算机视觉会议 (ICCV), 2015 年 12 月, 第 1476–1483 页.

[44] F. Lotte, "基于脑机接口的振荡活动的信号处理方法以最小化或抑制校准时间," IEEE 会议录, 卷. 103, 第. 6, pp. 871–890, 2015 年 6 月.

[45] R. Fazel-Rezai, B. Z. Allison, C. Guger, E. W. Sellers, S. C. Kleih, 和 A. K¨ubler, "P300 脑机接口：当前挑战和新兴趋势," 神经工程前沿, 卷. 5, 第. 14, 2012 年.

[46] G. Pfurtscheller 和 C. Neuper, "运动想象和直接的脑机通信," IEEE 会议录, 卷. 89, 第. 7, pp. 1123–1134, 2001 年 7 月. [47] S. Makeig, "脑电图频谱的听觉事件相关动力学及暴露于音调的影响," 脑电图与临床神经生理学, 卷. 86, 第. 4, pp. 283 – 293, 1993 年.

[48] J. Polich, "更新 P300：P3a 和 P3b 的整合理论," 临床神经生理学, 卷. 118, 第. 10, pp. 2128 – 2148, 2007 年.

[49] P. Sajda, E. Pohlmeyer, J. Wang, L. C. Parra, C. Christoforou, J. Dmochowski, B. Hanna, C. Bahlmann, M. K.

Singh, 和 S. F. Chang, "在眨眼之间和晶体管切换之间：皮层耦合的计算机视觉,"
IEEE 会刊，第 98 卷，第 3 期，第 462–478 页，2010 年 3 月。

[50] A. R. Marathe, V. J. Lawhern, D. Wu, D. Slayback, 和 B. J. Lance, "改进的神经信号分类
在一个使用主动学习的快速序列视觉呈现任务中，"IEEE 神经系统和康复工程汇刊，第 24 卷，第 3 期，第 333-343 页，2016 年 3 月。

[51] N. Waytowich, V. Lawhern, A. Bohannon, K. Ball, 和 B. Lance，"使用信息几何的频谱迁移学习，用于用户无关的脑机接口，" 神经科学前沿，第 10 卷，第 430 页，2016 年。

[在线]. 可获得：http://journal.frontiersin.org/article/10.3389/fnins.2016.00430 [52] A. Delorme 和 S. Makeig，"Eeglab：一个用于分析单次试验脑电图动力学的开源工具箱，包括独立成分分析，"神经科学方法杂志，第 134 卷，第 1 期，第 9-21 页，2004 年。

[53] W. H. R. Miltner, C. H. Braun, 和 M. G. H. Coles，"错误反馈后的事件相关脑电位"

在时间估计任务中："关于错误检测的'通用'神经系统的证据"，《认知神经科学杂志》，第 9 卷，第 6 期，第 788-798 页，1997 年。

[54] W. J. Gehring, B. Goss, M. G. H. Coles, D. E. Meyer, 和 E. Donchin, "一个用于错误
检测与补偿，" 心理科学，第 4 卷，第 6 期，第 385–390 页，1993 年。[在线]. 可获得：
http://pss.sagepub.com/content/4/6/385.abstract

[55] M. Falkenstein, J. Hohnsbein, J. Hoormann, 和 L. Blanke, "跨模态分配注意力对晚期..."
{ERP}组件。ii.选择反应任务中的错误处理，"脑电图与临床神经生理学，卷 78，第 6 期，第 447-455 页，1991 年。

[56] P. Margaux, M. Emmanuel, D. S´ebastien, B. Olivier,和 M. J´er´emie，"客观和主观评价
基于 p300 的拼写在线纠错，"人机交互进展，卷 2012，第 4 页，2012 年。

[57] T. O. Zander, C. Kothe, S. Welke, 和 M. Roetting, 利用被动脑机接口的二次输入
增强人机交互的接口。柏林，海德堡：Springer Berlin Heidelberg，2009 年，第 759–771 页。

[58] J. d. R. Millán, R. Rupp, G. Mueller-Putz, R. Murray-Smith, C. Giugliemma, M. Tangermann, C. Vidaurre, F. Cincotti, A. Kubler, R. Leeb, C. Neuper, K. R. Müeller, and D. Mattia, "Combining brain-computer interfaces and assistive technologies: State-of-the-art and challenges," *Frontiers in Neuroscience*, vol. 4, no. 161, 2010.

[59] M. Spüler, M. Bensch, S. Kleih, W. Rosenstiel, M. Bogdan, and A. Kübler, "Online use of error-related potentials in healthy users and people with severe motor impairment increases performance of a p300-bci," *Clinical Neurophysiology*, vol. 123, no. 7, pp. 1328 – 1337, 2012.

[60] D. J. Krusienski, E. W. Sellers, D. J. McFarland, T. M. Vaughan, and J. R. Wolpaw, "Toward enhanced p300 speller performance," *Journal of neuroscience methods*, vol. 167, no. 1, pp. 15–21, 2008.

[61] C. Toro, G. Deuschl, R. Thatcher, S. Sato, C. Kufta, and M. Hallett, "Event-related desynchronization and movement-related cortical potentials on the {ECoG} and {EEG}," *Electroencephalography and Clinical Neurophysiology/Evoked Potentials Section*, vol. 93, no. 5, pp. 380 – 389, 1994.

[62] G. Pfurtscheller and A. Aranibar, "Event-related cortical desynchronization detected by power measurements of scalp {EEG}," *Electroencephalography and Clinical Neurophysiology*, vol. 42, no. 6, pp. 817 – 826, 1977.

[63] G. Pfurtscheller and F. L. da Silva, "Event-related eeg/meg synchronization and desynchronization: basic principles," *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842 – 1857, 1999.

[64] K. Liao, R. Xiao, J. Gonzalez, and L. Ding, "Decoding individual finger movements from one hand using human eeg signals," *PLoS ONE*, vol. 9, no. 1, pp. 1–12, 01 2014.

[65] G. Barrett, H. Shibasaki, and R. Neshige, "Cortical potentials preceding voluntary movement: Evidence for three periods of preparation in man," *Electroencephalography and Clinical Neurophysiology*, vol. 63, no. 4, pp. 327 – 339, 1986. [Online]. Available: http://www.sciencedirect.com/science/article/pii/0013469486900179

[66] O. Yilmaz, N. Birbaumer, and A. Ramos-Murguialday, "Movement related slow cortical potentials in severely paralyzed chronic stroke patients," *Frontiers in Human Neuroscience*, vol. 8, p. 1033, 2015. [Online]. Available: https://www.frontiersin.org/article/10.3389/fnhum.2014.01033

[67] L. Deecke, P. Scheid, and H. H. Kornhuber, "Distribution of readiness potential, pre-motion positivity, and motor potential of the human cerebral cortex preceding voluntary finger movements," *Experimental Brain Research*, vol. 7, no. 2, pp. 158–168, Jan 1969. [Online]. Available: https://doi.org/10.1007/BF00235441

[68] E. C. Leuthardt, G. Schalk, D. Moran, and J. G. Ojemann, "The emerging world of motor neuroprosthetics: a neurosurgical perspective," *Neurosurgery*, vol. 59, no. 1, pp. 1–14, 2006.

[69] E. Yom-Tov and G. F. Inbar, "Detection of movement-related potentials from the electro-encephalogram for possible use in a brain-computer interface," *Medical and Biological Engineering and Computing*, vol. 41, no. 1, pp. 85–93, Jan 2003. [Online]. Available: https://doi.org/10.1007/BF02343543

[70] F. Karimi, J. Kofman, N. Mrachacz-Kersting, D. Farina, and N. Jiang, "Detection of movement related cortical potentials from eeg using constrained ica for brain-computer interface applications," *Frontiers in Neuroscience*, vol. 11, p. 356, 2017. [Online]. Available: https://www.frontiersin.org/article/10.3389/fnins.2017.00356

[71] S. Gordon, V. Lawhern, A. Passaro, and K. McDowell, "Informed decomposition of electroencephalographic data," *Journal of Neuroscience Methods*, vol. 256, pp. 41 – 55, 2015.

[72] N. Bigdely-Shamlo, T. Mullen, C. Kothe, K. M. Su, and K. A. Robbins, "The prep pipeline: Standardized preprocessing for large-scale eeg analysis," *Frontiers in Neuroinformatics*, vol. 9, no. 16, 2015.

[73] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. Miller, G. Mueller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, and B. Blankertz, "Review of the bci competition iv," *Frontiers in Neuroscience*, vol. 6, p. 55, 2012. [Online]. Available: http://journal.frontiersin.org/article/10.3389/fnins.2012.00055

[74] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiV*, vol. abs/1412.6980, 2014. [Online]. Available: http://arxiv.org/abs/1412.6980

[75] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, M. Kudlur, J. Levenberg, R. Monga, S. Moore, D. G. Murray, B. Steiner, P. Tucker, V. Vasudevan, P. Warden, M. Wicke, Y. Yu, and X. Zheng, "Tensorflow: A system for large-scale machine learning," in *Proceedings of the 12th USENIX Conference on Operating Systems Design and Implementation*, ser. OSDI'16. Berkeley, CA, USA: USENIX Association, 2016, pp. 265–283. [Online]. Available: http://dl.acm.org/citation.cfm?id=3026877.3026899

[58] J. d. R. Mill´an, R. Rupp, G. Mueller-Putz, R. Murray-Smith, C. Giugliemma, M. Tangermann, C. Vidaurre, F. Cincotti, A. Kubler, R. Leeb, C. Neuper, K. R. M¨ueller, 和 D. Mattia, “结合脑机接口和辅助技术：现状与挑战,” 前沿神经科学, 第 4 卷, 第 161, 2010.

[59] M. Sp¨uler, M. Bensch, S. Kleih, W. Rosenstiel, M. Bogdan, 和 A. K¨ubler, “在线使用错误相关电位在健康用户和严重运动障碍者中提高 p300-bci 的性能,” 临床神经生理学，第 123 卷，第 7 期，第 1328-1337 页，2012 年。

[60] D. J. Krusienski, E. W. Sellers, D. J. McFarland, T. M. Vaughan, 和 J. R. Wolpaw, “迈向增强的 p300 拼写器性能,” Journal of neuroscience methods, 卷. 167, 期. 1, 页. 15–21, 2008.

[61] C. Toro, G. Deuschl, R. Thatcher, S. Sato, C. Kufta,和 M. Hallett，“事件相关去同步和与运动相关的皮质电位在{ECoG}和{EEG}上，”脑电图和临床神经生理学/诱发电位分册，第 93 卷，第 5 期，第 380-389 页，1994 年。

[62] G. Pfurtscheller 和 A. Aranibar, “通过头皮 {EEG} 功率测量检测的事件相关皮层去同步,” 《脑电图学及临床神经生理学》, 第 42 卷, 第 6 期, pp. 817 – 826, 1977.

[63] G. Pfurtscheller 和 F. L. da Silva, “事件相关 EEG/MEG 同步与去同步: 基本原理,” 《临床神经生理学》, 第 110 卷, 第 11 期, pp. 1842 – 1857, 1999.

[64] K. Liao, R. Xiao, J. Gonzalez 和 L. Ding, “使用人体 EEG 信号解码单手手指运动,” 《PLoS ONE》, 第 9 卷, 第 1 期, pp. 1–12, 2014 年 1 月.

[65] G. Barrett, H. Shibasaki 和 R. Neshige, “自愿运动前的皮层电位: 人类准备过程中的三个时期证据,” 《脑电图学及临床神经生理学》, 第 63 卷, 第 4 期, pp. 327 – 339, 1986. [在线]. 获取方式：http://www.sciencedirect.com/science/article/pii/0013469486900179 [66] O. Yilmaz, N. Birbaumer, 和 A. Ramos-Murguialday, “与运动相关的慢皮质电位在严重“瘫痪性慢性中风患者，”《人类神经科学前沿》，第 8 卷，第 1033 页，2015 年。[在线]. 可获得：https://www.frontiersin.org/article/10.3389/fnhum.2014.01033 [67] L. Deecke, P. Scheid, 和 H. H. Kornhuber, “准备电位、运动前正电位的分布，”人脑皮层运动潜能在自主手指运动之前的实验脑研究，第 7 卷，第 2 期，第 158-168 页，1969 年 1 月。[在线]. 可获得：https://doi.org/10.1007/BF00235441 [68] E. C. Leuthardt, G. Schalk, D. Moran, 和 J. G. Ojemann, “运动神经假肢的新世界：神经外科视角,” 神经外科手术，第 59 卷，第 1 期，第 1-14 页，2006 年。

[69] E. Yom-Tov 和 G. F. Inbar，“从脑电图检测运动相关电位”可能在脑机接口中的应用，” 医疗与生物工程计算，第 41 卷，第 1 期，第 85-93 页，2003 年 1 月。[在线]. 可获得：https://doi.org/10.1007/BF02343543 [70] F. Karimi, J. Kofman, N. Mrachacz-Kersting, D. Farina, 和 N. Jiang, “运动相关皮质活动的检测”使用约束 ICA 从脑电图提取潜力用于脑机接口应用，”神经科学前沿，第 11 卷，第 356 页，2017 年。[在线]. 可获得: https://www.frontiersin.org/article/10.3389/fnins.2017.00356 [71] S. Gordon, V. Lawhern, A. Passaro,和 K. McDowell, “脑电图数据的知情分解,”神经科学方法杂志，第 256 卷，第 41-55 页，2015 年。

[72] N. Bigdely-Shamlo, T. Mullen, C. Kothe, K. M. Su, 和 K. A. Robbins, “The prep pipeline: 标准化预处理用于大规模脑电图分析,” Frontiers in Neuroinformatics, 卷 9, 第 16 期, 2015.

[73] M. 坦格曼, K.-R. 米勒, A. 艾尔森, N. 比尔鲍默, C. 布劳恩, C. 布鲁纳, R. 利布, C. 梅林 K. Miller, G. Mueller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, 和 B. Blankertz, “bci 竞赛 iv 的综述,” 前沿科学杂志, 第 6 卷, 第 55 页, 2012 年. [在线]. 可获得: http://journal.frontiersin.org/article/10.3389/fnins.2012.00055 [74] D. P. Kingma 和 J. Ba, “Adam: 一种随机优化方法,” arXiV, 第 abs/1412.6980 条目, 2014 年.
[在线]. 可获得: http://arxiv.org/abs/1412.6980 [75] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving,

M. Isard, M. Kudlur, J. Levenberg, R. Monga, S. Moore, D. G. Murray, B. Steiner, P. Tucker, V. Vasudevan, P. Warden, M. Wicke, Y. Yu, 和 X. Zheng, “Tensorflow: 一种大规模机器学习系统,” 在第 12 届 USENIX 操作系统设计与实现会议论文集中, OSDI’16 系列. 加利福尼亚州伯克利: USENIX 协会, 2016 年, 第 265-283 页. [在线]. 可获得: http://dl.acm.org/citation.cfm?id=3026877.3026899

[76] F. Chollet, "Keras," https://github.com/fchollet/keras, 2015.

[77] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter bank common spatial pattern algorithm on bci competition iv datasets 2a and 2b," *Frontiers in Neuroscience*, vol. 6, p. 39, 2012. [Online]. Available: http://journal.frontiersin.org/article/10.3389/fnins.2012.00039

[78] M. Dyrholm, C. Christoforou, and L. C. Parra, "Bilinear discriminant component analysis," *Journal of Machine Learning Research*, vol. 8, no. May, pp. 1097–1111, 2007.

[79] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," *arXiv*, vol. abs/1502.03167, 2015. [Online]. Available: http://arxiv.org/abs/1502.03167

[80] D. Clevert, T. Unterthiner, and S. Hochreiter, "Fast and accurate deep network learning by exponential linear units (elus)," *CoRR*, vol. abs/1511.07289, 2015. [Online]. Available: http://arxiv.org/abs/1511.07289

[81] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014. [Online]. Available: http://jmlr.org/papers/v15/srivastava14a.html

[82] J. T. Springenberg, A. Dosovitskiy, T. Brox, and M. A. Riedmiller, "Striving for simplicity: The all convolutional net," *arXiV*, vol. abs/1412.6806, 2014. [Online]. Available: http://arxiv.org/abs/1412.6806

[83] B. Rivet, A. Souloumiac, V. Attina, and G. Gibert, "xDAWN algorithm to enhance evoked potentials: Application to brain-computer interface," *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 8, pp. 2035–2043, Aug 2009.

[84] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Multiclass Brain-Computer Interface Classification by Riemannian Geometry," *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 4, pp. 920–928, Apr. 2012.

[85] A. Barachant and M. Congedo, "A Plug&Play P300 BCI Using Information Geometry," *arXiv:1409.0107 [cs, stat]*, Aug. 2014, arXiv: 1409.0107. [Online]. Available: http://arxiv.org/abs/1409.0107

[86] M. Congedo, A. Barachant, and A. Andreev, "A new generation of brain-computer interface based on riemannian geometry," *CoRR*, vol. abs/1310.8115, 2013. [Online]. Available: http://arxiv.org/abs/1310.8115

[87] A. Barachant and S. Bonnet, "Channel selection procedure using riemannian distance for bci applications," in *2011 5th International IEEE/EMBS Conference on Neural Engineering*, April 2011, pp. 348–351.

[88] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Classification of covariance matrices using a Riemannian-based kernel for BCI applications," *Neurocomputing*, vol. 112, pp. 172–178, Jul. 2013.

[89] A. Y. Ng, "Feature selection, l1 vs. l2 regularization, and rotational invariance," in *Proceedings of the Twenty-first International Conference on Machine Learning*, ser. ICML '04. New York, NY, USA: ACM, 2004, pp. 78–. [Online]. Available: http://doi.acm.org/10.1145/1015330.1015435

[90] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365 – 411, 2004.

[91] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 67, no. 2, pp. 301–320, 2005. [Online]. Available: http://dx.doi.org/10.1111/j.1467-9868.2005.00503.x

[92] C. A. Kothe and S. Makeig, "Bcilab: a platform for brain–computer interface development," *Journal of Neural Engineering*, vol. 10, no. 5, p. 056014, 2013. [Online]. Available: http://stacks.iop.org/1741-2552/10/i=5/a=056014

[93] D. Baehrens, T. Schroeter, S. Harmeling, M. Kawanabe, K. Hansen, and K.-R. MÃžller, "How to explain individual classification decisions," *Journal of Machine Learning Research*, vol. 11, no. Jun, pp. 1803–1831, 2010.

[94] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Computer Vision – ECCV 2014*, D. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Cham: Springer International Publishing, 2014, pp. 818–833.

[95] A. M. Nguyen, J. Yosinski, and J. Clune, "Deep neural networks are easily fooled: High confidence predictions for unrecognizable images," *CoRR*, vol. abs/1412.1897, 2014. [Online]. Available: http://arxiv.org/abs/1412.1897

可用：http://dl [76] F. Chollet, “Keras,” https://github.com/fchollet/keras, 2015.

[77] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, 和 H. Zhang, “Filter bank common spatial pattern algorithm on bci competition iv datasets 2a and 2b,” Frontiers in Neuroscience, 卷 6, 页 39, 2012. [在线]. 可用: http://journal.frontiersin.org/article/10.3389/fnins.2012.00039 [78] M. Dyrholm, C. Christoforou, 和 L. C. Parra, “Bilinear discriminant component analysis,” Journal of Machine Learning Research, 卷 8, 第 May 期, 页 1097–1111, 2007.

[79] S. Ioffe 和 C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” arXiv, 卷 abs/1502.03167, 2015. [在线]. 可用: http://arxiv.org/abs/1502.03167 [80] D. Clevert, T. Unterthiner, 和 S. Hochreiter, “Fast and accurate deep network learning by exponential linear

单位（elus）,” CoRR，卷 abs/1511.07289，2015 年。[在线]。可获取：http://arxiv.org/abs/1511.07289 [81] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, 和 R. Salakhutdinov, “Dropout: A simple way to

防止神经网络过拟合，”机器学习研究杂志，第 15 卷，第 1929-1958 页，2014 年。[在线]. 可获得：http://jmlr.org/papers/v15/srivastava14a.html [82] J. T. Springenberg, A. Dosovitskiy, T. Brox, 和 M. A. Riedmiller, “追求简单：全卷积网络，”arXiV，编号 abs/1412.6806，2014 年。[在线]. 可获得：http://arxiv.org/abs/1412.6806 [83] B. Rivet, A. Souloumiac, V. Attina, 和 G. Gibert, “xDAWN 算法增强诱发电位：应用

对脑机接口的介绍，”IEEE 生物医学工程汇刊，第 56 卷，第 8 期，第 2035–2043 页，2009 年 8 月。

[84] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, “基于黎曼几何的多类脑机接口分类,” IEEE 生物医学工程汇刊, 第 59 卷, 第 4 期, 第 920-928 页, 2012 年 4 月.

[85] A. Barachant 和 M. Congedo, “一个基于信息几何的即插即用 P300 BCI,” arXiv:1409.0107 [cs, stat], 2014 年 8 月, arXiv: 1409.0107. [在线]. 可获得: http://arxiv.org/abs/1409.0107 [86] M. Congedo, A. Barachant, 和 A. Andreev, “一种基于...的新一代脑机接口”

黎曼几何,” CoRR，第 abs/1310.8115 卷，2013 年。[在线]。可获取：http://arxiv.org/abs/1310.8115 [87] A. Barachant 和 S. Bonnet，“使用黎曼距离进行脑机接口应用的信道选择程序，”在 2011 年第五届国际 IEEE/EMBS 神经工程会议，2011 年 4 月，第 348-351 页。

[88] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, “使用基于黎曼的核对协方差矩阵进行分类，用于 BCI 应用,” Neurocomputing, vol. 112, pp. 172–178, Jul. 2013.

[89] A. Y. Ng, “特征选择、L1 与 L2 正则化、旋转不变性,” 在《会议录》中

第二十一届机器学习国际会议，系列 ICML '04。美国纽约，纽约州：ACM，2004 年，第 78 页至。[在线]。可获取：http://doi.acm.org/10.1145/1015330.1015435 [90] O. Ledoit 和 M. Wolf，“一种适用于高维协方差矩阵的良好条件估计量，”《多元统计分析杂志》，第 88 卷，第 2 期，第 365 页至 411 页，2004 年。

[91] H. Zou 和 T. Hastie, “通过弹性网络进行正则化和变量选择,”《皇家学会学报》

统计学会：B 系列（统计方法），第 67 卷，第 2 期，第 301-320 页，2005 年。[在线]. 可获得：http://dx.doi.org/10.1111/j.1467-9868.2005.00503.x [92] C. A. Kothe 和 S. Makeig，“Bcilab：一个脑机接口开发平台”，

神经工程杂志，第 10 卷，第 5 期，第 056014 页，2013 年。[在线]. 可获得：http: //stacks.iop.org/1741-2552/10/i=5/a=056014 [93] D. Baehrens, T. Schroeter, S. Harmeling, M. Kawanabe, K. Hansen 和 K.-R. M ˜Aˇzller，“如何解释

个体分类决策，”机器学习研究杂志，第 11 卷，第 6 月期，第 1803-1831 页，2010 年。

[94] M. D. Zeiler 和 R. Fergus，“可视化和理解卷积网络，”在计算机视觉 – ECCV 2014, D. Fleet, T. Pajdla, B. Schiele, 和 T. Tuytelaars 编. Cham: Springer International Publishing, 2014, 第 818–833 页.

[95] A. M. 阮, J. Yosinski 和 J. Clune, “深度神经网络很容易被欺骗：高对无法识别图像的置信度预测，”CoRR，卷 abs/1412.1897，2014。[在线]。可获取：http://arxiv.org/abs/1412.1897

[96] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22Nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '16. New York, NY, USA: ACM, 2016, pp. 1135–1144. [Online]. Available: http://doi.acm.org/10.1145/2939672.2939778

[97] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," *CoRR*, vol. abs/1704.02685, 2017. [Online]. Available: http://arxiv.org/abs/1704.02685

[98] M. Ancona, E. Ceolini, C. Öztireli, and M. Gross, "Towards better understanding of gradient-based attribution methods for deep neural networks," in *International Conference on Learning Representations*, 2018. [Online]. Available: https://openreview.net/forum?id=Sy21R9JAW

[99] G. Montavon, W. Samek, and K.-R. Müller, "Methods for interpreting and understanding deep neural networks," *Digital Signal Processing*, vol. 73, pp. 1 – 15, 2018. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S1051200417302385

[100] C. Torrence and G. P. Compo, "A practical guide to wavelet analysis," *Bulletin of the American Meteorological society*, vol. 79, no. 1, pp. 61–78, 1998.

[101] I. Sturm, S. Lapuschkin, W. Samek, and K.-R. Müller, "Interpretable deep neural networks for single-trial eeg classification," *Journal of Neuroscience Methods*, vol. 274, pp. 141 – 145, 2016. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0165027016302333

[102] A. Mazaheri and T. W. Picton, "Eeg spectral dynamics during discrimination of auditory and visual targets," *Cognitive Brain Research*, vol. 24, no. 1, pp. 81 – 96, 2005. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0926641004003519

[103] G. Johnson, N. Waytowich, and D. J. Krusienski, "The challenges of using scalp-eeg input signals for continuous device control," in *Foundations of Augmented Cognition. Directing the Future of Adaptive Systems*, D. D. Schmorrow and C. M. Fidopiastis, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 525–527.

[104] V. Lawhern, W. D. Hairston, K. McDowell, M. Westerfield, and K. Robbins, "Detection and classification of subject-generated artifacts in {EEG} signals using autoregressive models," *Journal of Neuroscience Methods*, vol. 208, no. 2, pp. 181 – 189, 2012.

[96] M. T. Ribeiro, S. Singh, 和 C. Guestrin, “为什么我应该信任你？：解释任何预测”

"分类器，”在《第 22 届 ACM SIGKDD 国际知识发现与数据挖掘会议论文集》中，KDD '16 系列。美国纽约：ACM，2016 年，第 1135-1144 页。[在线]. 可获得：http://doi.acm.org/10.1145/2939672.2939778 [97] A. Shrikumar、P. Greenside 和 A. Kundaje，“通过传播激活学习重要特征”

差异，”CoRR，第 abs/1704.02685 卷，2017 年。[在线]。可访问：http://arxiv.org/abs/1704.02685 [98] M. Ancona, E. Ceolini, C. ¨Oztireli 和 M. Gross，“为了更好地理解深度神经网络的基于梯度的归因方法，”在 2018 年国际学习表示会议。[在线]。

可用：https://openreview.net/forum?id=Sy21R9JAW [99] G. Montavon, W. Samek, and K.-R. M¨uller, “用于解释和理解深度

神经网络，” 数字信号处理，第 73 卷，第 1-15 页，2018 年。[在线]. 可获得：

http://www.sciencedirect.com/science/article/pii/S10512004173023852014 [100] C. Torrence 和 G. P. Compo, “一种实用的波动分析指南，” 美国气象学会公报，第 79 卷，第 1 期，第 61-78 页，1998 年。

[101] I. Sturm, S. Lapuschkin, W. Samek, and K.-R. M¨uller, “可解释的深度神经网络用于单次试验

脑电图分类，”神经科学方法杂志，第 274 卷，第 141-145 页，2016 年。[在线]. 可获得：

http://www.sciencedirect.com/science/article/pii/S0165027016302333 [102] A. Mazaheri 和 T. W. Picton，“听觉和...的脑电图频谱动力学”

视觉目标，”认知脑研究，第 24 卷，第 1 期，第 81-96 页，2005 年。[在线]. 可获得：

http://www.sciencedirect.com/science/article/pii/S0926641004003519 [103] G. Johnson, N. Waytowich, 和 D. J. Krusienski, “使用头皮脑电图输入信号进行连续设备控制面临的挑战，” 在《增强认知的基础。指导自适应系统的未来》中，D. D.

Schmorrow 和 C. M. Fidopiastis, 编. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, 第 525–527 页.

[104] V. Lawhern, W. D. Hairston, K. McDowell, M. Westerfield, and K. Robbins, “Detection and classification of 使用自回归模型在{EEG}信号中生成主题产生的伪影，”神经科学方法杂志，第 208 卷，第 2 期，第 181-189 页，2012 年。